

Singularities in Commutative Algebra Through Cohomological Methods

Notes for an SLMath Graduate Summer School

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Warning!

Proceed with caution. These notes are under construction and are 100% guaranteed to contain typos. If you find any typos or errors, we will be most grateful to you for letting us know.

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These notes take much inspiration from other sources, especially Avramov's *Infinite free resolutions* [Avr10].

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Chapter 0

Background

0.1 Setup

Throughout, unless otherwise stated, noetherian local rings are assumed to be commutative with identity 1. Moreover, whenever we talk about modules over a noetherian local ring, we assume them to be unital, and all ring homomorphisms between noetherian local rings send 1 to 1. Noncommutative rings will also play an important role when we talk about dg algebras and universal enveloping algebras; the commutativity assumption only applies in the local case.

We will be primarily be concerned with noetherian local rings and finitely generated modules over such rings. However, the reader may be more familiar with the case of graded rings and modules. There are many parallels between these two settings, where the homogeneous maximal ideal in the graded case plays an analogous role to the unique maximal ideal in the local case. There is a sort of metatheorem that says that *most* theorems that hold in the local case have an analogous result in the graded setting, by adding in the words *homogeneous* and *graded* in the appropriate places. However, this does not mean that it is sufficient to prove a theorem in one of the two settings to obtain the other: often one needs to adapt the techniques slightly, and in rare occasions we even need an entirely different argument in each case. We should view this analogy as informing what we expect to be true in each case, but still treat the two cases as separate.

While these notes focus on the local case, many of the theorems we will state hold in both settings, though we note that the proofs can be quite different. The table below indicates the translations between the two settings that may serve as motivation to any reader who feels more comfortable in the graded setting.

local setting	graded setting
$(R, \mathfrak{m}, k)$ noetherian local ring	$R = k[x_1, \dots, x_d]/I$ $k[x_1, \dots, x_d]$ standard graded, k field, I homogeneous
M is a finitely generated R -module	M is a finitely generated <u>graded</u> R -module
$\mathfrak{m}$ the unique maximal ideal	$\mathfrak{m} = (x_1, \dots, x_d)$ unique homogeneous maximal ideal

0.2 Free resolutions

Let R be a noetherian local ring. Given a finitely generated R -module M , we can completely describe M by giving a **presentation**, which consists of an R -module homomorphism

$$R^m \xrightarrow{\pi} R^n$$

with $\text{coker}(\pi) = M$. If $x_1, \dots, x_n \in M$ are the images of a basis for R^n under the canonical quotient $R^n \twoheadrightarrow \text{coker}(\pi) = M$, then $x_1, \dots, x_n$ form a generating set for M . Moreover, the images of a basis for R^m give generators for the **relations** among $x_1, \dots, x_n$. More precisely, $(a_1, \dots, a_n) \in \text{im}(\pi)$ if and only if

$$a_1 x_1 + \dots + a_n x_n = 0.$$

Note that since M is finitely generated and R is noetherian, we can choose such a presentation with m and n finite. In our local setting, we may even take such a presentation to be minimal, meaning we can choose m and n to be smallest possible. This is equivalent to choosing $x_1, \dots, x_n$ such that their images in $M/\mathfrak{m}M$ form a basis for $M/\mathfrak{m}M$ as a vector space of $k = R/\mathfrak{m}$, and to choose π such that $\text{im}(\pi) \subseteq \mathfrak{m}R^n$.

While such a presentation (minimal or not) determines M completely, we get a lot more information about M if we continue this process and construct a free resolution for M .

Definition 0.2.1. Let M be an R -module. A **free resolution** of M is a complex of free R -modules

$$F = \dots \longrightarrow F_n \longrightarrow \dots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow 0$$

$\qquad\qquad\qquad \text{\scriptsize n} \qquad\qquad\qquad \qquad\qquad\qquad \text{\scriptsize 1} \qquad\qquad\qquad \text{\scriptsize 0}$

together with an isomorphism $H_0(F) \cong M$ such that $H_i(F) = 0$ for all $i \neq 0$. We will abuse notation and carelessly identify a resolution for M with the **augmented resolution**

$$\dots \longrightarrow F_n \longrightarrow \dots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0.$$

$\qquad\qquad\qquad \text{\scriptsize n} \qquad\qquad\qquad \qquad\qquad\qquad \text{\scriptsize 1} \qquad\qquad\qquad \text{\scriptsize 0}$

Remark 0.2.2. We can also talk about projective resolutions, where instead we allow the F_i to be projective modules. Over a noetherian local ring, Kaplansky showed that every projective module is free [Kap58], so we will focus only on free resolutions.

Every R -module has a free resolution; we will see next that this is a consequence of the fact that every R -module is a quotient of a free module. One indicates this by saying that the category of R -modules has enough projectives.

Construction 0.2.3 (Free resolution). Let M be a finitely generated module over R with β_0 generators, say $m_1, \dots, m_{\beta_0}$. The R -module homomorphism induced by

$$\begin{array}{ccc} R^{\beta_0} & \xrightarrow{\pi_0} & M \\ \mathbf{e}_i & \longmapsto & m_i \end{array}$$

is surjective by construction.

If π_0 is an isomorphism, then $M \cong R^{\beta_0}$ is a **free module** of **free rank** β_0 . Otherwise, π_0 has a nonzero kernel $\ker(\pi_0)$, which must also be a finitely generated module since R is noetherian. Fix generators $s_1, \dots, s_{\beta_1}$ for $\ker(\pi_0)$. We now consider the surjective R -module homomorphism induced by

$$\begin{aligned} R^{\beta_1} &\xrightarrow{\pi_1} \ker(\pi_0) \\ \mathbf{e}_i &\longmapsto s_i. \end{aligned}$$

Composing π_1 with the inclusion of $\ker(\pi_0)$ into R^{β_0} , we get a map $\partial_1: R^{\beta_1} \rightarrow R^{\beta_0}$; this is a presentation for M , and we get an exact sequence

$$\begin{array}{ccccc} R^{\beta_1} & \xrightarrow{\quad \partial_1 \quad} & R^{\beta_0} & \xrightarrow{\pi_0} & M. \\ & \searrow \pi_1 & \swarrow & & \\ & \ker(\pi_0) & & & \end{array}$$

We can then continue this process and construct a free resolution for M :

$$\cdots \longrightarrow F_n \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_2} F_1 \xrightarrow{\pi_1} F_0 \xrightarrow{\pi_0} M \longrightarrow 0.$$

At each step, we find the kernel of $\partial_d: R^{\beta_d} \rightarrow R^{\beta_{d-1}}$. If $\ker(\partial_d) = 0$, then the resolution has stopped:

$$0 \longrightarrow F_d \xrightarrow{\partial_d} \cdots \xrightarrow{\partial_2} F_1 \xrightarrow{\pi_1} F_0 \xrightarrow{\pi_0} M \longrightarrow 0.$$

Otherwise, we find a surjection from a free module $\pi_{d+1}: R^{\beta_{d+1}} \twoheadrightarrow \ker(\pi_d)$ and compose it with the inclusion $\ker(\pi_d) \hookrightarrow R^{\beta_d}$ to obtain the next map $\partial_{d+1}: R^{\beta_{d+1}} \rightarrow R^{\beta_d}$.

At each step, as long as we are in the local setting, we can choose F_d to have the minimal number of generators possible; in that case, we say that F is a **minimal free resolution** for M . It turns out that no matter what choices we make along the way, the minimal number of generators for the free module in degree d is a well-defined invariant of M . In fact, the following hold:

- Every free resolution of M has a minimal free resolution of M as a direct summand.
- Any two minimal free resolutions of M are isomorphic complexes. From now on, we will refer to *the* minimal free resolution of M .
- As a consequence of the previous facts, the minimal free resolution of M must have the shortest length of any resolution for M , and thus M has a finite resolution if and only if the minimal free resolution of M is finite.
- A free resolution F of M with differential ∂ is minimal if and only if $\partial(F) \subseteq \mathfrak{m}F$. Thus if we fix bases for all the free modules F_i , the resolution is minimal if and only if all the entries in the matrices representing ∂ have all entries in $\mathfrak{m}$.

Definition 0.2.4 (Betti numbers). Let F be the minimal free resolution of M . The i th Betti number of M is the free rank of F_i :

$$\beta_i(M) := \text{rank}(F_i).$$

Remark 0.2.5. If F is the minimal free resolution for M , $F_i \cong R^{\beta_i(M)}$.

Exercise 0.2.6. Show that $\beta_i(M) = \dim_k(\text{Tor}_i^R(M, k)) = \dim_k(\text{Ext}_R^i(M, k))$.

Remark 0.2.7. Note that in particular $\beta_0(M) = \mu(M) = \dim_k(M/\mathfrak{m}M)$ is the minimal number of generators of M .

Such relations are called **syzygies**.

Definition 0.2.8. Consider a minimal free resolution F of M . The n th syzygy module of M ,¹ denoted $\text{Syz}_n(M)$, is defined to be the image of ∂_n , or equivalently the kernel of ∂_{n-1} .

Definition 0.2.9. Let M be a nonzero R -module. A finite free resolution

$$F = \cdots \longrightarrow F_c \longrightarrow \cdots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow 0$$

has length c if $F_c \neq 0$ and $F_i = 0$ for all $i \geq c$. A free resolution F has infinite length if $F_i \neq 0$ for all $i \geq 0$. The **projective dimension** of M is

$$\begin{aligned} \text{pdim}_R(M) &:= \inf \{c \mid M \text{ has a projective resolution of length } c\} \\ &= \text{length of any minimal free resolution for } M. \end{aligned}$$

We may write $\text{pdim}(M)$ whenever the underlying ring is clear from context.

Macaulay2. In Macaulay2, we can compute free resolutions using the command `res`. For an R -module M , `res(M)` will return a free resolution for M , which will by default be minimal as long as M is a graded R -module. Given an ideal I , `res(I)` outputs a resolution for R/I .

The projective dimension of a finitely generated module can be both finite and infinite.

Example 0.2.10. Let k be a field, $R = k[[x]]$, and consider the residue field $k = R/(x)$. This R -module is cyclic, and the kernel of the canonical map $R \twoheadrightarrow R/(x)$ has only one generator, (x) , so the resolution begins with

$$R \xrightarrow{x} R \longrightarrow R/(x) \longrightarrow 0.$$

Since R is a domain, x is a regular element, that is, not a zerodivisor, and thus the leftmost map is injective, completing our resolution. Since x is in the maximal ideal, we conclude that this is the minimal free resolution for k :

$$0 \longrightarrow R \xrightarrow{x} R \longrightarrow R/(x) \longrightarrow 0.$$

¹Fun fact: in astronomy, a syzygy is an alignment of three or more celestial objects.

Example 0.2.11. Let k be a field and $R = k[[x]]/(x^3)$. When we resolve the residue field $k \cong R/(x)$, we now note that multiplication by x on R does in fact have a nontrivial kernel, and thus the resolution is a bit more complicated. In fact, the resolution is infinite:

$$\cdots \longrightarrow R \xrightarrow{x} R \xrightarrow{x^2} R \xrightarrow{x} R \longrightarrow R/(x) \longrightarrow 0.$$

In particular, $\text{pdim}_R(k) = \infty$ and $\beta_i(k) = 1$ for all i .

The last two examples illustrate a very important theorem of Auslander–Buchsbaum and Serre. First, it is not a coincidence that the projective dimension of the residue field is finite in the first case and infinite in the second case: this is a reflection of the fact that $k[[x]]$ is a regular ring while $k[[x]]/(x^3)$ is singular. In fact, the residue field contains a lot of information.

Exercise 0.2.12. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. Show that for all finitely generated R -modules M ,

$$\text{pdim}_R(M) \leq \text{pdim}_R(k).$$

Exercise 0.2.13. Let $(R, \mathfrak{m})$ be a noetherian local ring and A and B be R -modules such that A is a direct summand of B . Show that $\text{pdim}_R(A) \leq \text{pdim}_R(B)$.

What can resolutions tell us about singularities? First, we recall the definition of a regular ring; these are the rings that are nonsingular.

Definition 0.2.14. A noetherian local ring $(R, \mathfrak{m}, k)$ is a **regular local ring** if $\mathfrak{m}$ is minimally generated by $\dim(R)$ many elements, that is

$$\mu(\mathfrak{m}) = \dim(R).$$

The minimal number of generators of $\mathfrak{m}$ is known as the **embedding dimension** of R , written $\text{embdim}(R)$. We will sometimes write **RLR** as shorthand for regular local ring.

Remark 0.2.15. Recall that by Krull’s Height Theorem, we always have

$$\dim(R) = \text{height}(\mathfrak{m}) \leq \mu(\mathfrak{m}) = \text{embdim}(R).$$

Thus a local ring is regular if its maximal ideal has the smallest possible number of generators.

I. S. Cohen [Coh46] proved that any complete regular local ring containing a field is of the form $k[[x_1, \dots, x_d]]$ for k a field. In fact, he proved more:

Theorem 0.2.16 (Cohen’s Structure Theorem). *Every complete noetherian local ring can be written as a quotient of a regular local ring.*

Exercise 0.2.17. Let $(R, \mathfrak{m}, k)$ be a regular local ring. Show that $R/(f)$ is a regular local ring if and only if $f \in \mathfrak{m} \setminus \mathfrak{m}^2$.

Thanks to Cohen's Structure Theorem and [Exercise 0.2.17](#), we can write any complete regular ring in the form $R = Q/I$ with $(Q, \mathfrak{m})$ a regular local ring and $I \subseteq \mathfrak{m}^2$.

Definition 0.2.18. Given a noetherian local ring R , a **minimal regular presentation** for R consists of a regular local ring $(Q, \mathfrak{m})$ and an ideal $I \subseteq \mathfrak{m}^2$ in Q such that $\widehat{R} \cong Q/I$, where $\widehat{R}$ is the completion of R at its maximal ideal.

Exercise 0.2.19. Show that if $\widehat{R} \cong Q/I$ is a minimal regular presentation for R , then

$$\text{embdim}(Q) = \text{embdim}(\widehat{R}) = \text{embdim}(R).$$

For those unfamiliar with completions or DVRs, a good starting point is to assume that from now on we will be talking about complete equicharacteristic rings: rings of the form $R = k[[x_1, \dots, x_e]]/I$, where k is a field and $I \subseteq (x_1, \dots, x_e)^2$.

Note that in such a minimal regular presentation, $I = 0$ if and only if R is a regular ring. When R is not regular, we will say R is a **singular ring**. The big goal of these lectures is to introduce homological tools that can be used to detect singularities. The next theorem is the first instance of this phenomenon.

Theorem 0.2.20 (Auslander–Buchsbaum, Serre [[AB57](#), [Ser56](#)]). *The following are equivalent for any noetherian local ring $(R, \mathfrak{m}, k)$:*

- 1) *The ring R is regular.*
- 2) *The residue field k has finite projective dimension.*
- 3) *Every finitely generated R -module has finite projective dimension.*

This theorem marks the advent of homological methods in commutative algebra. One amazing consequence of having such a homological characterization of regular rings is that it allows us to solve the Localization Problem for regular rings: if P is a prime ideal in a regular local ring R , is R_P a regular local ring? Many good properties of a local ring are preserved by localization, so this is a reasonable question. Note that a positive answer to the Localization Problem gives us a natural choice for a global definition of regular ring: a noetherian ring R is regular if R_P is a regular local ring for all primes P .

This problem is unwieldy via the embedding dimension definition, as it would require us to show that for all primes P ,

$$\mu(P_P) = \dim(R_P) = \text{height}(P).$$

But this now follows easily via the homological characterization of regularity.

Exercise 0.2.21 (The Localization Problem for Regular Rings). Let R be a regular local ring. Show that for all prime ideals P , the localization R_P is a regular local ring.

One big difference between [Example 0.2.10](#) and [Example 0.2.11](#) is whether x is a regular element on R . This leads us to the next topic: regular sequences.

0.3 Regular sequences and depth

While we assume that the reader is familiar with regular sequences and depth, we will now collect the most important definitions and facts we will use throughout these lectures. More details on these topics can be found in Bruns and Herzog's *Cohen-Macaulay rings* [BH98].

Definition 0.3.1 (Regular sequence). Let R be a ring and M be an R -module. An element $x \in R$ is **regular** on M if $xM \neq 0$ and for any $m \in M$

$$xm = 0 \Rightarrow m = 0.$$

Definition 0.3.2. A sequence of elements $x_1, \dots, x_n$ is a **regular sequence on M** if

- $(x_1, \dots, x_n)M \neq 0$, and
- for each i , the element x_i is regular on $M/(x_1, \dots, x_{i-1})M$.

When $M = R$, we may drop the *on M* and say $x_1, \dots, x_n$ is a regular sequence.

In general, whether $x_1, \dots, x_n$ is a regular sequence depends on the order of the elements. However, in the local setting, this is independent of the order of the elements. One can also show that all maximal regular sequences on M have the same length.

Definition 0.3.3. Let $(R, \mathfrak{m})$ be a noetherian local ring and M a finitely generated R -module. The maximal length n of $\underline{x} = x_1, \dots, x_n \in \mathfrak{m}$ a regular sequence on M is the **depth** of M , denoted $\text{depth}(M)$.

Exercise 0.3.4. Let $(R, \mathfrak{m})$ be a noetherian local ring and M a finitely generated R -module. Show that $\text{depth}(M) = 0$ if and only if $\mathfrak{m} \in \text{Ass}(M)$.

Finding a maximal regular sequence on a module M boils down to identifying associated primes of M and some of its quotients.

Remark 0.3.5. Here is a recipe for finding the depth of M , and for constructing an explicit maximal regular sequence on M .

First, we determine $\text{Ass}(M)$. If $\mathfrak{m} \in \text{Ass}(M)$, then $\text{depth}(M) = 0$ and we are done. If not, then we find an element $x_1 \in \mathfrak{m}$ that is not in any associated prime of M , which exists by Prime Avoidance. Such an x_1 is necessarily regular on M , given that the union of the associated primes of M coincides with the set of zerodivisors on M .

We then proceed analogously, now on the module $M/(x_1)M$. At each step, we determine whether $\mathfrak{m}$ is associated to $M/(x_1, \dots, x_i)M$, and if not then we find a new element $x_{i+1} \in \mathfrak{m}$ that is not in any of the associated primes of $M/(x_1, \dots, x_i)M$. At each stage, the depth drops by one, and thus this process must eventually stop.

We can also measure depth via the (non)vanishing certain Ext modules.

Theorem 0.3.6. *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring and M a finitely generated R -module. Then*

$$\text{depth}(M) = \min\{i \mid \text{Ext}_R^i(k, M) \neq 0\}.$$

Theorem 0.3.7 (Auslander–Buchsbaum Formula). *Let R be local and let M be a finitely generated (graded) R -module of finite projective dimension. Then*

$$\text{pdim}_R(M) + \text{depth}(M) = \text{depth}(R).$$

Given a noetherian local ring R , one can show that we always have

$$\text{depth}(R) \leq \dim(R).$$

It is natural to ask when equality holds; this leads to a very important class of rings.

Definition 0.3.8. A noetherian local ring R is **Cohen-Macaulay** if $\text{depth}(R) = \dim(R)$.

Mel Hochster famously wrote that *life is really worth living in a (...) Cohen-Macaulay ring*.² Hochster was likely referring to the vast wealth of nice properties that Cohen-Macaulay rings enjoy. These are often properties one would hope are true in general, but that (sometimes surprisingly) can fail for rings that are not Cohen-Macaulay. For example, over a Cohen-Macaulay ring we have the following desirable dimension formula:

Theorem 0.3.9. *Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring, and let I be any proper ideal in R . Then*

$$\dim(R/I) = \dim(R) - \text{height}(I).$$

For our purposes, we will often use this property in the special case when R is a regular local ring; as a consequence of [Theorem 0.3.12](#), all regular local rings are Cohen-Macaulay.

Over any Cohen-Macaulay ring (so for example, over any regular ring), we have a nice simple characterization of ideals generated by a regular sequence:

Corollary 0.3.10. *Let $(R, \mathfrak{m})$ be a Cohen-Macaulay local ring, and let I be a proper ideal in R . The following are equivalent:*

- 1) *The ideal I is generated by a regular sequence.*
- 2) *Every minimal generating set for I is a regular sequence.*
- 3) $\mu(I) = \text{height}(I)$.

Remark 0.3.11. Recall that by Krull’s Height Theorem, we always have

$$\mu(I) \geq \text{height}(I).$$

Thus an ideal is generated by a regular sequence exactly when it has the smallest possible number of generators for its height.

This should remind us of [Remark 0.2.15](#). Indeed:

Theorem 0.3.12. *A noetherian local ring $(R, \mathfrak{m})$ is regular if and only if $\mathfrak{m}$ is generated by a regular sequence.*

²“Life is really worth living in a noetherian ring R when all the local rings have the property that every system of parameters is an R -sequence. Such a ring is called Cohen-Macaulay (C-M for short).”

Chapter 1

Differential graded algebras

1.1 The Koszul complex

Definition 1.1.1 (Koszul complex). The **Koszul complex** on $x_1, \dots, x_n \in R$, denoted $\text{Kos}^R(x_1, \dots, x_n)$, or $\text{Kos}(x_1, \dots, x_n)$ if R is clear from context, is a complex of free R -modules, defined as follows.

- In degree d , we have the R -module

$$\wedge^d(R^n) = R^{\binom{n}{d}}$$

with basis

$$e_{i_1} \wedge \cdots \wedge e_{i_d} \quad \text{where } 1 \leq i_1 < \cdots < i_d \leq n.$$

In particular, note that $\text{Kos}(x_1, \dots, x_n)$ lives in homological degrees 0 through n .

- The differential is defined on basis elements as follows:

$$\partial(e_{i_1} \wedge \cdots \wedge e_{i_s}) = \sum_{1 \leq p \leq s} (-1)^{p-1} x_{i_p} e_{i_1} \wedge \cdots \wedge \widehat{e_{i_p}} \wedge \cdots \wedge e_{i_s}.$$

Example 1.1.2. In the case of two elements, say f and g in R ,

$$\text{Kos}(x, y) = 0 \longrightarrow \wedge^2 R^2 \longrightarrow \wedge^1 R^2 \longrightarrow R \longrightarrow 0$$

with $\partial(e_1) = f$, $\partial(e_2) = g$, $\partial(e_1 \wedge e_2) = fe_2 - ge_1$, so

$$\text{Kos}(x, y) = 0 \longrightarrow R \xrightarrow{\begin{pmatrix} -g \\ f \end{pmatrix}} R^2 \xrightarrow{\begin{pmatrix} f & g \end{pmatrix}} R \longrightarrow 0.$$

Exercise 1.1.3. Write the Koszul complex on 3 elements f_1, f_2, f_3 .

Note, however, that the Koszul complex may have nontrivial homology.

Theorem 1.1.4. *Let R be local, and let $f_1, \dots, f_n \in R$ be nonunits. The following are equivalent:*

- 1) *The elements $f_1, \dots, f_n$ form a regular sequence.*
- 2) *The Koszul complex $\text{Kos}(f_1, \dots, f_n)$ is a free resolution of $R/(f_1, \dots, f_n)$, that is, $H_i(\text{Kos}(f_1, \dots, f_n)) = 0$ for all $i \neq 0$.*
- 3) *The first Koszul homology vanishes: $H_1(\text{Kos}(f_1, \dots, f_n)) = 0$.*

Remark 1.1.5. The nonzero entries in the differentials of $\text{Kos}(\underline{f})$ are all of the form $\pm f_i$, and thus when $\underline{f}$ is a regular sequence $\text{Kos}(\underline{f})$ is a minimal free resolution for $R/(\underline{f})$.

Remark 1.1.6. By [Theorem 0.3.12](#), when R is regular, the maximal ideal is generated by a regular sequence. The Koszul complex on such a regular sequence is thus a minimal free resolution for k over R .

One can also define the Koszul complex inductively, by taking successive tensor products.

Notation 1.1.7. Given any complex A , we write $|a| := i$ to indicate that $a \in A_i$.

Definition 1.1.8. Let A and B be complexes of R -modules. The **tensor product** of A and B is the complex $A \otimes_R B$ with

$$(A \otimes_R B)_n = \bigoplus_{i+j=n} A_i \otimes_R B_j$$

and

$$\partial_{A \otimes_R B}(a \otimes b) = \partial_A(a) \otimes b + (-1)^{|a|} a \otimes \partial_B(b).$$

Remark 1.1.9. The tensor product of two complexes A and B is the totalization of the double complex with

$$(A \otimes B)_{p,q} = A_p \otimes B_q \quad d^h = \partial_A \otimes \text{id}_B, \quad \text{and} \quad d^v = (-1)^p \text{id}_A \otimes \partial_B.$$

We are now ready for an alternative definition of the Koszul complex.

Construction 1.1.10 (The Koszul complex). The **Koszul complex** on one element $x \in R$ is the complex

$$\text{Kos}(x) := 0 \longrightarrow R \xrightarrow{x} R \longrightarrow 0.$$

1 0

More generally, given $\underline{x} = x_1, \dots, x_n \in R$, the **Koszul complex** with respect to $\underline{x}$ is the complex $\text{Kos}^R(\underline{x}) = \text{Kos}(x_1, \dots, x_n)$ defined inductively as

$$\text{Kos}(x_1, \dots, x_n) := \text{Kos}(x_1, \dots, x_{n-1}) \otimes_R \text{Kos}(x_n).$$

Example 1.1.11. The Koszul complex on $f, g \in R$ is given by

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & R & \xrightarrow{f} & R & \longrightarrow & 0 \\
 & & \downarrow -g & & \downarrow g & & \\
 0 & \longrightarrow & R & \xrightarrow{f} & R & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & &
 \end{array}$$

$\text{Kos}(f, g) = \text{Totalization of}$

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & R & \xrightarrow{f} & R & \longrightarrow & 0 \\
 & & \downarrow -g & & \downarrow g & & \\
 0 & \longrightarrow & R & \xrightarrow{f} & R & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & &
 \end{array}$$

which is

$$0 \longrightarrow R \xrightarrow{\begin{pmatrix} -g \\ f \end{pmatrix}} R^2 \xrightarrow{\begin{pmatrix} f & g \end{pmatrix}} R \longrightarrow 0.$$

Exercise 1.1.12. Let $I = (x_1, \dots, x_n)$ be an ideal in a noetherian ring R , and $\underline{x} = x_1, \dots, x_n$.

- 1) $H_i(\text{Kos}(\underline{x})) = 0$ whenever $i < 0$ or $i > n$.
- 2) $H_0(\text{Kos}(\underline{x})) = R/I$.
- 3) $H_n(\text{Kos}(\underline{x})) = (0 : I) = \text{ann}(I)$.
- 4) The Koszul homology modules $H_i(\text{Kos}(\underline{x}))$ are finitely generated for every i .
- 5) Every Koszul homology module $H_i(\text{Kos}(\underline{x}))$ is killed by I .

Hint: show that for all $a \in I$, multiplication by a on $\text{Kos}(\underline{x})$ is nullhomotopic.

Remark 1.1.13. Set $C := \text{Kos}(x_1, \dots, x_n)$, and note that

$$[\text{Kos}(x_1, \dots, x_{n+1})]_i = C_{i-1} \otimes_R R \oplus C_i \otimes_R R \cong C_{i-1} \oplus C_i.$$

Given $a \in C_{i-1}$, $b \in C_i$, $r \in R$ (in homological degree 1) and $s \in R$ (in homological degree 0),

$$\partial(a \otimes r + b \otimes s) = \partial(a) \otimes r + (-1)^{i-1} a \otimes (x_{n+1} r) + \partial(b) \otimes s + (-1)^i b \otimes 0,$$

so

$$\partial_i := \begin{pmatrix} \partial_C & 0 \\ (-1)^{i-1} x_{n+1} & \partial_C \end{pmatrix} : \begin{array}{ccc} C_{i-1} & \xrightarrow{\partial_C} & C_{i-2} \\ \oplus & \searrow^{(-1)^{i-1} x_{n+1}} & \oplus \\ C_i & \xrightarrow{\partial_C} & C_{i-1} \end{array}$$

In fact, this is the cone of the map $\text{Kos}(x_1, \dots, x_n) \rightarrow \text{Kos}(x_1, \dots, x_n)$ given by multiplication by x_{n+1} in every degree. Note that there are different sign conventions on what the differential of a cone is, so this might not match your favorite definition. For the reader unfamiliar with cones, we will discuss this topic in more detail later.

1.2 Differential graded algebras

When we first introduced the Koszul complex, we used the suggestive notation $\wedge^d(R^n)$ when talking about the free modules in each homological degree. Indeed, the Koszul complex has more structure than simply being a complex: it is the exterior algebra on n elements. In fact, the Koszul complex is an example of a differential graded algebra, which we will abbreviate to dg algebra.

Definition 1.2.1. Let R be a commutative ring. A dg (differential graded) algebra over R is a complex (A, ∂) of R -modules equipped with an associated algebra structure compatible with the differential, as follows:

- 1) The underlying graded object

$$A = \bigoplus_{i \in \mathbb{Z}} A_i$$

is a unital associative graded R -algebra.

- 2) The differential ∂ satisfies the **Leibniz rule**: for all $a, b \in A$, we have

$$\partial(ab) = \partial(a)b + (-1)^{|a|}a\partial(b).$$

Therefore, the multiplication $A \otimes_R A \rightarrow A$ is a map of complexes.

Remark 1.2.2. We will follow the convention that whenever A is a dg algebra and we write $a \in A$, we mean that $a \in A_i$ for some i , meaning that a is a homogeneous element. Strictly speaking, an element in the algebra A is in fact a *sum* of homogeneous elements. However, most facts we would ever want to prove about elements in A reduce to the homogeneous case, which justifies our convention.

Definition 1.2.3. A dg algebra is a **(strictly) graded commutative dg algebra** if for all homogeneous elements a and b ,

$$ab = (-1)^{|a| \cdot |b|}ba \quad \text{and} \quad a^2 = 0 \quad \text{whenever } a \text{ has odd degree.}$$

Remark 1.2.4. Note that in characteristic other than 2, the condition

$$ab = (-1)^{|a| \cdot |b|}ba$$

automatically implies that $a^2 = 0$ whenever $|a|$ is odd. However, this does not follow in characteristic 2, and thus we add this condition as well. The *strictly* in strictly graded commutative refers to this added condition that $a^2 = 0$ for all $|a|$ odd. While some authors refer to graded commutative and strictly graded commutative dg algebras separately, we will not consider one condition without the other, so we may shorten it to graded commutative.

Remark 1.2.5. Note that the fact that $A_0 A_0 \subseteq A_0$ implies that A_0 is a (noncommutative) ring. Moreover, A_0 is a commutative ring if A is graded commutative.

Example 1.2.6. Given a ring R , the complex with R in degree 0 and zeroes elsewhere is a dg R -algebra. More generally, an R -algebra is exactly a dg R -algebra concentrated in degree 0.

The canonical example of a dg algebra is the Koszul complex.

Example 1.2.7 (the Koszul complex revisited). Given elements $\underline{f} = f_1, \dots, f_n$ in a commutative ring R , the Koszul complex $E = \text{Kos}(\underline{f})$ is a graded commutative R -algebra with the wedge product induced by

$$(e_{i_1} \wedge \dots \wedge e_{i_s}) \cdot (e_{j_1} \wedge \dots \wedge e_{j_t}) = e_{i_1} \wedge \dots \wedge e_{i_s} \wedge e_{j_1} \wedge \dots \wedge e_{j_t}.$$

Note that $\text{Kos}(\underline{f})$ is the exterior algebra on $e_1, \dots, e_n$. The differential in [Definition 1.1.1](#) is the unique differential with $\partial(e_i) = f_i$ that satisfies the Leibniz rule with respect to this product. We also use the alternative notation

$$\text{Kos}(\underline{f}) = R[e_1, \dots, e_n \mid \partial(e_i) = f_i],$$

which indicates that the Koszul complex was obtained from R by *adjoining* exterior variables $e_1, \dots, e_n$ that kill the cycles $f_1, \dots, f_n$ in degree 0 into boundaries, giving us a dg algebra with

$$H_0(R[e_1, \dots, e_n \mid \partial(e_i) = f_i]) = R/(f_1, \dots, f_n).$$

Notation 1.2.8. Given a complex, we write $[x]$ to denote the homology class of a cycle x .

Remark 1.2.9. Let A be a dg R -algebra. Note that the product of two cycles is a cycle, and thus the cycles $Z(A)$ of A form a subalgebra of A . Moreover, the boundaries $B(A)$ of A form an ideal of $Z(A)$. We conclude that the homology of A

$$H(A) = \bigoplus_i H_i(A)$$

forms a graded R -algebra with multiplication induced by the multiplication on A , meaning that

$$[a] \cdot [b] := [a \cdot b].$$

We leave the details as an exercise (see below). Note that if A is graded commutative, then so is $H(A)$.

Exercise 1.2.10. For a dg algebra A , prove that the multiplication $[a] \cdot [b] := [a \cdot b]$ is well-defined and makes $H(A)$ a graded algebra.

Definition 1.2.11. Let A and B be dg algebras over R . A **homomorphism of dg algebras** $f: A \rightarrow B$ is a (degree 0) map of complexes such that

$$f(xy) = f(x)f(y) \quad \text{and} \quad f(1_A) = 1_B.$$

A map of dg algebras is an isomorphism whenever the underlying map of complexes is an isomorphism. Moreover, $f: A \rightarrow B$ is a **quasiisomorphism of dg algebras** if f is a map of dg algebras and $H(f): H(A) \rightarrow H(B)$ is an isomorphism.

Notation 1.2.12. The symbol $\simeq$ denotes a quasiisomorphism (of either complexes or dg algebras, according to context).

Remark 1.2.13. Let F be a free resolution for an R -module M , and consider the canonical surjection $\pi: F_0 \rightarrow H_0(F) \cong M$. We can view M as a complex concentrated in homological degree 0, and extend π to a map of complexes $F \rightarrow M$, which is a quasiisomorphism of complexes. We will often denote this by

$$F \xrightarrow{\simeq} M.$$

If $M = R/I$ and F has a dg algebra structure, note that this is a quasiisomorphism of dg algebras.

Exercise 1.2.14. Let A and B be dg algebras over R . Check that the tensor product complex $A \otimes_R B$ is also a dg algebra, under the product induced by

$$(a_1 \otimes b_1)(a_2 \otimes b_2) = (-1)^{|a_2| \cdot |b_1|} (a_1 a_2) \otimes (b_1 b_2).$$

Moreover, show that

$$\begin{array}{ccc} A & \longrightarrow & A \otimes_R B \longleftarrow B \\ a & \longmapsto & a \otimes 1 \\ & & 1 \otimes b \longleftarrow b \end{array}$$

are dg algebra maps.

Definition 1.2.15. Given a dg algebra A , an ideal I of the (noncommutative) ring A is a **dg ideal** of A if $\partial_A(I) \subseteq I$.

Example 1.2.16. Let $f \in R$ and consider the Koszul complex $A = \text{Kos}^Q(f)$. Note that the ideal $I = A_{\geq 1}$, which as an A -ideal is generated by $e_1, \dots, e_n$, is not a dg ideal, since $\partial(e_i) = f_i \notin I$. However, the A -ideal

$$J = (e_1, \dots, e_n, f_1, \dots, f_n)$$

is a dg ideal. In fact, J is the dg ideal generated by $e_1, \dots, e_n$.

Remark 1.2.17. Given a dg algebra A and elements $a_1, \dots, a_n \in A$, the dg ideal generated by $a_1, \dots, a_n$, by which we mean the smallest dg ideal containing $a_1, \dots, a_n$, is the ideal

$$(a_1, \dots, a_n, \partial(a_1), \dots, \partial(a_n)).$$

Remark 1.2.18. Suppose that A is a dg algebra and I is a dg ideal in A . Then A/I is a dg algebra, and the canonical quotient map $A \twoheadrightarrow A/I$ is a dg algebra map.

Exercise 1.2.19. Let I be a dg ideal of the dg algebra A . Show that the canonical map $A \twoheadrightarrow A/I$ is a quasiisomorphism if and only if $H(I) = 0$.

Definition 1.2.20. A dg algebra A is a **local dg algebra** if A has a unique maximal dg ideal. We may write $\mathfrak{m}_A$ to indicate the unique maximal dg ideal.

Example 1.2.21. Let $(R, \mathfrak{m})$ be a local ring and let $\underline{f} \in R$. The Koszul complex $A = \text{Kos}(\underline{f})$ is a local dg algebra with unique maximal dg ideal

$$\mathfrak{m}_R \oplus \left(\bigoplus_{i=1}^n Re_i \right) \oplus \left(\bigoplus_{i < j} Re_i e_j \right) \oplus \cdots \oplus Re_1 \cdots e_n = \mathfrak{m}_R \oplus A_{\geq 1}.$$

Remark 1.2.22. More generally, let A be a nonnegatively graded dg algebra, meaning that $A_i = 0$ for all $i < 0$. If $A_0 = R$ is a local ring, then A is a local dg algebra with unique maximal dg ideal

$$\mathfrak{m}_A = \mathfrak{m}_R \oplus A_{\geq 1}.$$

Definition 1.2.23. A map of local dg algebras $\varphi: A \rightarrow B$ is a **local map of dg algebras** if $\varphi(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$.

Example 1.2.24. Let F and G be complexes of R -modules, and let $\text{Hom}_R(F, G)$ denote the complex with

$$\text{Hom}_R(F, G)_n := \prod_i \text{Hom}_R(F_i, G_{i+n})$$

and differential

$$\partial(\varphi) = \partial_G \circ \varphi - (-1)^{|\varphi|} \varphi \circ \partial_F.$$

We leave it as an exercise to check that this is indeed a complex of R -modules; in fact, when $F = G$, this is a dg R -algebra with the product given by composition.

Exercise 1.2.25. Let M and N be R -complexes.

- 1) Check that $\text{Hom}_R(M, N)$ is a complex.
- 2) Describe $Z_0(\text{Hom}_R(M, N))$ and $H_0(\text{Hom}_R(M, N))$.
- 3) Set $\text{End}_R(M) = \text{Hom}_R(M, M)$. Prove that $\text{End}_R(M)$ is a dg R -algebra.

Example 1.2.26. Let F be any complex. Then $\text{End}_R(F) = \text{Hom}_R(F, F)$ is a dg algebra, though it is not graded commutative and not local.

Definition 1.2.27. A (left) **dg module** M over a dg R -algebra A is a complex of R -modules with the structure of a graded module over the underlying ring A that satisfies the Leibniz rule, that is, for all $a \in A$ and $m \in M$,

$$\partial_M(am) = \partial_A(a)m + (-1)^{|a|} a \partial_M(m).$$

Notation 1.2.28. In general, given a dg A -module M , we will write ∂_M for the differential on M .

Example 1.2.29. Given any dg algebra A , if we forget the differential, any dg module over A is in particular a left A -module.

Notation 1.2.30. The **suspension** or **shift** of a complex C is the complex ΣC , sometimes also denoted $C[1]$, with

$$(\Sigma C)_n = C_{n-1} \quad \text{and} \quad \partial^{\Sigma C} = -\partial^C.$$

More generally, for any ℓ , the ℓ **th suspension** of C is the complex $\Sigma^\ell C$ with

$$(\Sigma^\ell C)_n := C_{n-\ell} \quad \text{with} \quad \partial^{\Sigma^\ell C} = (-1)^\ell \partial^C.$$

In particular, $\Sigma^{-1} C$ is the complex with

$$(\Sigma^{-1} C)_n = C_{n+1} \quad \text{and} \quad \partial^{\Sigma^{-1} C} = -\partial^C.$$

Remark 1.2.31. Note that

$$\Sigma^\ell C := \underbrace{\Sigma \cdots \Sigma}_{\ell \text{ times}} C.$$

Example 1.2.32. If A is a dg algebra, then ΣA is a dg A -module via

$$a \cdot (\Sigma b) := (-1)^{|a|} \Sigma(ab).$$

More generally, given an arbitrary dg A -module M , $\Sigma^\ell M$ is a dg A -module:

Exercise 1.2.33. Let A be a dg algebra and M a (left) dg A -module. Prove that for each ℓ , the complex $\Sigma^\ell M$ has a (left) dg A -module structure where the A -action is given by

$$a \cdot \Sigma^\ell m := (-1)^{|a|\ell} \Sigma^\ell(am).$$

Example 1.2.34. Let F and G be complexes of R -modules. Then $\text{Hom}_R(F, G)$ is a left dg $\text{End}_R(G)$ -module and a right dg $\text{End}_R(F)$ -module in a compatible way (i.e., a dg bimodule).

Example 1.2.35. If $\varphi: A \rightarrow B$ is a map of dg algebras, then it endows B with the structure of a dg A -module, akin to restricting scalars along an ordinary ring homomorphism.

1.3 Adjoining variables to kill cycles

From now on, we will often have not one but two rings of interest: a noetherian local ring Q , and a quotient ring $R = Q/I$. We will often reserve Q for the case when Q is a regular local ring, though for now Q is only assumed to be a noetherian local ring.

When we build the Koszul complex on $f_1, \dots, f_n$, we adjoin n exterior variables $e_1, \dots, e_n$ to kill the cycles $f_1, \dots, f_n$, and the only relations we impose on these variables are the ones that are forced by the definition of a strict graded commutative algebra. Indeed, the Koszul $\text{Kos}(\underline{f}) = \wedge_R(e_1, \dots, e_n)$ is the *free* strictly graded commutative dg algebra generated by $e_1, \dots, e_n$. We will now discuss a higher order version of this idea: at each stage, we impose only the minimal relations required among our new variables.

Construction 1.3.1. In general, given a dg algebra A and a variable x , we write $A[x]$ to denote the free strictly graded commutative A -algebra on x , which is given by

$$A[x] = \begin{cases} A \oplus Ax & \text{the exterior algebra on } x, \text{ if } |x| \text{ is odd} \\ A \oplus Ax \oplus Ax^2 \oplus \dots & \text{the polynomial algebra on } x, \text{ if } |x| \text{ is even.} \end{cases}$$

Exercise 1.3.2 (Killing cycles). Let A be a dg algebra and $z \in A_n$ a cycle. Let x be a variable with $|x| = |z| + 1 = n + 1$. Show that there exists a unique strictly graded commutative dg algebra structure on $A[x]$ extending the differential in A and such that $\partial(x) = z$. Moreover,

$$H_i(A[x]) = \begin{cases} H_i(A) & \text{if } i < n \\ H_n(A)/([z]) & \text{if } i = n. \end{cases}$$

Note that [Exercise 1.3.2](#) says nothing about $H_{>n}(A[x])$. Since

$$H_n(A[x]) = H_n(A)/H_0(A)([z]),$$

we say the cycle z is *killed*.

Inspired by ideas from topology, Tate [\[Tat57\]](#) constructed dg algebra resolutions over local rings, using this idea of adjoining variables to kill cycles.

Construction 1.3.3. Let Q be a noetherian local ring and $R = Q/I$. We will construct a dg algebra resolution for R in steps, by successively adding variables in each degree to kill homology in the degree below.

Step 0: Consider the complex Q concentrated in degree 0. Our goal is to change the homology of this complex so that it becomes R in degree 0, and zeroes elsewhere.

Step 1: Fix a generating set $f_1, \dots, f_n$ for I and adjoin variables $x_1, \dots, x_n$ of homological degree 1 so that $\partial(x_i) = f_i$. We write

$$Q[x_1, \dots, x_n \mid \partial(x_i) = f_i]$$

to represent the resulting complex, or $Q[X_1]$ with $X_1 = \{x_1, \dots, x_n\}$ for short.

The resulting complex starts with

$$\begin{array}{ccc} \bigoplus_{i=1}^n Q \cdot x_i & \xrightarrow{\partial} & Q \\ 1 & & 0 \end{array}$$

just as we would normally start with when building a resolution for R over Q , but these x_i are elements in a dg algebra, so we need to consider their products as well, which live in higher degrees. We take these to be exterior variables, so that the only relations among them are the ones necessary to satisfy the definition of a graded commutative dg algebra: we set

$$x_i x_j = -x_j x_i \quad \text{and} \quad x_i^2 = 0.$$

The differential on any other element of $Q[X_1]$ is now completely determined by linearity and the Leibniz rule. In fact, $Q[X_1]$ is simply the Koszul complex on $f_1, \dots, f_n$:

$$0 \longrightarrow Q \cdot x_1 \cdots x_n \longrightarrow \cdots \longrightarrow \bigoplus_{i < j} Q \cdot x_i x_j \longrightarrow \bigoplus_{i=1}^n Q \cdot x_i \xrightarrow{\partial} Q.$$

So far, we have managed to fix the homology in degree 0 to be R . If $H_1(Q[X_1]) = 0$, then in fact by [Theorem 1.1.4](#) the Koszul complex must be exact, and we have finished constructing a resolution for R . Otherwise, we proceed to step 2.

Before we proceed, note that since R is a noetherian ring and we are dealing with finitely generated modules, the R -module $H_1(Q[X_1])$ is finitely generated.

Step 2: Fix cycles $z_1, \dots, z_s \in Q[X_1]$ of degree 1 whose homology classes $[z_1], \dots, [z_s]$ generate $H_1(Q[X_1])$, and adjoin polynomial variables $x_{n+1}, \dots, x_{n+s}$ of degree 2 to kill the homology of degree 1, meaning that we set

$$\partial(x_{n+i}) = z_i.$$

We add no additional relations except for the fact that any two variables of degree 2 commute with each other and with all variables of degree 1. The differential of the resulting complex is completely determined by Q -linearity and the Leibniz rule.

Setting $X_2 = \{x_{n_1}, \dots, x_{n+s}\}$, we have

$$H_0(Q[X_1, X_2]) = R \quad \text{and} \quad H_1(Q[X_1, X_2]) = 0.$$

We continue in this fashion, repeating this process in every degree. Note once more that the fact that R is noetherian and that at each stage we add finitely many variables guarantees that the successive homology modules $H_{d-1}(Q[X_1, X_2, \dots, X_{d-1}])$ are finitely generated. More precisely, here is what we do at each stage:

Step d: Given sets of variables $X_1, \dots, X_{d-1}$ such that

$$H_0(Q[X_1, X_2, \dots, X_{d-1}]) = R \quad \text{and} \quad H_i(Q[X_1, X_2, \dots, X_{d-1}]) = 0 \text{ for all } i < d-1,$$

we fix cycles $u_1, \dots, u_t$ of degree $d-1$ in $Q[X_1, X_2, \dots, X_{d-1}]$ whose classes in homology generate $H_{d-1}(Q[X_1, X_2, \dots, X_{d-1}])$, and add new variables $v_1, \dots, v_t$ of degree d to kill the homology in degree $d-1$:

$$\partial(v_i) = u_i.$$

We set $X_d = \{v_1, \dots, v_t\}$ and proceed with $Q[X_1, X_2, \dots, X_d] = Q[X_{\leq d}]$.

Our new variables satisfy only the relations they must:

- When d is odd, we take all v_i to be exterior variables, so that $v_i v_j = -v_j v_i$ and $v_i^2 = 0$.
- When d is even, we take all v_i to be polynomial variables, which commute with all other variables.

Finally, we set

$$X := \bigcup_{i \geq 1} X_i.$$

The resulting complex $Q[X]$ is a free resolution for R over Q with a dg algebra structure.

Remark 1.3.4. For each fixed i , once we are past the stage where we have added the variables in X_{i+1} to correct homology in degree i , we never again change the homology in degrees i and below. Thus

$$H_i(Q[X]) = H_i(Q[X_{\leq i+1}]) = \begin{cases} R & \text{if } i = 0 \\ 0 & \text{if } i \neq 0. \end{cases}$$

This shows that $Q[X]$ is indeed a free resolution for R over Q , and in fact that every cyclic module over a noetherian local ring has a dg algebra resolution.

However, note that the process might never terminate: we might need to add variables in infinitely many degrees. In fact, as we will see below, we often add variables in *every* degree.

Remark 1.3.5. When I is generated by a regular sequence, we may stop at step 1, since the Koszul complex is a resolution for $R = Q/I$. On the other hand, if the minimal generators for I do not form a regular sequence, by [Theorem 1.1.4](#) the Koszul complex is not exact in degree 1, and thus we must add variables of degree 2.

Definition 1.3.6. Given a dg algebra A and a cycle $z \in Z(A)$, a **simple semifree extension** of A is a dg algebra $B = A[x \mid \partial(x) = z]$ obtained by adjoining a variable x to kill the cycle z , where x is taken to be an exterior variable if $|x| = |z| + 1$ is odd and a polynomial variable if $|x| = |z| + 1$ is even. Any dg algebra B obtained from A by a sequence of simple semifree extensions of A is a **semifree extension** of A . If X is the set of variables we adjoined to form B , we write $B = A[X]$.

Remark 1.3.7. Note that the resolution $Q[X]$ for $R = Q/I$ over Q in [Construction 1.3.3](#) is a semifree extension $Q[X]$ of Q with

$$H(Q[X]) = R.$$

Notation 1.3.8. Given a semifree extension $A[X]$ of A , we will write X_i to denote the variables of homological degree i in X , and

$$X_{\geq d} = \bigcup_{i \geq d} X_i, \quad X_{\leq d} = \bigcup_{i \leq d} X_i.$$

Moreover,

$$X_{i_1} \cdots X_{i_s} := \{x_{i_1} \cdots x_{i_s} \mid x_{i_j} \in X_{i_j}\}$$

and $AX_{i_1} \cdots X_{i_s}$ denotes the free graded A -module on the set $X_{i_1} \cdots X_{i_s}$. Analogously, we may write X^d or AX^d .

There are two instances of this construction that will be of special interest to us.

Definition 1.3.9. Let $\widehat{R} \cong Q/I$ be a minimal regular presentation for a local ring R . A **minimal model** $Q[X]$ for R over Q is a semifree extension over Q that resolves Q/I over Q , where we adjoin exterior variables in odd degrees and polynomial variables in even degrees, and take the smallest number possible of variables in each degree.

Macaulay2. The Macaulay2 package `DGAlgebras` allows one to compute minimal modules.

```
o1 = DGAlgebras

o1 : Package

i2 : Q = QQ[x,y];

i3 : I = ideal"x2,xy";

o3 : Ideal of Q

i4 : A = minimalModel(I, EndDegree => 3);
```

The `EndDegree => 3` instruction tells Macaulay to compute $Q[X_{\leq 4}]$, so that we get zero homology up to degree 3. Macaulay2 creates a ring whose variables are indexed as $T_{i,j}$, with i indicating homological degree.

We can extract the differential of $R\langle Y_{\leq 4} \rangle$ via

```
i5 : d = A.diff;
```

and then apply it to a particular variable to find its image:

```
i6 : d(T_(1,2))
```

```
o6 = x*y
```

```
o6 : Q[T_{1,1}..T_{1,2}, T_{2,1}, T_{3,1}, T_{4,1}..T_{4,2}]
```

Warning: we cannot use `d` calculate the differential of a product of variables directly, due to the way that is coded into Macaulay2. However, we can compute it by calculating the image of each variable individually, and using the Leibniz rule.

Exercise 1.3.10. Let $Q = k[[x, y]]$, $I = (x^2, xy)$, and $R = Q/I$.

- 1) Write the first 3 steps to construct a minimal model for R over Q .
- 2) Check your work with the `DGAlgebras` package.

Example 1.3.11. Let $Q = k[[x, y, z]]$ and $I = (xy, zx, yz)$. Let us compute the first few steps in a minimal model for $R = Q/I$ over Q . Set

$$f_1 = xy \quad f_2 = xz \quad f_3 = yz.$$

First, we take $X_1 = \{T_{1,1}, T_{1,2}, T_{1,3}\}$ with

$$\partial(T_{1,1}) = xy \quad \partial(T_{1,2}) = xz \quad \partial(T_{1,3}) = yz.$$

The boundaries of degree 1 in $Q[X_1]$ correspond to the koszul relations on R , which are generated by $f_i f_j - f_j f_i$ with $i \neq j$. The nonkoszul relations on f_1, f_2, f_3 are generated by

$$zf_1 - yf_2 \quad yf_2 - xf_3 \quad zf_1 - xf_3.$$

In fact, note that any two of these generate the module of relations on I . More precisely, $zT_1 - yT_2$ and $zT_1 - zT_3$ generate $H_1(Q[X_1])$. Thus we take $X_2 = \{T_{2,1}, T_{2,2}\}$ with

$$\partial(T_{2,1}) = zT_{1,1} - yT_{1,2} \quad \text{and} \quad \partial(T_{2,2}) = zT_{1,1} - xT_{1,3}.$$

Now while we can compute $H_2(Q[X_{\leq 2}])$ by hand, we can also get a little help from Macaulay2. We can directly set up the matrices involve, or set up the complex by hand ourselves, or use the `DGAlgebras` package to build it. Let us use the `DGAlgebras` package. First, we set up $Q[X_{\leq 2}]$:

```
i1 : needsPackage "DGAlgebras";
```

```
i2 : Q = QQ[x,y,z];
```

```
i3 : I = ideal"xy,xz,yz";
```

```
i4 : R = Q/I;
```

```

i5 : A = minimalModel(I, EndDegree => 1)

o5 = {Ring => Q
      Underlying algebra => Q[T1,1 ..T1,3 , T2,1 ..T2,2 ]
      Differential => {x*y, x*z, y*z, - z*T1,1 + y*T1,2 , - z*T1,1 + x*T1,3 }

o5 : DGAalgebra

```

Let us check which the choices that Macaulay2 has made:

```

i6 : d = A.diff;

o6 : RingMap Q[T1,1 ..T1,3 , T2,1 ..T2,2 ] <-- Q[T1,1 ..T1,3 , T2,1 ..T2,2 ]

i7 : d(T_(2,1))

o7 = - z*T1,1 + y*T1,2

o7 : Q[T1,1 ..T1,3 , T2,1 ..T2,2 ]

i8 : d(T_(2,2))

o8 = - z*T1,1 + x*T1,3

o8 : Q[T1,1 ..T1,3 , T2,1 ..T2,2 ]

```

We can see that Macaulay2 has made the same choice we made, though with opposite signs. Before we compute $H_2(Q[X_{\leq 2}])$, we should also check what order Macaulay2 is using for the elements in $QX_1^2 \oplus QX_2$. For example, we can see this by turning $A = Q[X]$ into the form of a complex, and asking for the differential ∂_2 , as follows:

```

i9 : C = toComplex (A, 3)

o9 = Q1 <-- Q3 <-- Q5 <-- Q7
      0      1      2      3

```

o9 : Complex

i10 : C.dd_2

```
o10 = {2} | -xz -yz 0   -z -z |
      {2} | xy  0   -yz y  0  |
      {2} | 0   xy  xz  0  x  |
```

```

      3      5
o10 : Matrix Q  <-- Q
```

We see that the basis in degree 2 has been chosen in the following order:

$$\{T_{1,1}T_{1,2}, \quad T_{1,1}T_{1,3}, \quad T_{1,2}T_{1,3}, \quad T_{2,1}, \quad T_{2,2}\}.$$

Now we are ready to compute $H_2(Q[X_{\leq 2}])$:

i9 : mingens HH_2(A)

```
o9 = {4} | -1 0  0  |
      {4} | 0  0 -1  |
      {4} | 0 -1 0  |
      {3} | x  -z 0  |
      {3} | 0  z  y  |
```

```

      5      3
o9 : Matrix Q  <-- Q
```

We now set $X_3 = \{T_{3,1}, T_{3,2}, T_{3,3}\}$ with

$$\partial(T_{3,1}) = T_{1,1}T_{1,2} - xT_{2,1} \quad \partial(T_{3,2}) = T_{1,2}T_{1,3} + zT_{2,1} - zT_{2,2} \quad \partial(T_{3,3}) = T_{1,1}T_{1,3} - yT_{2,2}.$$

We can check by hand that these three elements are in fact cycles but not boundaries in $Q[X_{\leq 2}]$; the trickier part would have been to see that they do form a minimal generating set for the second homology.

A minimal model is typically not a minimal free resolution. However, analogously to what happens with minimal free resolutions, we can characterize minimal models according to the shape of the differential.

Exercise 1.3.12. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring, and fix a minimal regular presentation $\widehat{R} \cong Q/I$ for R . Let $(Q[X], \partial)$ be a minimal model for R over Q .

- 1) Show that $\partial(X_1) \subseteq \mathfrak{m}^2$ and $\partial(X_2) \subseteq \mathfrak{m}X_1$.
- 2) Give an example showing that $\partial(X_{n+1}) \subseteq \mathfrak{m}X_n$ might fail in general.

Remark 1.3.13. Given a minimal model $A := Q[X]$ for R over the regular local ring $(Q, \mathfrak{m}, k)$, we will use X_0 to denote a minimal set of generators for $\mathfrak{m}$. Note that $Q[X]$ is a local dg algebra, and this notation allows us to write the maximal ideal of $Q[X]$ as (X) , where we now view $X = X_{\geq 0}$.

Similarly, $k[X] = Q[X] \otimes_Q k$ is also a local dg algebra, with maximal ideal $(X) = (X_{\geq 1})$.

Definition 1.3.14. Let $(R, \mathfrak{m})$ be a noetherian local ring, and let A be a dg R -algebra with $A_0 = R$. We say that A is a **minimal dg algebra** if the maximal ideal $J = \mathfrak{m} + (A_{\geq 1})$ satisfies

$$\partial(J) \subseteq J^2.$$

Note that this minimality condition on a dg algebra is far from requiring that it be a minimal complex. Recall that in general, a complex of modules over a local ring $(R, \mathfrak{m})$ is minimal if

$$\partial(F) \subseteq \mathfrak{m}F.$$

Theorem 1.3.15. *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring and $\widehat{R} \cong Q/I$ a minimal regular presentation. Let $Q[X]$ be the minimal model for R over Q , with maximal ideal $(X_{\geq 0})$. Then*

$$\partial_{Q[X]}(X) \subseteq (X)^2.$$

As a consequence, in $k[X] = Q[X] \otimes_Q k$, the maximal ideal $(X_{\geq 1})$ satisfies

$$\partial_{k[X]}(X) \subseteq (X)^2.$$

Proof. First, note that $\partial_{k[X]}(X_1) = 0$ by [Exercise 1.3.12](#). Fix $d \geq 1$. For each $y \in X_{d+1}$, write $\partial(y) \in Q[X]$ as

$$\partial(y) = w + \sum_{x \in X_d} a_x x$$

for some $a_x \in Q$ and $w \in (X)^2$. Applying ∂ again, we get

$$0 = \partial(w) + \sum_{x \in X_d} a_x \partial(x) \in Z_d(Q[X_{<d}]).$$

Note that by degree reasons, $w \in Q[X_{<d}]$, so $\partial(w)$ is a boundary in $Q[X_{<d}]$. Therefore,

$$0 = \sum_{x \in X_d} a_x [\partial(x)] \in H_d(Q[X_{<d}]).$$

By assumption, $Q[X]$ is a minimal model for R , and thus $\{[\partial(x)] \mid x \in X_d\}$ forms a minimal generating set for $H_d(Q[X_{<d}])$. By NAK, $a_x \in \mathfrak{m}$ for all $x \in X_d$, and therefore

$$\partial(y) = w + \sum_{x \in X_d} a_x x \in (X)^2 + \mathfrak{m}X. \quad \square$$

In fact, this condition characterizes minimal models.

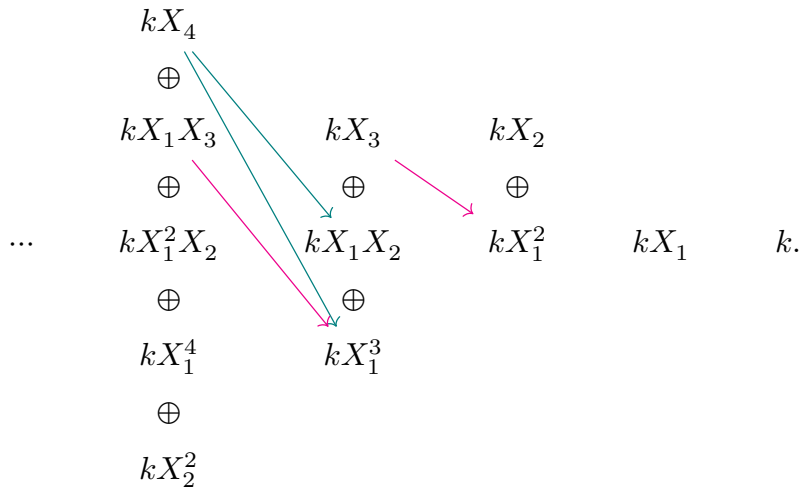
Exercise 1.3.16. Let $Q[X]$ be a semifree extension of Q . Show that $Q[X]$ is a minimal model for $R = Q/I$ over Q if and only if

$$H_n(Q[X]) = \begin{cases} R & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\partial(X) \subseteq (X)^2.$$

Remark 1.3.17. In a picture, here is what a minimal dg algebra $k[X]$ looks like, where we indicate with arrows the only components that might be nonzero.



Our next goal is to show that minimal models are unique up to isomorphism. To build towards that, we need to better understand the construction. As we briefly discussed earlier, the semifree extensions are free strictly graded commutative algebras, and thus they come equipped with the following universal property:

Theorem 1.3.18. Let $\varphi: A \rightarrow B$ be a map of dg algebras and let Z be a set of cycles in A whose image $\varphi(Z)$ is a set of boundaries in B . For each $z \in Z$, let $b_z \in B$ be such that $\partial(b_z) = \varphi(z)$. Given a set of variables indexed by Z

$$X = \{x_z \mid z \in Z, |x_z| = |z| + 1\}$$

there exists a unique map of dg algebras $\tilde{\varphi}: A[X \mid \partial(x_z) = z] \rightarrow B$ extending φ and such that

$$\tilde{\varphi}(x_z) = b_z.$$

Proof. We can construct the extension inductively, one variable at a time, so it is sufficient to give a proof in the case of one variable.

Let $z \in Z(A)$ be such that $\varphi(z) = \partial(b)$ for some $b \in B$. Note that $A[x \mid \partial(x) = z]$ is a free graded commutative A -algebra, so φ extends uniquely to a map of graded commutative algebras

$$\tilde{\varphi}: A[x \mid \partial(x) = z] \longrightarrow B$$

with $\tilde{\varphi}(x) = b$. Moreover, we claim that $\tilde{\varphi}(x)$ is a map of complexes. By linearity, it is sufficient to consider elements of the form ax^i for some $i \geq 1$. We have

$$\begin{aligned} \partial \circ \tilde{\varphi}(ax^i) &= \partial(\varphi(a)b^i) \\ &= \partial(\varphi(a))b^i + (-1)^{|a|}\varphi(a)\partial(b^i) && \text{by the Leibniz rule} \\ &= \varphi(\partial(a))b^i + (-1)^{|a|}\varphi(a)\partial(b^i) && \text{since } \varphi\partial = \partial\varphi \\ &= \varphi(\partial(a))b^i + (-1)^{|a|}i\varphi(a)\partial(b)b^{i-1} && \text{by Exercise 1.5.2} \\ &= \varphi(\partial(a))\varphi(x^i) + (-1)^{|a|}i\varphi(a)\varphi(\partial(x))\tilde{\varphi}(x^{i-1}) && \text{since } \varphi(\partial(x)) = \varphi(z) = \partial(b) \\ &= \tilde{\varphi}(\partial(a)x^i + (-1)^{|a|}iaz\varphi(\partial(x))x^{i-1}) \\ &= \tilde{\varphi}(\partial(ax^i)) && \text{by Exercise 1.5.2.} \end{aligned}$$

We conclude that $\tilde{\varphi}$ is indeed a map of dg algebras. $\square$

In colloquial terms, [Theorem 1.3.18](#) tells us that to give a map of dg algebras $A[X] \longrightarrow B$ is to give a map of dg algebras $A \longrightarrow B$ together with the images of each $x \in X$, such that the image of each $x \in X$ are that $\partial(x)$ needs to be sent to a boundary with $\varphi(\partial(x)) = \partial(\varphi(x))$.

Remark 1.3.19. Let A be a dg algebra and consider a semifree extension

$$C = A[x \mid \partial(x) = y].$$

When $|x|$ is odd, $C = A \oplus Ax$, and the inclusion of A into C induces a natural short exact sequence of complexes

$$0 \longrightarrow A \longrightarrow A[x] \longrightarrow \Sigma^{|x|} A \longrightarrow 0.$$

More generally, for any x , the truncations

$$\bigoplus_{i \leq n} Ax^i$$

obtained by quotienting $A[x]$ by (x^{n+1}) give us short exact sequences of dg A -modules

$$0 \longrightarrow \bigoplus_{i < n} Ax^i \longrightarrow \bigoplus_{i \leq n} Ax^i \longrightarrow Ax^n \longrightarrow 0.$$

Note that when $n = 1$, this is the same as the first short exact sequence above for $|x|$ odd. Note moreover that

$$A[x] = \bigoplus_i Ax^i$$

and that whenever $\ell \leq n - 1$,

$$H_\ell(A[x]) = H_\ell\left(\bigoplus_{i \leq n} Ax^i\right).$$

Corollary 1.3.20. *Let $\alpha: A \rightarrow B$ be a map of dg algebras and let Z be a set of cycles in A . Then given a set of variables indexed by Z*

$$X = \{x_z \mid z \in Z\}$$

there exists a unique map of dg algebras

$$\begin{array}{ccc} A[X \mid \partial(x_z) = z] & \xrightarrow{\tilde{\alpha}} & B[X \mid \partial(x_z) = \alpha(z)] \\ x_z & \longmapsto & x_z \end{array}$$

extending α . Moreover, if α is surjective, a quasiisomorphism, or a local map, then so is $\tilde{\alpha}$.

Proof. Consider the composition of α with the natural inclusion of B into $B[X]$:

$$\beta: A \xrightarrow{\alpha} B \hookrightarrow B[X].$$

For each $z \in Z$, since z is a cycle and α is a map of dg algebras, $\alpha(z)$ is also a cycle, which becomes a boundary in $B[X]$. By [Theorem 1.3.18](#), there exists a unique map of dg algebras $\gamma: A[X] \rightarrow B[X]$ extending β and such that $\gamma(x_z) = x_z$ for each $z \in Z$. By construction, γ extends α . Note moreover that any map of dg algebras $A[X] \rightarrow B[X]$ extending α must also extend β , and thus γ is the unique extension of α .

Now assume that α is a quasiisomorphism. To show that $\gamma = \tilde{\alpha}$ is also a quasiisomorphism, it suffices to consider the case of one variable x . Consider the restrictions of $\tilde{\alpha}$ given by

$$\bigoplus_{i \leq n} Ax^i \xrightarrow{\alpha_n} \bigoplus_{i \leq n} Bx^i.$$

By [Remark 1.3.19](#), we have following commutative diagram of short exact sequences of complexes:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \bigoplus_{i < n} Ax^i & \longrightarrow & \bigoplus_{i \leq n} Ax^i & \longrightarrow & Ax^n \longrightarrow 0 \\ & & \downarrow \alpha_{n-1} & & \downarrow \alpha_n & & \downarrow \cong \\ 0 & \longrightarrow & \bigoplus_{i < n} Bx^i & \longrightarrow & \bigoplus_{i \leq n} Bx^i & \longrightarrow & Bx^n \longrightarrow 0. \end{array}$$

Note that the map on the right is a quasiisomorphism since it is given by a shift of α . Following an argument by induction, we may assume that α_n is a quasiisomorphism; note that $\alpha_0 = \alpha$ is a quasiisomorphism by assumption. This gives rise to the following diagram of long exact sequences in homology:

$$\begin{array}{ccccccccccc} \cdots & \rightarrow & H_{\ell-1}(A) & \rightarrow & H_{\ell}\left(\bigoplus_{i < n} Ax^i\right) & \rightarrow & H_{\ell}\left(\bigoplus_{i \leq n} Ax^i\right) & \rightarrow & H_{\ell}(A) & \rightarrow & H_{\ell+1}\left(\bigoplus_{i < n} Ax^i\right) & \rightarrow \cdots \\ & & \downarrow \cong & & \downarrow \cong & & \downarrow H(\alpha_n) & & \downarrow \cong & & \downarrow \cong & \\ \cdots & \rightarrow & H_{\ell-1}(B) & \rightarrow & H_{\ell}\left(\bigoplus_{i < n} Bx^i\right) & \rightarrow & H_{\ell}\left(\bigoplus_{i \leq n} Bx^i\right) & \rightarrow & H_{\ell}(B) & \rightarrow & H_{\ell+1}\left(\bigoplus_{i < n} Bx^i\right) & \rightarrow \cdots \end{array}$$

By the Five Lemma, $H_\ell(\alpha_n)$ must be an isomorphism. Note that for all $\ell \leq n$,

$$H_\ell(\alpha_n) = H_\ell(\alpha).$$

Proceeding by induction, we conclude that $\tilde{\alpha}$ is a quasiisomorphism.

Finally, note that $B[X]$ is generated by B and X , and that the image of $\tilde{\alpha}$ is generated by $\alpha(A)$ and (X) . Therefore, if α is surjective then so is $\tilde{\alpha}$, and if α is local then so is $\hat{\alpha}$. $\square$

Exercise 1.3.21. Let F be a complex of free R -modules with $F_n = 0$ for all $n < 0$ and let $f: A \rightarrow B$ be a quasiisomorphism of R -complexes.

- 1) Consider the truncation $F_{\leq n}$ of F that has F_d in homological degrees $d \leq n$, and zeroes elsewhere. Show that the natural inclusion of $F_{\leq n}$ into $F_{\leq n+1}$ induces a split short exact sequence of complexes.

- 2) Show that

$$f \otimes_R \text{id}_F: A \otimes_R F \rightarrow B \otimes_R F$$

is a quasiisomorphism.

- 3) Show that

$$f_* := \text{Hom}_R(F, f): \text{Hom}_R(F, A) \rightarrow \text{Hom}_R(F, B)$$

is a quasiisomorphism.

Exercise 1.3.22. Let $\pi: A \rightarrow B$ be a map of complexes such that $\pi_* = \text{Hom}(C, \pi)$ is a quasiisomorphism. Show that if $\alpha \in \ker(\pi_*)$ then α is nullhomotopic.

Theorem 1.3.23. Let $(Q, \mathfrak{m}, k)$ be a commutative ring and $R = Q/I$. Let $Q[X]$ be a semifree extension of Q that is a free resolution for R over Q , and let $\pi: A \rightarrow B$ be a surjection quasiisomorphism of dg algebras. Every dg algebra map $\beta: Q[X] \rightarrow B$ lifts to a dg algebra map $\alpha: Q[X] \rightarrow A$ making the following diagram commute:

$$\begin{array}{ccc} & & A \\ & \nearrow \alpha & \downarrow \pi \\ Q[X] & \xrightarrow{\beta} & B \end{array}$$

Moreover, any such lift α is unique up to Q -linear homotopy.

Proof. Since Q is a free Q -module, there exists map of Q -complexes making

$$\begin{array}{ccc} & & A_0 \\ & \nearrow \alpha_0 & \downarrow \pi_0 \\ Q & \xrightarrow{\beta_0} & B_0 \end{array}$$

commute. Note that α_0 determines a map of dg algebras $\alpha_0: Q \rightarrow A$.

Now we construct $\alpha_n: Q[X_{\leq n}] \rightarrow A$ inductively. Suppose that α_n has been set, and fix a variable $x \in X_{n+1}$. Since π is surjective, there exists an element $w_x \in A$ such that $\pi(w_x) = \beta(x)$. Note that

$$\alpha_n(\partial(x)) = \partial(w_x)$$

is a boundary. By [Theorem 1.3.18](#), there exists a unique map of dg algebras

$$\alpha_{n+1}: Q[X_{\leq n+1}] \rightarrow A$$

extending α_n and such that $\alpha(x) = z_x$. Note that α_n extends β by construction. Proceeding inductively leads us to $\alpha: Q[X] \rightarrow A$.

Now suppose that $\gamma: Q[X] \rightarrow A$ is another lift of β . Since $\pi \circ (\alpha - \gamma) = 0$, $\alpha - \gamma \in \ker(\pi_*)$. By [Exercise 1.3.21](#), π_* is a quasiisomorphism, so $\alpha - \gamma$ is nullhomotopic by [Exercise 1.3.22](#). Thus α and γ are homotopic. $\square$

Notation 1.3.24. Given two maps of complexes $f, g: A \rightarrow B$, we write $f \sim g$ to denote that f and g are homotopic.

Definition 1.3.25. A map of complexes $\alpha: A \rightarrow B$ is a **homotopy equivalence** if there exists a map of complexes $\beta: B \rightarrow A$ such that $\alpha \circ \beta \sim \text{id}_B$ and $\beta \circ \alpha \sim \text{id}_A$. We say A and B are **homotopy equivalent** if there exists a homotopy equivalence between them. Whenever A and B are dg algebras, a **homotopy equivalence of dg algebras** is a homotopy equivalence such that α and β as above are maps of dg algebras.

Remark 1.3.26. Any homotopy equivalence is a quasiisomorphism, since homology is a functor. Note that any two semifree extensions of Q resolving $R = Q/I$ over Q are quasiisomorphic: by [Theorem 1.3.23](#), there are maps of dg algebras α and β making the following diagrams commute:

$$\begin{array}{ccc} & Q[Y] & \\ \alpha \nearrow & \downarrow \simeq & \\ Q[X] & \xrightarrow{\simeq} & R \end{array} \quad \begin{array}{ccc} & Q[X] & \\ \beta \nearrow & \downarrow \simeq & \\ Q[Y] & \xrightarrow{\simeq} & R. \end{array}$$

By [Theorem 1.3.23](#), α and β are quasiisomorphisms since they are lifts of quasiisomorphisms.

Lemma 1.3.27. *Any two semifree extensions A and B of Q resolving $R = Q/I$ over Q are homotopy equivalent. Moreover, any quasiisomorphism $A \rightarrow B$ is a homotopy equivalence.*

Proof. We saw in [Remark 1.3.26](#) that there exist quasiisomorphisms $\alpha: A \rightarrow B$ and $\beta: B \rightarrow A$, so it suffices to prove the second statement.

When we compose α and β , we get commutative diagrams

$$\begin{array}{ccc} & Q[X] & \\ \beta \circ \alpha \nearrow & \downarrow \simeq & \\ Q[X] & \xrightarrow{\simeq} & R \end{array} \quad \text{and} \quad \begin{array}{ccc} & Q[Y] & \\ \alpha \circ \beta \nearrow & \downarrow \simeq & \\ Q[Y] & \xrightarrow{\simeq} & R. \end{array}$$

Note that the identity on $Q[X]$ or $Q[Y]$ are also maps that makes each of these diagrams commute, so by the uniqueness in [Theorem 1.3.23](#) we conclude that $\beta \circ \alpha$ is homotopic to $\text{id}_{Q[X]}$ and $\alpha \circ \beta$ is homotopic to $\text{id}_{Q[Y]}$. $\square$

Exercise 1.3.28. Let A be a dg algebra and assume $\alpha: M \rightarrow N$ is a homotopy equivalence of dg A -modules. Prove that for any dg ideal J of A the induced map $\bar{\alpha}: M/JM \rightarrow N/JN$ is a homotopy equivalence of dg A/J -modules.

Theorem 1.3.29. Let R be a noetherian local ring, and $\widehat{R} \cong Q/I$ a minimal regular presentation. Any two minimal models for R over Q are isomorphic dg algebras.

Proof. Let $Q[X]$ and $Q[Y]$ be two minimal models for R over Q . We claim that it suffices to show that any homotopy equivalence $Q[X] \rightarrow Q[X]$ is an isomorphism. Indeed, by [Lemma 1.3.27](#), $Q[X]$ and $Q[Y]$ are homotopy equivalent, so given homotopy equivalences $\alpha: Q[X] \rightarrow Q[Y]$ and $\beta: Q[X] \rightarrow Q[Y]$, we will have shown that $\alpha \circ \beta$ and $\beta \circ \alpha$ are isomorphisms, which implies that α and β are isomorphisms.

Let $\varphi: Q[X] \rightarrow Q[X]$ be a homotopy equivalence. We will work in $k[X] = Q[X] \otimes_Q k$. By [Theorem 1.3.15](#), $J = (X)$ is a dg ideal in $k[X]$. Consider the inverse limit

$$k[X]/J \longleftarrow k[X]/J^2 \longleftarrow k[X]/J^3 \longleftarrow \dots$$

We claim that¹

$$\varprojlim k[X]/J^n \cong k[X].$$

Indeed, this inverse limit can be calculated degreewise, so fix a degree d . Note that since J lives only in positive degrees, then for $n \gg 0$ we have $(J^n)_d = 0$, so $(k[X]/J^n)_d$ is eventually constant and equal to $k[X]_d$.

Consider the induced maps

$$\varphi_n: k[X]/J^n \rightarrow k[X]/J^n$$

which are still homotopy equivalences by [Exercise 1.3.28](#). We claim that all of these induced maps are isomorphisms. To see that, first consider $n = 1$, and note that the induced map is simply the identity on k , which is in fact an isomorphism.

Now we proceed by induction. Assume that φ_n is an isomorphism. Since φ respects the algebra structure, $\varphi(J^n) \subseteq J^n$ for all n , so φ_{n+1} induces a map $J^n/J^{n+1} \rightarrow J^n/J^{n+1}$, which must also be a homotopy equivalence. Consider the commutative diagram short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & J^n/J^{n+1} & \longrightarrow & k[X]/J^{n+1} & \longrightarrow & k[X]/J^n \longrightarrow 0 \\ & & \downarrow \varphi_{n+1} & & \downarrow \varphi_{n+1} & & \downarrow \varphi_n \\ 0 & \longrightarrow & J^n/J^{n+1} & \longrightarrow & k[X]/J^{n+1} & \longrightarrow & k[X]/J^n \longrightarrow 0. \end{array}$$

¹Warning: for those familiar with completions, this will be very disconcerting. The key point to keep in mind is that the grading plays an important role here.

By the Five Lemma, it suffices to show that the restriction of φ_{n+1} to J^n/J^{n+1} is also an isomorphism.

By [Theorem 1.3.15](#) $\partial(J) \subseteq J^2$, so it follows that J^n/J^{n+1} has trivial differential for all $n \geq 1$. Therefore, $H(J^n/J^{n+1}) = J^n/J^{n+1}$, and thus the map induced in homology by

$$J^n/J^{n+1} \xrightarrow{\varphi_n} J^n/J^{n+1}$$

can be identified with itself. By assumption, φ_n is a homotopy equivalence, and thus φ_n is an isomorphism. Therefore, $\varphi \otimes_Q \text{id}_k$ is an isomorphism. Applying NAK degreewise, we conclude that φ is an isomorphism. $\square$

Similarly, acyclic closures are also unique.

Exercise 1.3.30. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. Show that any two acyclic closures $R\langle Y \rangle$ and $R\langle Z \rangle$ for k over R are isomorphic dg algebras. More precisely, there is an isomorphism of dg algebras $\alpha: R\langle Y \rangle \rightarrow R\langle Z \rangle$ such that the induced map $\tilde{\alpha}: k\langle Y \rangle \rightarrow k\langle Z \rangle$ is an isomorphism of graded k -vector spaces.

Lemma 1.3.31. Let R be a noetherian local ring and let $\widehat{R} \cong Q/I$ be a minimal regular presentation. The following are equivalent:

- 1) The ideal I is generated by a regular sequence.
- 2) We have $X = X_1$, so $Q[X]$ is the Koszul complex on a minimal generating set for I .
- 3) The minimal model $Q[X]$ for R is a finite free resolution for Q/I over Q .
- 4) The minimal model $Q[X]$ for R is a minimal free resolution for Q/I over Q .

Proof. Let $\underline{f} = f_1, \dots, f_n \in I$ be such that $\partial(X_1) = \{f_1, \dots, f_n\}$. Note that $\underline{f}$ is a minimal generating set for I and $Q[X_1] = \text{Kos}(\underline{f})$. Then

$$\begin{aligned} H_1(Q[X_1]) = 0 &\iff \underline{f} \text{ is a regular sequence} && \text{by Theorem 1.1.4} \\ &\iff I \text{ is generated by a regular sequence} && \text{by Corollary 0.3.10} \\ &\iff H_i(Q[X_1]) = 0 \text{ for all } i \geq 1 && \text{by Corollary 0.3.10.} \end{aligned}$$

This shows that $1) \Leftrightarrow 2)$, $1) \Rightarrow 3)$, and $1) \Rightarrow 4)$. Now we prove $4) \Rightarrow 3) \Rightarrow 2)$.

Suppose $Q[X]$ is a minimal free resolution for Q/I . By [Theorem 0.2.20](#), since Q is a regular local ring, $Q[X]$ must be a finite free resolution. Then $X_2 = \emptyset$, since adjoining any polynomial variable of even degree results in an infinite resolution. Thus $H_1(Q[X_1]) = 0$, so $Q[X_1]$ is a resolution for R/I , by [Theorem 1.1.4](#), and therefore $X = X_1$. $\square$

Complete intersections are the rings satisfying the equivalent properties in [Lemma 1.3.31](#). Throughout these lectures, we will gain an understanding of complete intersections and give many equivalent definitions. Throughout, a pattern will emerge: there is a dichotomy between complete intersections and all other noetherian local rings, where complete intersections are the nicely behaved rings, and once R is not a complete intersection, then its behavior with respect to many interesting properties becomes completely wild. But first, let us dive deeper into the definition of our new favorite class of rings.

1.4 Complete intersections

Among singular rings, complete intersections are the mildest singularities. One of our main goals is to make this precise, and to give several characterizations of this class of rings.

Definition 1.4.1. A noetherian local ring R is a **complete intersection** of **codimension** c if for some regular presentation $\widehat{R} \cong Q/I$ for R , the ideal I is generated by a regular sequence of length c .

It is not clear from the definition that this notion does not depend on the choice of minimal regular presentation; we will show soon that this is indeed the case. First, let us see some examples.

Example 1.4.2. Any regular local ring is a complete intersection: we will see below in [Remark 1.4.16](#) that $\widehat{R}$ must also be complete, so in this case, our usual defining ideal I is simply $I = 0$.

Example 1.4.3. A **hypersurface** is a complete intersection of codimension 1, meaning there is some nonzero $f \in \mathfrak{m}_Q^2$ such that $\widehat{R} \cong Q/(f)$. Conversely, note that any (complete) regular local ring $(Q, \mathfrak{m})$ is a domain, and thus any nonzero $f \in \mathfrak{m}^2$ is a regular element, so that $Q/(f)$ is a complete intersection.

Example 1.4.4. Let k be a field and take $R = Q/I$ with $Q = k[[x, y]]$, and $I = (x^2, xy)$. Note that $I = (x^2, xy)$ has two minimal generators but height 1; in fact, $\text{Min}(I) = \{(x)\}$. Therefore, $\text{height}(I) = 1$ and I is not generated by a regular sequence. The ring $R = Q/I$ is not a complete intersection.

Exercise 1.4.5. Let k be any field. Show that

$$R = k[[x, y, z]]/(x^2, y^2, z^2, xyz)$$

is not a complete intersection ring.

We can give a more intrinsic definition of complete intersections via Koszul homology.

Definition 1.4.6. Let R be a noetherian local ring, and let $\underline{x}$ be any minimal generating set for the maximal ideal of R . We set

$$K^R := \text{Kos}^R(\underline{x}).$$

The **Koszul homology of R** is the graded k -algebra $H(K^R)$.

Exercise 1.4.7. Show that $H(K^R)$ is a finite dimensional graded commutative k -algebra with

$$H_0(K^R) = k \quad \text{and} \quad H_{\text{embdim}(R)}(K^R) = \text{ann}_R(\mathfrak{m}).^2$$

²You might have met $\text{ann}_R(\mathfrak{m})$ as the **socle** of R .

Remark 1.4.8. Depth sensitivity is one of the many nice properties of the Koszul complex we did not have a chance to discuss. In this setting, it tells us that

$$\sup\{i \mid H_i(K^R) \neq 0\} = \text{embdim}(R) - \text{depth}(R).$$

This number is often called the **codepth** of R , written $\text{codepth}(R)$.

Exercise 1.4.9. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring, and let $\underline{x}$ and $\underline{y}$ be two minimal generating sets for the same ideal. Show that $\text{Kos}(\underline{x})$ and $\text{Kos}(\underline{y})$ are isomorphic dg algebras.

Remark 1.4.10. By [Exercise 1.4.9](#), any two minimal generating sets for the maximal ideal of R lead to isomorphic Koszul complexes. In particular, K^R is only well-defined up to isomorphism of dg algebras.

Lemma 1.4.11. *A noetherian local ring $(R, \mathfrak{m}, k)$ is regular if and only if $H_1(K^R) = 0$.*

Proof. Fix a minimal generating set for $\mathfrak{m}$, say $\underline{x}$. Then

$$\begin{aligned} R \text{ is regular} &\iff \underline{x} \text{ is a regular sequence} && \text{by Theorem 0.3.12} \\ &\iff H_1(\text{Kos}(\underline{x})) = 0 && \text{by Theorem 1.4.21} \\ &\iff H_1(K^R) = 0 && \text{by definition. } \square \end{aligned}$$

Exercise 1.4.12. Describe the graded k -algebra structure of $H(K^R)$ for $R = k[[x, y]]/(x^2, y^2)$ and $R = k[[x, y]]/(x^2, xy)$.

Example 1.4.13. Let $R = k[[x, y]]/(x^2, y^2)$. One can show (see [Exercise 1.4.12](#)) that

$$H_i(K^R) = \begin{cases} k & \text{if } i = 0 \\ ka \oplus kb & \text{if } i = 1 \\ kc & \text{if } i = 2 \\ 0 & \text{if } i \notin \{1, 2, 3\} \end{cases}$$

where we take $a = [xe_1]$, $b = [ye_2]$, and $c = [xye_1e_2]$. Note that

$$ab = [xe_1] \cdot [ye_2] = [xye_1e_2] = c.$$

Thus

$$H(K^R) \cong \wedge_k(ka \oplus kb).$$

In contrast, consider $S = k[[x, y]]/(x^2, xy)$. On the one hand, the k -algebra structure is the same as R : we have

$$H_i(K^S) = \begin{cases} k & \text{if } i = 0 \\ ka \oplus kb & \text{if } i = 1 \\ kc & \text{if } i = 2 \\ 0 & \text{if } i \notin \{1, 2, 3\}. \end{cases}$$

In this case, we take $a = [xe_1]$, $b = [xe_2] = [ye_1]$, and $[c = xe_1e_2]$. But now

$$ab = [xye_1^2] = 0 \neq 0.$$

In fact, we have an isomorphism of dg algebras

$$H(K^S) \cong k[a, b, c]/(ab, ac, bc, c^2) \cong k \ltimes k\{a, b, c\}.$$

Thus while these two k -algebras $H(K^R)$ and $H(K^S)$ agree, the dg algebra structure is quite different.

Remark 1.4.14 (Survival guide to completion). In what follows,

$$\varinjlim_n M/\mathfrak{m}^n M$$

we will use the following well-known properties of $\mathfrak{m}$ -adic completion:

- The completion $\widehat{R}$ is a flat R -algebra.
- For any finitely generated R -module M , its completion $\widehat{M}$ satisfies $\widehat{M} \cong M \otimes_R \widehat{R}$.
- The completion of a noetherian local ring $(R, \mathfrak{m})$ is a noetherian local ring $(\widehat{R}, \widehat{\mathfrak{m}})$ with

$$\dim(\widehat{R}) = \dim(R) \quad \text{embdim}(\widehat{R}) = \text{embdim}(R) \quad \text{depth}(\widehat{R}) = \text{depth}(R).$$

- For all $n \geq 1$, we have $\mathfrak{m}^n/\mathfrak{m}^{n+1} \cong \widehat{\mathfrak{m}}^n/\widehat{\mathfrak{m}}^{n+1}$.

For those new to the topic of completion, we recommend focusing on the case of complete noetherian local rings, and more concretely on the special case of those containing a field, which are of the form $R = k[[x_1, \dots, x_e]]/I$ with $I \subseteq (x_1, \dots, x_e)^2$. Nevertheless, we include here some of the technical details that allow us to reduce to the complete case.

Remark 1.4.15. Let M be a finitely generated R -module. We can take a minimal free resolution of M over R and tensor it with the flat R -module $\widehat{R}$ to obtain a free resolution over $\widehat{M}$, which is still minimal. Thus

$$\beta_i^R(M) = \beta_i^{\widehat{R}}(\widehat{M})$$

for all i . In particular,

$$\mu_R(M) = \beta_0^R(M) = \beta_0^{\widehat{R}}(\widehat{M}) = \mu_{\widehat{R}}(\widehat{M}).$$

Remark 1.4.16. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. By [Remark 1.4.15](#), $\mu(\widehat{\mathfrak{m}}) = \mu(\mathfrak{m})$, so $\text{embdim}(\widehat{R}) = \text{embdim}(R)$. Since $\dim(R) = \dim(\widehat{R})$, we conclude that R is regular if and only if $\widehat{R}$ is regular.

Theorem 1.4.17. *Let R be a noetherian local ring. Then there is a quasiisomorphism of dg algebras*

$$K^R \longrightarrow K^{\widehat{R}}.$$

In particular, for all i ,

$$\mu_R(H_i(K^R)) = \dim_k(H_i(K^R)) = \dim_k(H_i(K^{\widehat{R}})) = \mu_{\widehat{R}}(H_i(K^{\widehat{R}})).$$

Proof. Fix a minimal generating set $\underline{x}$ for the maximal ideal of R . Let $\widehat{x}$ be the image of $\underline{x}$ in $\widehat{R}$, and consider the map of complexes obtained by tensoring K^R with the canonical inclusion $R \longrightarrow \widehat{R}$:

$$K^R \cong K^R \otimes_R R \longrightarrow K^R \otimes_R \widehat{R}.$$

Note that $\widehat{x}$ is a minimal generating set for the maximal ideal $\widehat{m}$ of $\widehat{R}$, and there are natural isomorphisms

$$K^R \otimes_R \widehat{R} \cong \text{Kos}^R(\underline{x}) \otimes_R \widehat{R} \cong \text{Kos}^{\widehat{R}}(\widehat{x}) \cong K^{\widehat{R}}.$$

Thus we get a map of complexes $K^R \longrightarrow K^{\widehat{R}}$ making the following diagram commute:

$$\begin{array}{ccc} K^R \otimes_R R & \longrightarrow & K^R \otimes_R \widehat{R} \\ & \searrow & \downarrow \\ & & K^{\widehat{R}} \end{array}$$

We claim this map $K^R \longrightarrow K^{\widehat{R}}$ is a quasiisomorphism.

First, note that since $\widehat{R}$ is a flat R -module, tensoring with $\widehat{R}$ commutes with taking homology. In particular,

$$H_i(K^R) \otimes_R \widehat{R} = H_i(\text{Kos}(\underline{x})) \otimes_R \widehat{R} = H_i(\text{Kos}(\underline{x}) \otimes_R \widehat{R}) = H_i(K^{\widehat{R}}).$$

By flatness of $\widehat{R}$ over R , the induced map $H(K^R) \otimes_R R \longrightarrow H(K^R) \otimes_R \widehat{R}$ is an isomorphism. Thus we have a commutative diagram

$$\begin{array}{ccccc} & & H(K^R) \otimes_R \widehat{R} & & \\ & \nearrow \cong & \downarrow \cong & & \\ H(K^R) \otimes_R R & \longrightarrow & H(K^R \otimes_R \widehat{R}) & & \\ & \searrow & \downarrow \cong & & \\ & & H(K^{\widehat{R}}) & & \end{array}$$

and therefore the remaining maps must also be isomorphisms. We conclude that the canonical map $K^R \longrightarrow K^{\widehat{R}}$ is a quasiisomorphism and that the finite dimensional k -vector spaces $H(K^R)$ and $H(K^{\widehat{R}})$ are isomorphic. $\square$

More generally, [Theorem 1.4.17](#) is a corollary of the following:

Lemma 1.4.18. *Let R be a noetherian local ring and M be a complex of R -modules. If $H_n(M)$ is a finite length R -module for each n , then the natural map*

$$M \longrightarrow M \otimes_R \widehat{R}$$

is a quasiisomorphism.

Sketch of proof. We follow the same proof strategy of [Theorem 1.4.21](#), but this time using the commutative diagram

$$\begin{array}{ccc} & & H(M) \otimes_R \widehat{R} \\ & \nearrow \cong & \downarrow \cong \\ H(M) \otimes_R R & \longrightarrow & H(M \otimes_R \widehat{R}). \quad \square \end{array}$$

Theorem 1.4.19. *Let $(Q, \mathfrak{m}, k)$ a regular ring and $R = Q/I$ for some ideal $I \subseteq \mathfrak{m}^2$. Fix a minimal generating set $\underline{y}$ for $\mathfrak{m}$. There is an isomorphism of dg Q -algebras*

$$R \otimes_Q \text{Kos}(\underline{y}; Q) \longrightarrow K^R.$$

In particular, for all i we have

$$\text{Tor}_i^Q(R, k) = H_i(K^R).$$

Proof. Let $\underline{x}$ be the image of $\underline{y}$ in R . The natural isomorphisms $Q^n \otimes_Q R \cong R^n$ induce an isomorphism of complexes

$$\text{Kos}^Q(\underline{y}) \otimes_Q R \longrightarrow K^R = \text{Kos}^R(\underline{x})$$

that preserves the product structure. Taking homology on both sides, we conclude that

$$\text{Tor}_i^Q(R, k) = H_i(K^R). \quad \square$$

Exercise 1.4.20. Let $(Q, \mathfrak{m}, k)$ be a noetherian local ring and $R = Q/I$. Show that

$$\text{Tor}_1^Q(R, k) \cong I/\mathfrak{m}I.$$

We can now show that the minimal number of defining equations for $\widehat{R}$ over any minimal regular presentation is an invariant of R .

Theorem 1.4.21. *Let R be a noetherian local ring and $\widehat{R} \cong Q/I$ a minimal regular presentation for R . There is natural identification*

$$H_1(K^R) \cong I/\mathfrak{m}I.$$

In particular,

$$\mu(H_1(K^R)) = \mu(I).$$

Proof. By [Theorem 1.4.17](#), we can reduce to the case when R is complete, so we may assume that $R = Q/I$. By [Theorem 1.4.19](#) and [Exercise 1.4.20](#),

$$H_1(K^R) = \mathrm{Tor}_1^Q(R, k) \cong I/\mathfrak{m}I. \quad \square$$

Theorem 1.4.22. *For any noetherian local ring R ,*

$$\mu(H_1(K^R)) \geq \mathrm{embdim}(R) - \dim(R).$$

Moreover, R is a complete intersection if and only if

$$\mu(H_1(K^R)) = \mathrm{embdim}(R) - \dim(R).$$

Proof. Let $\widehat{R} \cong Q/I$ be a minimal regular presentation for I , meaning that $(Q, \mathfrak{m})$ is a regular local ring and $I \subseteq \mathfrak{m}^2$. By [Exercise 0.2.19](#), $\dim(Q) = \mathrm{embdim}(Q) = \mathrm{embdim}(R)$.

Since Q is a regular ring and $\dim(R) = \dim(\widehat{R})$, the dimension formula from [Theorem 0.3.9](#) gives us

$$\mathrm{height}(I) = \dim(Q) - \dim(Q/I) = \mathrm{embdim}(R) - \dim(R).$$

Therefore,

$$\begin{aligned} \mu(H_1(K^R)) &= \mu(I) && \text{by Theorem 1.4.21} \\ &\geq \mathrm{height}(I) && \text{by Krull's Height Theorem} \\ &= \mathrm{embdim}(R) - \dim(R). \end{aligned}$$

On the other hand, by [Corollary 0.3.10](#)

$$I \text{ is generated by a regular sequence} \iff \mu(I) = \mathrm{height}(I).$$

Thus I is generated by a regular sequence if and only if $\mu(H_1(K^R)) = \mathrm{embdim}(R) - \dim(R)$. $\square$

Corollary 1.4.23. *Let R be a noetherian local ring and consider two regular local rings $(Q, \mathfrak{m})$ and $(S, \mathfrak{n})$ such that*

$$Q/I \cong \widehat{R} \cong S/J.$$

Then I is generated by a regular sequence if and only if J is generated by a regular sequence.

Proof. First, we claim that we can reduce to the case when both regular presentations are minimal. Indeed, note that if $I = L + (f)$ for some $f \in \mathfrak{m} \setminus \mathfrak{m}^2$, then f is a regular element in Q . Note moreover that such an f is necessarily a minimal generated of I . Thus I is generated by a regular sequence if and only if L is generated by a regular sequence. Finally, $Q/(f)$ is a regular local ring by [Exercise 0.2.17](#).

By [Exercise 0.2.19](#), our assumptions on I and J guarantee that

$$\dim(Q) = \text{embdim}(Q) = \text{embdim}(\widehat{R}) = \text{embdim}(S) = \dim(S).$$

Since Q and S are both regular rings, the dimension formula from [Theorem 0.3.9](#) gives us

$$\text{height}(I) = \dim(Q) - \dim(R) = \dim(S) - \dim(R) = \text{height}(J).$$

By [Theorem 1.4.21](#),

$$\mu(I) = \mu(H_1(K^R)) = \mu(J).$$

Therefore, applying [Corollary 0.3.10](#),

$$\begin{aligned} I \text{ is generated by a regular sequence} &\iff \mu(I) = \text{height}(I) \\ &\iff \mu(J) = \text{height}(J) \\ &\iff J \text{ is generated by a regular sequence.} \quad \square \end{aligned}$$

This settles the question of well-definedness of our definition of complete intersection.

Our next goal is to give another characterization of complete intersections. To do that, we will need some auxiliary tools.

The following idea is very helpful, and we might use variations of it on various occasions:

Remark 1.4.24. Let $(Q, \mathfrak{m}, k)$ be a regular local ring and let $R = Q/I$ for some ideal $I \subseteq \mathfrak{m}^2$. Fix minimal generators $\underline{y}$ for the maximal ideal of Q . Note that since Q is regular, the Koszul complex $\text{Kos}^Q(\underline{y})$ is a minimal free resolution for k over Q , by [Theorem 0.3.12](#). Let $Q[X]$ be a minimal model for R over Q . Since both $Q[X]$ and $\text{Kos}^Q(\underline{y})$ are bounded below complexes of free modules, by [Exercise 1.3.21](#) we get quasiisomorphisms of dg Q -algebras

$$K^R \xleftarrow{\simeq} \text{Kos}(\underline{y}; Q) \otimes_Q R \xleftarrow{\simeq} \text{Kos}^Q(\underline{y}) \otimes_Q Q[X] \xrightarrow{\simeq} k \otimes_Q Q[X] =: k[X].$$

Once we take homology, these all become isomorphisms, and in particular

$$H(K^R) \cong H(k[X]).$$

If we do not assume that R is complete, combining this argument with the quasiisomorphism $K^R \longrightarrow K^{\widehat{R}}$ from [Theorem 1.4.17](#) gives us $H(K^R) \cong H(K^{\widehat{R}}) \cong H(k[X])$.

On the other hand, the minimality condition on the minimal model (see [Theorem 1.3.15](#)) gives us an inclusion of kX_1 into $H(k[X])$, which induces a map

$$\wedge_k(kX_1) = k[X_1] \longrightarrow H(k[X]).$$

Moreover, note that $\wedge_k(\Sigma H_1(K^R)) = k[e_1, \dots, e_n]$, where $e_1, \dots, e_n$ is a basis for $H_1(K^R)$. The universal property of semifree extensions in [Corollary 1.3.20](#) gives us a natural map

$$\wedge_k(\Sigma H_1(K^R)) \longrightarrow H(K^R).$$

Therefore, the zigzag above gives us the commutative diagram

$$\begin{array}{ccc} H(K^R) & \xrightarrow{\cong} & H(k[X_1]) \\ \uparrow & & \uparrow \\ \wedge_k(\Sigma H_1(K^R)) & \xrightarrow{\cong} & k[X_1]. \end{array}$$

We saw in [Remark 1.2.9](#) that given a dg algebra, its homology inherits an algebra structure. In particular, we can consider the algebra $H(K^R)$. The following characterization of complete intersections is part of the PhD thesis of E. F. Assmus Jr, advised by John Tate.

Theorem 1.4.25 (Assmus). *Let R be a noetherian local ring. The following are equivalent:*

- 1) *The ring R is a complete intersection.*
- 2) *The natural map $H(K^R) \longrightarrow \wedge_k(\Sigma H_1(K^R))$ is an isomorphism of graded k -algebras.*
- 3) *There is a quasiisomorphism $K^R \simeq \wedge_k(\Sigma H_1(K^R))$ of local dg algebras.*
- 4) *The k -vector space $H_2(K^R)$ is generated by $H_1(K^R)$, meaning that $H_2(K^R) = H_1(K^R)^2$.*

Proof. By [Theorem 1.4.17](#), we can reduce to the complete case. Throughout, fix a regular presentation $(Q, \mathfrak{m}, k)$ for R , that is, assume $R = Q/I$ with $I \subseteq \mathfrak{m}^2$, and fix minimal generators $\underline{y}$ for the maximal ideal of Q . Let $Q[X]$ be a minimal model for R over Q , and let $\partial(X_1) = \{f_1, \dots, f_n\}$.

Suppose that R is a complete intersection. Then $\underline{f} = f_1, \dots, f_n$ is a regular sequence, and by [Lemma 1.3.31](#), $Q[X] = Q[X_1] = \text{Kos}^Q(\underline{f})$. Identifying $\underline{f}$ with its image in R and in k ,

$$K^R \cong \text{Kos}^Q(\underline{f}) \cong R \otimes_Q \text{Kos}^Q(\underline{f})$$

and

$$k[X_1] = k[X] = k \otimes_Q \text{Kos}^Q(\underline{f}) \cong \text{Kos}^k(\underline{f})$$

has trivial differential. Therefore, $H(k[X]) = k[X_1]$. Therefore, in the diagram we discussed in [Remark 1.4.24](#), we have

$$\begin{array}{ccc} H(K^R) & \xrightarrow{\cong} & H(k[X]) \\ \uparrow & & \cong \uparrow \\ \wedge_k(\Sigma H_1(K^R)) & \xrightarrow{\quad} & k[X_1] \end{array}$$

We conclude that $H(K^R) \cong \wedge(H_1(K^R))$ and the map on the left is an isomorphism.

Note that 2) $\implies$ 3) is immediate. Moreover, 3) $\implies$ 4) follows from applying homology and using that the exterior algebra $\wedge(H_1(K^R))$ is generated in degree 1.

Now suppose that $H_2(K^R) = H_1(K^R)^2$. By [Remark 1.4.24](#),

$$H(K^R) \cong H(k[X]).$$

Therefore, $H_2(k[X]) = H_1(k[X])^2$. By [Exercise 1.3.12](#), in $Q[X]$ we have $\partial(X_2) \subseteq \mathfrak{m}X_1$, and thus all variables in X_2 become cycles in $k[X]$. Similarly, all variables in X_1 are cycles in $k[X]$. We conclude that $H_1(k[X]) = k \cdot X_1$.

By [Theorem 1.3.15](#), none of the variables in X_2 is a boundary in $k[X]$. In particular, any $x \in X_2$ contributes a nontrivial class in $H_2(k[X])$. On the other hand, $(H_1(k[X]))^2 = k \cdot X_1^2$, and $k \cdot X_1^2$ contains no variables of degree 2. We conclude that $X_2 = \emptyset$, and thus the minimal model $Q[X]$ for R over Q has no variables of degree 2. By [Lemma 1.3.31](#), R must be a complete intersection. $\square$

Exercise 1.4.26. Let $R = k[[x, y]]/(x^2, xy)$. Find an explicit element in $H_2(K^R)$ that is not in $(H_1(K^R))^2$.

Problem 1.4.27 (Localization Problem for complete intersections). Let R be a noetherian local ring. If R is a complete intersection, is R_P a complete intersection for all primes P ?

Exercise 1.4.28. Let R be a *complete* noetherian local ring. Show that if R is a complete intersection, must R_P be a complete intersection for all primes P .

However, when R is not complete, the problem is quite delicate. The main issue is that completion and localization do not commute. Just like with the Localization Problem for regular local rings, we can solve this issue by giving an appropriate homological characterization of complete intersections. This problem was first solved by Avramov in 1977 [[Avr77](#)]. We will see an alternative proof later in these lectures.

Exercise 1.4.29 (Wiebe's Theorem). Let R be a local ring and set

$$n = \text{embdim}(R) - \text{depth}(R).$$

Show that R is a complete intersection if and only if $H_1(K^R)^n \neq 0$.

Hint for ($\Leftarrow$): First, reduce to the case that R has depth zero. Next, prove this assertion holds when R is artinian; in this case, factor a minimal regular presentation $Q \rightarrow S \rightarrow R$ with S artinian complete intersection and $1(K^S)^n \rightarrow 1(K^R)^n$ is nonzero. The identification

$$n(K^S) = \text{soc}(S) \quad \text{and} \quad n(K^R) = \text{soc}(R)$$

will also be important. Finally, for the case that $\dim(R) > 0$, consider the ideal J of R corresponding to $1(K^R)^n \subseteq n(K^R) = \text{soc}(R)$ and apply Krull's Intersection Theorem to produce $c > 0$ where $J \not\subseteq \mathfrak{m}^c$. Consider the sequence

$$R \rightarrow R/\mathfrak{m}^{c+1} \rightarrow R/\mathfrak{m}^c,$$

and find a way to apply the depth zero case to show R is actually artinian, giving a contradiction.

1.5 Deviations

The minimal model construction also applies to resolve any quotient of a noetherian local ring R . For example, we may wonder what happens if we use it to resolve the residue field.

Example 1.5.1. If we follow the minimal model algorithm to resolve the residue k over $R = k[[t]]/(t^2)$, when $k = \mathbb{Q}$ we will adjoin a variable e of degree 1 and a variable x of degree 2: we get

$$R[e, x \mid \partial(e) = t, \partial(x) = te]$$

which looks like

$$\dots \xrightarrow{4t} Rex^3 \xrightarrow{3t} Rex^2 \xrightarrow{2t} Rex \xrightarrow{t} Rx \xrightarrow{t} Re \longrightarrow R \longrightarrow 0.$$

However, if instead we take $k = \mathbb{F}_2$, $R[e, x \mid \partial(e) = t, \partial(x) = te]$ is no longer a minimal model for k over R : it looks like

$$\dots \xrightarrow{0} Rex^3 \xrightarrow{3t} Rex^2 \xrightarrow{0} Rex \xrightarrow{t} Rx \xrightarrow{t} Re \longrightarrow R \longrightarrow 0.$$

In fact, one can show that the minimal model of k over R is much larger, and we will need to keep adjoining variables in higher degrees.

This characteristic dependance is explained by the following:

Exercise 1.5.2. Let x be an element of even degree in a dg algebra with differential ∂ . Show that

$$\partial(x^n) = n\partial(x)x^{n-1} \quad \text{for all } n \geq 1.$$

Remark 1.5.3 (Divided power variables). The potential disadvantage of polynomial variables is only visible in prime characteristic. Each time we add a new variable x of even degree, its ripple effect is felt forever, as all the powers x^n are nonzero. This is sometimes an advantage: by the time we get to fixing the homology in some degree $d - 1$, we might already have elements of degree d , made out of products of variables of smaller degrees, that turn those cycles into boundaries. But in prime characteristic p , the lower degree variables might have created new cycles as well: if x has even degree, then by [Exercise 1.5.2](#)

$$\partial(x^p) = p\partial(x)x^{p-1} = 0.$$

To avoid this, when in prime characteristic, rather than adding one variable x in even degree, we add an infinite collection of variables $x = x^{(1)}$ and $x^{(i)}$ for all $i \geq 1$, satisfying the following rules:

$$x^{(i)}x^{(j)} = \binom{i+j}{i} x^{(i+j)} \quad \text{and} \quad \partial(x^{(i+1)}) = x^{(i)}\partial(x).$$

We say that $x^{(i)}$ are **divided power variables**. Note, however, that over a field of characteristic 0, this recipe coincides with adding polynomial variables, as

$$x^{(i)} = \frac{1}{i!} x^i.$$

We will write $S\langle x \rangle$ for the dg S -algebra obtained by adjoining the divided power variables $x^{(i)}$ to S , to distinguish from $S[x]$, obtained by adjoining the polynomial variable x . More generally, $S\langle X \rangle$ is a semifree extension of S obtained by adjoining exterior variables in odd degrees and divided power variables in even degrees.

Exercise 1.5.4. Let A be a dg algebra and x a divided power variable. Show that the differential

$$\partial(x^{(i+1)}) = x^{(i)}\partial(x)$$

satisfies the Leibniz rule.

Hint: use an appropriate binomial identity.

Definition 1.5.5. Let $\widehat{R} \cong Q/I$ be a minimal regular presentation for a local ring R . An **acyclic closure** $R\langle Y \rangle$ for k over R is a semifree extension over R that resolves k over R , where we adjoin exterior variables in odd degrees and divided power variables in even degrees, and take the smallest number possible of variables in each degree.

Exercise 1.5.6. Let $Q = k[[x, y]]$, $I = (x^2, xy)$, and $R = Q/I$.

- 1) Write the first 3 steps to construct an acyclic closure for k over R .
- 2) Check your work with the `DGAlgebras` package.

Some of the basic properties of minimal models we saw before have analogous versions for acyclic closures.

Exercise 1.5.7. Let $\alpha: A \rightarrow B$ be a map of dg algebras and let Z be a set of cycles in A . Show that given a set of variables X indexed by Z , there exists a unique map of dg algebras

$$\begin{array}{ccc} A\langle X \mid \partial(x_z) = z \rangle & \xrightarrow{\tilde{\alpha}} & B\langle X \mid \partial(x_z) = \alpha(z) \rangle \\ x_z \longmapsto & & x_z \end{array}$$

extending α . Show that if α is surjective, local, or a quasiisomorphism, then so is $\tilde{\alpha}$.

While the Betti numbers count the number of generators in each homological degree, there is a dg algebra analogue that counts the number of algebra generators we add in each degree. These are especially important for the acyclic closure of the residue field: as we will how in [Theorem 1.5.25](#), the acyclic closure of is in fact the minimal free resolution for of the residue field.

Definition 1.5.8. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring and $R\langle Y \rangle$ be an acyclic closure of k . The **deviations** of R count the number of variables in each degree:

$$\varepsilon_i(R) := |Y_i|.$$

Remark 1.5.9. Let us compute the first few deviations of a noetherian local ring $(R, \mathfrak{m}, k)$. By definition,

$$\varepsilon_1(R) = \mu(\mathfrak{m}) = \text{embdim}(R).$$

By construction, Y_2 maps bijectively onto $H_1(R\langle Y_1 \rangle) \cong H_1(K^R)$, so by [Theorem 1.4.21](#)

$$\varepsilon_2(R) = \mu(I).$$

Remark 1.5.10. Let $R\langle Y \rangle$ be an acyclic closure k over R . By [Lemma 1.4.18](#), the completion map induces quasiisomorphisms

$$R\langle Y_{\leq n} \rangle \longrightarrow R\langle Y_{\leq n} \rangle \otimes_R \widehat{R}.$$

Since $R\langle Y_{\leq n} \rangle$ are complexes of finitely generated R -modules, by [Exercise 1.5.7](#) we can inductively show that we have quasiisomorphisms

$$R\langle Y_{\leq n} \rangle \otimes_R \widehat{R} \cong \widehat{R}\langle Y_{\leq n} \rangle.$$

Therefore, we get isomorphisms

$$H_n(R\langle Y_{\leq n} \rangle) \cong H_n(\widehat{R}\langle Y_{\leq n} \rangle).$$

In particular,

$$\varepsilon_n(R) = \varepsilon_n(\widehat{R}).$$

Remark 1.5.11. Given the deviations of a local ring R , we can compute the Betti numbers of k . Let us illustrate this by computing the first few. We know that $\beta_0(k) = 1$. As we will see in [Theorem 1.5.25](#), the acyclic closure $R\langle Y \rangle$ is a minimal resolution for k , and $R\langle Y \rangle$ has the form

$$\begin{array}{ccccccc} & & RY_4 & & & & \\ & & \oplus & & & & \\ & & RY_1Y_3 & & RY_3 & & \\ & & \oplus & & \oplus & & \\ \cdots & \longrightarrow & RY_1^2Y_2 & \longrightarrow & RY_1Y_2 & \longrightarrow & RY_2 \\ & & \oplus & & \oplus & & \\ & & RY_1^4 & & RY_1^3 & & \\ & & \oplus & & & & \\ & & RY_2^{(2)} & & & & \end{array} \longrightarrow RY_1 \longrightarrow R.$$

Thus

$$\beta_1(k) = \varepsilon_1(R) \quad \beta_2(k) = \varepsilon_2(R) + \binom{\varepsilon_1(R)}{2} \quad \beta_3(k) = \varepsilon_3(R) + \varepsilon_2(R)\varepsilon_1(R) + \binom{\varepsilon_1(R)}{3}.$$

This strategy allows us to find formulas for $\beta_i(k)$ in terms of $\varepsilon_j(R)$ for all i . Note that in these formulas, all the coefficients on $\varepsilon_j(R)$ are positive.

The deviations of R are closely related with the Poincaré series of k , the generating function for the Betti sequence of k .

Definition 1.5.12. Let R be a noetherian local ring and M a finitely generated R -module. The Poincaré series of M is the power series with integer coefficients given by

$$P_M^R(t) := \sum_{d=0}^{\infty} \beta_d(M) t^d.$$

Remark 1.5.13. When $M = R/I$ is a cyclic R -module, then $\beta_0(R/I) = 1$, and the Poincaré series of R/I has the form

$$1 + \sum_{i=1}^{\infty} b_i t^i.$$

Any power series of this form can be written uniquely as a (possibly infinite) product of the form

$$1 + \sum_{i=1}^{\infty} b_i t^i = \frac{\prod_{i=1}^{\infty} (1 + t^{2i-1})^{e_{2i-1}}}{\prod_{i=1}^{\infty} (1 - t^{2i})^{e_{2i}}}$$

that converges in the (t) -adic topology of $\mathbb{Z}[[t]]$. This can be shown via a quick induction, going modulo (t^n) for each successive n to find e_n , which we leave as an exercise.

We claim that when we write the Poincaré series of the residue field k in this form, say

$$P_k^R(t) = \frac{\prod_{i=1}^{\infty} (1 + t^{2i-1})^{e_{2i-1}}}{\prod_{i=1}^{\infty} (1 - t^{2i})^{e_{2i}}},$$

these exponents e_n are precisely the deviations $\varepsilon_n(R)$ of R .

To see this, let $R\langle Y \rangle$ be an acyclic closure for k . By [Theorem 1.5.25](#), this is a minimal free resolution for k , so the differential in

$$k\langle Y \rangle := R\langle Y \rangle \otimes_R k$$

vanishes. Note that

$$k\langle Y \rangle = \bigotimes_{y \in Y} k\langle y \rangle.$$

Fix a particular variable $y \in Y$. If y has odd degree $2i - 1$, then $k\langle y \rangle$ has a copy of k in degree 0 and another in degree $2i - 1$, and nothing else, so

$$\sum_{n=0}^{\infty} \dim_k(k\langle y \rangle_n) \cdot t^n = 1 + t^{2i-1}.$$

If y has even degree $2i$, then $k\langle y \rangle = k\langle y^{(i)} \mid i \geq 1 \rangle$ has one copy of k in every degree that is a multiple of $2i$, and

$$\sum_{n=0}^{\infty} \dim_k(k\langle y \rangle_n) \cdot t^n = \sum_{\ell=0}^{\infty} t^{(2i)\ell} = \frac{1}{1-t^{2i}}.$$

To count the rank of $k\langle Y \rangle$ in degree n , we need only to count the number of monomials in the variables of Y of total degree n . Thus

$$P_k^R(t) = \frac{\prod_{i=1}^{\infty} (1+t^{2i-1})^{|Y_{2i-1}|}}{\prod_{i=1}^{\infty} (1-t^{2i})^{|Y_{2i}|}} = \frac{\prod_{i=1}^{\infty} (1+t^{2i-1})^{\varepsilon_{2i-1}(R)}}{\prod_{i=1}^{\infty} (1-t^{2i})^{\varepsilon_{2i}(R)}}.$$

Lemma 1.5.14. *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. The following are equivalent:*

- 1) R is regular.
- 2) $\varepsilon_n(R) = 0$ for all $n \geq 2$.
- 3) $\varepsilon_2(R) = 0$.

Proof. If R is regular, by [Theorem 0.3.12](#) the maximal ideal $\mathfrak{m}$ is generated by a regular sequence, and an acyclic closure of k is just the Koszul complex on a minimal generating set of $\mathfrak{m}$. Therefore, $\varepsilon_n(R) = 0$ for all $n \geq 2$.

Note that 2) $\Rightarrow$ 3) is obvious.

If $\varepsilon_2(R) = 0$, then the Koszul complex $R\langle Y_1 \rangle$ on a minimal generating set for $\mathfrak{m}$ has $H_1(R\langle Y_1 \rangle) = 0$, so $\mathfrak{m}$ is generated by a regular sequence by [Theorem 1.1.4](#). Therefore, R is regular by [Theorem 0.3.12](#).

Alternatively, we can see that $I = 0$ by [Remark 1.5.9](#), so $\widehat{R} \cong Q$ is a regular ring. This completes the proof since R is regular if and only if $\widehat{R}$ is regular. $\square$

Exercise 1.5.15. Let A be a dg algebra. Show that the natural map

$$A[x]\langle y \mid \partial(y) = x \rangle \longrightarrow A$$

is a homotopy equivalence of dg algebras. Hint: consider the cases when $|x|$ is even and odd separately.

Construction 1.5.16 (Avramov). We saw in [Remark 1.4.24](#) that there is a quasiisomorphism of dg algebras

$$K^Q \otimes_Q Q[X] \xrightarrow{\simeq} k[X].$$

Let us focus on the right hand side and construct an acyclic closure of k over $k[X]$.

By [Remark 1.4.24](#), we have

$$H_0(k[X]) = k \quad \text{and} \quad H_1(k[X]) = kX_1.$$

Set $Y_2 = \{y_x \mid x \in X_1\}$ to be a set of degree 2 variables, and consider $k[X]\langle y_x \mid \partial(y_x) = x \rangle$. By [Exercise 1.5.15](#),

$$k[X]\langle \partial(y_x) = x \rangle \xrightarrow{\cong} k[X]/(X_1) \cong k[X_{\geq 2}].$$

Hence

$$H_n(k[X]\langle \partial(y_x) = x \rangle) = \begin{cases} k & \text{if } n = 0 \\ 0 & \text{if } n = 1 \\ kX_2 & \text{if } n = 2. \end{cases}$$

Now we repeat this idea in degree 3: by [Exercise 1.5.7](#), there is a quasiisomorphism

$$k[X]\langle Y_2, Y_3 \rangle \xrightarrow{\cong} k[X_{\geq 2}]\langle Y_3 \mid \partial(y_x) = x \rangle.$$

Fix $x \in X_2$. Lifting $\partial(y_x)$ under this quasiisomorphism, we see that in $k[X]\langle Y_2, Y_3 \rangle$

$$\partial(y_x) = x + a + y$$

for some $a \in kX_1^2$ and $y \in kY_2$. We claim that $y = 0$. Indeed, applying ∂ again, we get

$$0 = \partial^2(y_x) = \partial(x) + \partial(a) + \partial(y).$$

Note that $\partial(x) = 0$, since $\partial(X_2) = 0$ in $k[X]$ by [Exercise 1.3.12](#). Moreover, $\partial(a) = 0$ since by [Exercise 1.3.12](#) and the Leibniz rule $\partial(X_1^2) = 0$ in $k[X]$. Therefore, $\partial(y) = 0$. However, note that kX_2 maps bijectively onto kX_1 under the differential:

$$\partial: kY_2 \xrightarrow{\cong} kX_1.$$

Thus $y = 0$. We now conclude that

$$\partial(y_x) = x + a_x \quad \text{for some } a_x \in (X)^2.$$

We now continue inductively, and construct an acyclic closure over $k[X]$, which has the form

$$k[X]\langle Y_{\geq 2} \rangle \xrightarrow{\cong} k$$

with

$$Y_n = \{y_x \mid x \in X_{n-1}\} \quad \text{and} \quad |y_x| = n \text{ for each } n \geq 2$$

and

$$\partial(y_x) = x + a_x \quad \text{with} \quad a_x \in (X)(X, Y^{(n)} \mid n \geq 1).$$

Theorem 1.5.17. *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. Fix an acyclic closure $R\langle Y \rangle$ for k and a minimal model $Q[X]$ for R . There are quasiisomorphisms of local dg algebras*

$$R\langle Y_{\leq n} \rangle \xrightarrow{\cong} k[X_{\geq n}] =: k[X]/(X_{< n}).$$

Proof. We saw in [Remark 1.4.24](#) that there is a zigzag of quasiisomorphisms of dg algebras

$$K^R \xleftarrow{\cong} K^Q \otimes_Q Q[X] \xrightarrow{\cong} k[X].$$

On the right hand side, the quaiisomorphism we built in [Construction 1.5.16](#) together with [Exercise 1.5.7](#) give us a quasiisomorphism

$$K^Q \otimes_Q Q[X]\langle Y_{\geq 2} \rangle \xrightarrow{\simeq} k[X]\langle Y_{\geq 2} \rangle \xrightarrow{\simeq} k.$$

Moreover, using [Exercise 1.5.7](#) on the right hand side of our zigzag, we also get a quasiisomorphism, leading to a new zigzag of quasiisomorphisms

$$K^R\langle Y_{\geq 2} \rangle \xleftarrow{\simeq} K^Q \otimes_Q Q[X]\langle Y_{\geq 2} \rangle \xrightarrow{\simeq} k[X]\langle Y_{\geq 2} \rangle \xrightarrow{\simeq} k. \quad \square$$

Remark 1.5.18. Let R be a noetherian local ring, and $\widehat{R} \cong Q/I$ with $(Q, \mathfrak{m}, k)$ a minimal regular presentation. Let $Q[X]$ be a minimal model for R over Q , set $k[X] := Q[X] \otimes_Q k$, and consider the quotient complexes $k[X_{\geq n}]$ attained by modding out by the dg ideal $(X_{\leq n-1})$. Since $Q[X]$ is a minimal model, in $k[X_{\geq n}]$ we have $\partial(X_{n+1}) = 0$ and $\partial(X_n) = 0$. Moreover,

$$k[X_{\geq n}]_n = kX_n.$$

Therefore,

$$H_n(k[X_{\geq n}]) = kX_n.$$

Theorem 1.5.19 (Avramov, 1984 [[Avr84](#)]). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. Fix an acyclic closure $R\langle Y \rangle$ for k and a minimal model $Q[X]$ for R . Then for all $i \geq 2$,*

$$\varepsilon_i(R) = |Y_i| = |X_{i-1}|.$$

Proof. By [Theorem 1.5.17](#), we have quasiisomorphisms

$$R\langle Y_{\leq n} \rangle \xrightarrow{\simeq} k[X_{\geq n}] =: k[X]/(X_{< n}).$$

By [Remark 1.5.18](#)

$$H_n(k[X_{\geq n}]) = kX_n.$$

Therefore,

$$H_n(R\langle Y_{\leq n} \rangle) \cong H_n(k[X_{\geq n}]) = kX_n$$

is minimally generated by $|X_n|$ many elements. Since the variables in Y_{n+1} are added to kill degree n cycles in $R\langle Y_{\leq n} \rangle$, we conclude that $|Y_{n+1}| = |X_n|$. $\square$

We can now think about deviations in two ways: via the acyclic closure of k or via the minimal model of R .

Remark 1.5.20. We saw in [Remark 1.5.9](#) that $\varepsilon_1(R) = \text{embdim}(R)$ and $\varepsilon_2(R) = \mu(I)$. We can now also compute $\varepsilon_3(R)$.

Since $Q[X_1]$ is the Koszul complex on a minimal generating set $\underline{f}$ for I , the number of variables in X_2 is the minimal number of generators for the first Koszul homology on $\underline{f}$. Since the Koszul homology is independent of the choice of minimal generators for I , we simply write this as $H_1(\text{Kos}(I))$. Thus

$$\varepsilon_3(R) = \mu(H_1(\text{Kos}(I))).$$

Theorem 1.5.21. *A noetherian local ring R is a complete intersection if and only if*

$$\varepsilon_n(R) = 0 \quad \text{for all} \quad n \geq 3.$$

Proof. Let $\widehat{R} \cong Q/I$ with $(Q, \mathfrak{m}, k)$ a minimal regular presentation for R , and let $Q[X]$ be a minimal model for R over Q .

If I is generated by a regular sequence, then $Q[X] = Q[X_1]$ by [Lemma 1.3.31](#). By [Theorem 1.5.19](#), we can read the deviations of R from the minimal model. Therefore, for all $n \geq 3$ we have

$$\varepsilon_n(R) = |X_{n-1}| = 0. \quad \square$$

Conversely, if $\varepsilon_n(R) = 0$ for all $n \geq 3$, then by [Theorem 1.5.19](#) we have $X_n = \emptyset$. By [Lemma 1.3.31](#), R must be a complete intersection.

Tate [[Tat57](#)] gave a very different proof of [Theorem 1.5.21](#) by showing directly that the acyclic closure $R\langle Y \rangle$ of k over R has $Y_n = 0$ for all $n \geq 3$.

Exercise 1.5.22. Let $R = Q/I$ with $(Q, \mathfrak{m}, k)$ a regular local ring and $I \subseteq \mathfrak{m}^2$ generated by a regular sequence. Use the quasiisomorphisms of local dg algebras (with residue field k)

$$K^R \xleftarrow{\simeq} Q[X_1] \otimes_Q K^Q \xrightarrow{\simeq} k[X_1].$$

and [Exercise 1.5.15](#) to write down an explicit acyclic closure of k over R in terms of the following data:³ $\mathfrak{m}$ is generated by $t_1, \dots, t_n$, the ideal I is minimally generated the regular sequence $f_1, \dots, f_c$, and

$$f_i = \sum_{j=1}^n a_{ij} t_j \quad \text{for some} \quad a_{ij} \in \mathfrak{m}.$$

In fact, the vanishing of the higher deviations characterizes complete intersections. The characterization of complete intersections we will state next puts together the work of several authors. The cumulative theorem tells us that various conditions are equivalent to being a complete intersection: Assmus [[Ass59](#)] showed the equivalence with (3) and Gulliksen showed the equivalence with conditions (4) [[Gul71](#)] and (5) [[Gul80](#)]. The last condition, due to Halperin [[Hal87](#)] and known as Halperin's Rigidity Theorem, is the most amazing: as long as at least one deviation vanishes, then R must be a complete intersection. This tells us that as long as R is not a complete intersection, then when constructing a minimal model for R over Q or an acyclic closure for k over R , we must add new variables in *every* degree.

³This description is due to Tate.

Theorem 1.5.23 (Assmus, 1959 [Ass59], Gulliksen, 1971 [Gul71] and 1980 [Gul80], Halperin, 1987 [Hal87]). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. The following are equivalent:*

- 1) R is a complete intersection.
- 2) $\varepsilon_n(R) = 0$ for all $n \geq 3$.
- 3) $\varepsilon_3(R) = 0$.
- 4) $\varepsilon_n(R) = 0$ for all $n \gg 0$.
- 5) $\varepsilon_{2n}(R) = 0$ for all $n \gg 0$.
- 6) $\varepsilon_n(R) = 0$ for some $n \geq 1$.

Remark 1.5.24. Let $Q[X]$ be a minimal model for R over Q and $R\langle Y \rangle$ an acyclic closure for k over R . By Theorem 1.5.23 and Theorem 1.5.19, if R is not a complete intersection, then $X_i \neq \emptyset$ and $Y_i \neq \emptyset$ for all $i \geq 1$.

Theorem 1.5.25 (Gulliksen, 1968 [Gul68], Schoeller, 1967 [Sch67]). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. An acyclic closure for k is a minimal free resolution for k .*

Chapter 2

The homotopy Lie algebra

2.1 Graded Lie algebras

Lie algebras arise throughout mathematics as spaces that parametrize infinitesimal automorphisms. In this section we will introduce graded Lie algebras and discuss basic examples and constructions concerning them. We work over a field k of arbitrary characteristic.

Definition 2.1.1. A **graded Lie algebra** is a graded k -vector space L equipped with a bilinear bracket operation

$$[-, -]: L \times L \longrightarrow L$$

and a quadratic operation

$$(-)^{[2]}: L^{2i+1} \longrightarrow L^{4i+2}$$

defined on elements of odd degree, where quadratic means that $(\alpha u)^{[2]} = \alpha^2 u^{[2]}$ for all $\alpha \in k$ and $u \in L$ of odd degree $|u|$. These operations must satisfy the following axioms:

- 1) (Compatibility) For any odd degree elements u and v in L ,

$$(u + v)^{[2]} = u^{[2]} + [u, v] + v^{[2]}.$$

- 2) (Antisymmetry) For any x, y, z in L with z even

(a) $[x, y] + (-1)^{|x||y|}[y, x] = 0$, and

(b) $[z, z] = 0$.

- 3) (Jacobi identities) For any x, y, z, u, v in L with u odd:

(a) $[x, [y, z]] = [[x, y], z] + (-1)^{|x||y|}[y, [x, z]],$

(b) $[u^{[2]}, v] = [u, [u, v]],$ and

(c) $[u, [u, u]] = 0$.

We say that L has **finite type** if L_n is finite dimensional for all n .

Throughout, we will assume that all our Lie algebras are finite dimensional in each degree (finite type) and positively graded.

Remark 2.1.2. When 2 is invertible in k , the compatibility axiom implies that if u has odd degree, then

$$u^{[2]} = \frac{1}{2}[u, u].$$

It follows that the reduced square can be removed from the definition, and moreover the axioms (1), (2b), and (3b) are redundant in this case.

When 3 is invertible in k , axiom (3c) is redundant. In particular, if the characteristic of k is at least 5, then a graded Lie algebra is simply a graded vector space with an antisymmetric bracket, in the sense of (2a), satisfying the usual graded Jacobi identity (3a).

Remark 2.1.3. Any Lie algebra L can be viewed as a graded Lie algebra with $L = L_0$.

Example 2.1.4. Fix a field k and $n \geq 1$, and consider the set $\text{Mat}_n(k)$ of all $n \times n$ matrices with entries in k , which is a Lie algebra with the Lie bracket given by

$$[A, B] := AB - BA.$$

Definition 2.1.5. A graded Lie algebra L is **abelian** if its bracket and its reduced square operation are both equal to zero.

Example 2.1.6. Any graded associative algebra A over k gives rise to a graded Lie algebra A^{Lie} with the same underlying graded vector space. The bracket is given by the graded commutator

$$[x, y] = xy - (-1)^{|x||y|}yx$$

while the reduced square is simply $u^{[2]} = u^2$ for any x, y, u in A with u of odd degree. This assignment yields a functor $\mathbf{Alg}_k \rightarrow \mathbf{Lie}_k$ from the category of associative graded (noncommutative) k -algebras to the category of graded Lie algebras over k .

Exercise 2.1.7. Check that the recipe in Example 2.1.6 does indeed give a graded Lie algebra, by checking that the axioms hold.

Example 2.1.8. When $A = \text{Mat}_n(k)$ is the k -algebra of all $n \times n$ matrices, one can show that $A^{\text{Lie}} = \text{GL}_n(k)$ concentrated in degree 0.

Definition 2.1.9. Let A be a graded k -algebra. A degree d k -linear **derivation** on A is a degree $|\alpha| = d$ homomorphism $\alpha: A \rightarrow A$ of graded k -modules such that

$$\alpha(xy) = \alpha(x)y + (-1)^{|x| \cdot |\alpha|}x\alpha(y)$$

for any elements $x, y \in A$. The graded set of all such derivations is $\text{Der}_k(A, A)$.

Example 2.1.10. Let A be a graded k -algebra. If α and β are derivations on A , then so is

$$[\alpha, \beta] = \alpha \circ \beta - (-1)^{|\alpha||\beta|}\beta \circ \alpha.$$

Moreover, if γ is a derivation on A with odd degree, then $\gamma^{[2]} = \gamma \circ \gamma$ is a derivation on A as well. It follows that $\text{Der}_k(A, A)$ is a Lie subalgebra of $\text{End}_k(A)^{\text{Lie}}$.

Exercise 2.1.11. Let L be a graded Lie algebra and A a strictly graded commutative algebra. Show that $L \otimes_k A$ is a graded Lie algebra with

$$[u \otimes a, v \otimes b] := (-1)^{|a||v|} [u, v] \otimes ab.$$

Definition 2.1.12. Let L be a graded Lie algebra. A **Lie subalgebra** A of L is a graded subspace of L that is closed under the two Lie algebra operations, meaning that

$$[A, A] \subseteq A \quad \text{and} \quad u^{[2]} \in A \text{ for all } u \in A \text{ of odd degree.}$$

A **Lie ideal** I of L is a Lie subalgebra of L such that $[L, I] \subseteq I$. Given a set of elements $X \subseteq L$, the **Lie ideal generated by** X is the intersection of the Lie ideals containing X .

Remark 2.1.13. Notice that any Lie subalgebra of L is a Lie algebra in its own right (under the same Lie operations).

Example 2.1.14. Fix a field k and consider the set of traceless $n \times n$ matrices, meaning those with trace zero. This is a Lie subalgebra of $\text{Mat}_n(k)$.

Example 2.1.15. Let L be any graded Lie algebra. The truncation $L_{\geq n}$ is a Lie ideal of L .

Exercise 2.1.16. Let A be an associative graded k -algebra and let $[-, -]$ denote the graded commutator on A . The set of **inner derivations** on A is

$$\text{Inn}_k(A) := \{\alpha : A \rightarrow A \mid \alpha = [a, -] \text{ for some } a \in A\}.$$

Show that $\text{Inn}_k(A)$ is a Lie ideal of $\text{Der}_k(A)$.

Given a graded Lie algebra L , and an element $z \in L$, the Jacobi identity 3a asserts that $[z, -]$ is a graded derivation on L . This derivation will take on a central role in our story.

Definition 2.1.17. Let L be a graded Lie algebra and fix an element $z \in L$. The **adjoint map** of z , denoted $\text{ad}(z)$ is the map $\text{ad}(z) : L \rightarrow L$ given by

$$\text{ad}(z) = [z, -].$$

More precisely, for each $y \in L$, $\text{ad}(z)(y) = [z, y]$.

Notation 2.1.18. Let L be a graded Lie algebra and fix an element $z \in L$. We write

$$\text{ad}^n(z) := \underbrace{\text{ad}(z) \circ \cdots \circ \text{ad}(z)}_{n \text{ times}} = \underbrace{[z, [z, \cdots, [z, -] \cdots]]}_{n \text{ times}}.$$

Our main focus will be on a graded Lie algebra, $\pi^*(R)$, that arises from a noetherian local ring R , and more generally, from minimal semifree extensions. Over $\pi^*(R)$, we will be particularly interested in elements for which $\text{ad}(z)$ is close to zero.

Definition 2.1.19. Let L be a graded Lie algebra and fix an element $z \in L$.

- We say z is **central** if $\text{ad}(z) = 0$.
- We say z is **radical** if there exists n such that the restriction

$$\text{ad}(z): L^{\geq n} \longrightarrow L^{\geq n}$$

is zero.

- We say z is **nilpotent** if there exists n such that $\text{ad}^n(z) = 0$.
- We say z is **Engel** if for each $\alpha \in L$ there exists N_α such that $\text{ad}^{N_\alpha}(z)(\alpha) = 0$.

In general, our definition of radical element is not equivalent to the standard definition, though the two notions do match in the setting we are interested in. In general, radical elements are defined in terms of the derived and central series of our graded Lie algebra. However, these series will not be strictly necessary for the applications we will focus on.

Definition 2.1.20. Let L be a graded Lie algebra. Consider the Lie ideals

$$L^{[0]} := L \quad \text{and} \quad L^{[n+1]} := [L^{[n]}, L^{[n]}] \text{ for all } n \geq 0.$$

The **derived series** of L is the sequence

$$L^{[0]} \supseteq L^{[1]} \supseteq L^{[2]} \supseteq \dots.$$

We say L is **solvable** if $L^{[n]} = 0$ for some n . A ideal of a graded Lie algebra is **solvable** if it is solvable as a Lie algebra in its own right.

Moreover, for each $n \geq 0$ let

$$L^{(0)} := L \quad \text{and} \quad L^{(n+1)} = [L^{(n)}, L] \text{ for all } n \geq 0.$$

The **lower central series** of L is the sequence

$$L^{(0)} \supseteq L^{(1)} \supseteq L^{(2)} \supseteq \dots.$$

We say L is **nilpotent** if $L^{(n)} = 0$ for some n .

Exercise 2.1.21. Show that every nilpotent Lie algebra is solvable.

Remark 2.1.22. Let z be an element in a graded Lie algebra. Note that

$$z \text{ is central} \implies z \text{ is radical} \implies z \text{ is nilpotent} \implies z \text{ is Engel}.$$

Definition 2.1.23. Let L be a graded Lie algebra. The **center** of L , denoted $\zeta(L)$, is the set of all central elements in L .

Exercise 2.1.24. Show that the center of a graded Lie algebra L is an abelian Lie subalgebra of L .

Definition 2.1.25. Let L be a graded Lie algebra. The **radical** of L , denoted $\text{rad}(L)$, is the set of all radical elements.

One can show that in the setting we are interested in, radical elements are precisely those that generate a solvable ideal. In particular, the radical is the sum of all the solvable ideals. This equivalence is by no means obvious, and in fact follow from a hard theorem (see [FHJ⁺88, Theorem C]).

Notation 2.1.26. Let L be a graded Lie algebra. We will write

$$\text{rad}^n(L) := \text{rad}(L) \cap L^n \quad \text{and} \quad \zeta^n(L) := \zeta(L) \cap L^n.$$

Exercise 2.1.27. Let k be a field and consider the graded k -vector space

$$L = k\{u_i, v_i \mid i \geq 0\}$$

where $|u_i| = 2i$ and $|v_i| = 2i + 1$, so that $\dim_k(L_i) = 1$ for all $i \geq 1$. We claim that L has a graded Lie algebra structure with

$$[v_i, v_j] = 0 \quad v_i^{[2]} \quad [u_i, u_j] = 0 \quad [u_i, v_j] = v_{i+j}.$$

- 1) Check that L is a graded Lie algebra.
- 2) What are the central elements in L ?
- 3) What are the nilpotent elements in L ?

Definition 2.1.28. A **map of graded Lie algebras** is a map of graded vector spaces $\varphi: L \rightarrow L'$ that respects the two Lie algebra operations, so that

$$\varphi[x, y] = [\varphi(x), \varphi(y)] \quad \text{and} \quad \varphi(u^{[2]}) = (\varphi(u))^{[2]}$$

for any $x, y, u \in L$ with u of odd degree. We write $\mathbf{Lie}_k$ for the resulting category of graded Lie algebras over k .

Example 2.1.29. Let L be a Lie algebra and A a Lie subalgebra of L . The inclusion map $A \hookrightarrow L$ is a map of graded Lie algebras.

Exercise 2.1.30. Let L be a graded Lie algebra and A be a Lie ideal of L . Show that there exists a unique Lie algebra structure the vector space quotient L/A such that the quotient map $L \twoheadrightarrow L/A$ is a map of Lie algebras.

Definition 2.1.31. Let V be a graded vector space over k . The **free Lie algebra** on V is the Lie algebra $\text{Lie}(V)$ with the following universal property: given any basis B for V consisting of homogeneous elements, any graded Lie algebra L , and any function $\varphi: B \rightarrow L$ that preserves the grading, there exists a unique map of graded Lie algebras $\widehat{\varphi}: \text{Lie}(V) \rightarrow L$ extending φ .

2.2 Universal enveloping algebras

Definition 2.2.1. Let L be a graded Lie algebra. The **universal enveloping algebra** of L is an associative k -algebra $U(L)$ together with a map of Lie algebras $L \rightarrow U(L)^{\text{Lie}}$ with the following universal property: if A is an algebra, and $L \rightarrow A^{\text{Lie}}$ is a map of graded Lie algebras, then there is a unique map of algebras $U(L) \rightarrow A^{\text{Lie}}$ extending $L \rightarrow A^{\text{Lie}}$, meaning that the following diagram commutes:

$$\begin{array}{ccc} U(L) & \xrightarrow{\exists! \varphi} & A \\ & \nwarrow \quad \nearrow & \\ & L & \end{array}$$

We can describe $U(L)$ explicitly using the tensor algebra.

Remark 2.2.2. Recall that the tensor algebra on a graded vector space V is the graded k -algebra

$$T(V) = \bigoplus_{n \geq 0} V^{\otimes n}$$

with the product

$$(u_1 \otimes \cdots \otimes u_i) \cdot (v_1 \otimes \cdots \otimes v_j) = u_1 \otimes \cdots \otimes u_i \otimes v_1 \otimes \cdots \otimes v_j$$

and with unit the empty tensor $1 \in k = V^{\otimes 0}$. Note that each of the tensor products above is a graded tensor product, with

$$|v_1 \otimes \cdots \otimes v_n| = |v_1| + \cdots + |v_n|.$$

In fact, $T(V)$ is even bigraded: besides the grading we just described, there is also a grading coming from taking $V^{\otimes n}$ in degree n .

One can show that $T(V)$ is the free unital, associative, graded algebra on V . This means that for any graded algebra A and any map $V \rightarrow A$ of graded vector spaces, there is a unique map of graded algebras $T(V) \rightarrow A$ factoring $V \rightarrow A$, meaning that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\quad} & A \\ \downarrow & \nearrow & \\ T(V) & & \end{array}$$

Exercise 2.2.3. Show that there is an isomorphism of graded k -algebras

$$U(L) \cong \frac{T(L)}{(xy - (-1)^{|x||y|}yx = [x, y], z^2 = z^{[2]} \mid x, y, z \in L, |z| \text{ odd})}.$$

Example 2.2.4. If $L = L_{\text{even}}$ is an abelian graded Lie algebra, then

$$U(L) = \frac{T(V)}{(xy - yx = [x, y] \mid x, y \in L)} = \text{Sym}(L).$$

If $L = L_{\text{odd}}$ is an abelian graded Lie algebra, then

$$U(L) = \frac{T(V)}{(xy + yx, x^2 \mid x, y \in L)} = \wedge(L).$$

Theorem 2.2.5 (Poincaré–Birkoff–Witt Theorem). *Fix a homogeneous basis $\{x_i\}_{i \in I}$ for L indexed by a totally ordered set I . Then $U(L)$ has a basis of monomials*

$$\{x_{i_1} \cdots x_{i_n} \mid i_1 \leq \cdots \leq i_n \in I \text{ and } i_j \neq i_{j+1} \text{ if } |x_{i_j}| \text{ is odd}\}.$$

Let us illustrate the idea we will use to show that these monomials form a spanning set with an example.

Example 2.2.6. Suppose that $x < y < z$ are basis elements of L of even degree. To express $y(xz)$ as a linear combination of the monomials in [Theorem 2.2.5](#), we first write

$$y(xz) = xyz - (xy - yx)z = xyz - \underbrace{[x, y]}_{\in L} z.$$

Now $[x, y] \in L$ can also be written in terms of the basis elements, say

$$[x, y] = a_1 w_1 + \cdots + a_n w_n$$

for some $a_i \in k$ and $w_1 < \cdots < w_n$ in our chosen basis for L . For each of those basis elements where $w = w_i > z$, we again employ the same trick, and rewrite

$$wz = zw + [w, z].$$

Finally, $[w, z] \in L$ is a linear combination of the basis elements.

In general, one can show by induction on the length of monomials that any monomial is a linear combination of admissible monomials. Indeed, for a monomial of length 1, there is nothing to show, as any monomial of length 1 is admissible. Now note that by successive applications of the relations

$$xy - (-1)^{|x||y|}yx = [x, y], z^2 = z^{[2]}$$

we can rewrite any monomial of length ℓ as a linear combination of admissible monomials and monomials of length $\ell - 1$, which by induction is a linear combination of admissible monomials.

It is more difficult to show that the admissible monomials are linearly independent. We will finish the proof of [Theorem 2.2.5](#) later, using homological methods.

Exercise 2.2.7. Show that $U(L)$ is finite dimensional if and only if L is finite dimensional and $L = L_{\text{odd}}$.

Exercise 2.2.8. Show that if $U(L)$ is infinite dimensional, then $U(L)$ is torsionfree, meaning that

$$\text{tors}(U(L)) := \{y \in U(L) \mid yx_1 \cdots x_n = 0 \text{ for all } n \gg 0 \text{ and all } x_i \in L\}$$

satisfies $\text{tors}(U(L)) = 0$.

Exercise 2.2.9. Let L be a graded Lie algebra. Show that $U(\zeta(L))$ contains the symmetric algebra on $\zeta(L)$ as a central subalgebra.

Remark 2.2.10. Let G be a graded Lie ideal of a graded Lie algebra L . The short exact sequence of Lie algebras

$$0 \longrightarrow G \longrightarrow L \longrightarrow L/G \longrightarrow 0$$

induces a map of graded algebras $U(G) \rightarrow U(L)$ and an isomorphism of graded $U(G)$ -modules

$$U(L) \cong U(G) \otimes_k U(L/G)$$

where we view $U(L)$ as a graded $U(G)$ -module via restriction of scalars along $U(G) \rightarrow U(L)$. To see this, note that [Theorem 2.2.5](#) gives us a basis of $U(L)$, which we can decompose as a tensor product the bases for $U(G)$ and $U(L/G)$ provided by [Theorem 2.2.5](#).

Exercise 2.2.11. Show that $\text{Lie}(V)$ is isomorphic to the subalgebra of $T(V)^{\text{Lie}}$ generated by V .

Hint 1: Use the universal properties of the tensor algebra, the free Lie algebra, and the universal envelope to show that $U(\text{Lie}(V)) \cong T(V)$.

Hint 2: [Theorem 2.2.5](#) implies that $L \rightarrow U(L)^{\text{Lie}}$ is injective.

Theorem 2.2.12 (Assmus, Levin, Schoeller, Milnor–Moore, André, Sjöden). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. There is a graded Lie algebra L with $\dim_k(L^n) = \varepsilon_n(R)$, and there is a canonical isomorphism*

$$U(L) \cong \text{Ext}_R(k, k).$$

The proof of this result will be given later. The Lie algebra in question is known as the homotopy Lie algebra of R ; to construct it we will need to first go back to minimal models.

Before doing that, we note some applications of this and [Theorem 2.2.5](#):

Corollary 2.2.13. *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. The Yoneda algebra $\text{Ext}_R(k, k)$ is torsion if R is regular, and torsion free if R is not regular.*

Proof. Let $E = \text{Ext}_R(k, k)$. By torsion we mean that $\text{tors}(E) = E$, and by torsion free we mean that $\text{tors}(E) = 0$; see [Exercise 2.2.8](#).

By [Lemma 1.5.14](#), if R is regular, then $\varepsilon_n(R) = 0$ for all $n \geq 2$. Therefore, the Lie algebra L in [Theorem 2.2.12](#) is finite dimensional and concentrated in odd degrees, so by [Exercise 2.2.7](#), $E \cong U(L)$ is finite dimensional and $\text{Tors}(E) = E$ is torsion.

By [Lemma 1.5.14](#), if R is not regular then $\varepsilon_2(R) \neq 0$, so there exists a nonzero $x \in L^2$. From the Poincaré–Birkhoff–Witt Theorem [2.2.5](#) we learn that $E \cong U(L)$ is infinite dimensional, and from [Exercise 2.2.8](#) we learn that $\text{Tors}(E) = 0$. $\square$

Theorem 2.2.14 (Martsinkovsky). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring and M a finitely generated R -module of finite projective dimension. If there exists n and R -module maps such that the composition*

$$\text{Syz}_R^n(k) \longrightarrow M \longrightarrow k$$

is nonzero, then R is a regular local ring.

Remark 2.2.15. Suppose that M is a finitely generated R -module of finite projective dimension, and that there exists a surjection $\text{Syz}_R^n(k) \twoheadrightarrow M$. We claim that the hypothesis of [Theorem 2.2.14](#) are satisfied whenever there exists a surjection $\pi: \text{Syz}_R^n(k) \twoheadrightarrow M$. Indeed, note that we can compose π with the canonical surjection $M \twoheadrightarrow M/\mathfrak{m}M \cong k^{\mu(M)}$, giving us a surjection

$$\text{Syz}_R^n(k) \xrightarrow{\pi} M \twoheadrightarrow M/\mathfrak{m}M \cong k^{\mu(M)}.$$

Now we can compose with any of the many surjections onto k , to obtain a surjection $\text{Syz}_R^n(k) \longrightarrow k$ that factors through M .

In fact, we need only to assume that the map $\pi: \text{Syz}_R^n(k) \twoheadrightarrow M$ has a minimal generator m of M in its image; we can compose with a map $M \twoheadrightarrow k$ that sends m to 1 to obtain a nontrivial composition $\text{Syz}_R^n(k) \longrightarrow M \longrightarrow k$.

Proof of Theorem 2.2.14. Call the maps in the theorem $f: M \longrightarrow k$ and $g: \text{Syz}_R^n(k) \longrightarrow M$. Our assumption is that $fg \neq 0$.

As we will see in the next chapter, in [Theorem 3.2.2](#), if $M \longrightarrow N$ is a map of complexes of R -modules, then $\text{Ext}_R^*(N, k) \longrightarrow \text{Ext}_R^*(M, k)$ is a map of graded $E = \text{Ext}_R^*(k, k)$ -modules.

We will use this in two ways. For the first, let $P \xrightarrow{\sim} k$ be a projective resolution. Note that $P_{\geq n}$ is a projective resolution of $\text{Syz}_R^n(k)$ (shifted into degree n), and there is a surjective map of complexes $P \longrightarrow P_{\geq n}$ inducing the map of graded E -modules

$$\text{Ext}_R^*(\text{Syz}_R^n(k), k) \longrightarrow \text{Ext}_R^{*+n}(k, k).$$

Note that we can identify this with the inclusion $\text{Ext}_R^{\geq n}(k, k) \subseteq \text{Ext}_R^*(k, k)$.

The second way we apply [Theorem 3.2.2](#) to the map $g: \text{Syz}_R^n(k) \longrightarrow M$. This produces a diagram of graded E -modules:

$$\begin{array}{ccccc} \text{Ext}_R^*(M, k) & \longrightarrow & \text{Ext}_R^*(\text{Syz}_R^n(k), k) & \hookrightarrow & \text{Ext}_R^{*+n}(k, k) \\ \Psi & & \Psi & & \\ f & \longmapsto & fg & & \end{array}$$

Assume towards a contradiction that R is not regular. By [Corollary 2.2.13](#) $E = \text{Ext}_R^*(k, k)$ is torsion free, and so its submodule $\text{Ext}_R^*(\text{Syz}_R^n(k), k) \cong \text{Ext}_R^{\geq n}(k, k)$ is torsion free as well.

We assumed that the projective dimension of M is finite, so $\text{Ext}_R^{\gg 0}(M, k) = 0$. It follows that f is a torsion element, that is $x_{i_1} \cdots x_{i_n} \cdot f = 0$ for all $n \gg 0$ and $x_{i_j} \in E^{>0}$. Since $\text{Ext}_R^*(M, k) \longrightarrow \text{Ext}_R^*(\text{Syz}_R^n(k), k)$ is an E -module homomorphism, this implies that fg is a nonzero torsion element of $\text{Ext}_R^*(\text{Syz}_R^n(k), k) \cong \text{Ext}_R^{\geq n}(k, k)$.

This contradiction implies that R is regular. □

2.3 Quadratic models

A minimal model can have a complicated differential, though writing down concrete examples is also be quite challenging. The simplest kind of minimal model is one with a quadratic differential.

Definition 2.3.1. A **quadratic model** over k is a semifree extension $A = k[X]$ where each X_i is a finite set of variables and such that the differential satisfies

$$\partial(X) \subseteq kX^2.$$

Exercise 2.3.2. If $k[X]$ is a quadratic model, show that

$$\partial(X^n) \subseteq kX^{n+1}.$$

Exercise 2.3.3. Let R be a noetherian local ring and fix a minimal regular presentation $\widehat{R} \cong Q/I$ where $(Q, \mathfrak{m}, k)$ is a regular local ring and $I \subseteq \mathfrak{m}^2$. Consider a minimal model $Q[X]$ for R over Q . Show that the maximal ideal $J = \mathfrak{m} + (X)$ satisfies

$$\partial(J^n) \subseteq J^{n+1}.$$

Definition 2.3.4. A **map of quadratic models** $\varphi: k[X] \rightarrow k[Y]$ is a homomorphism of dg algebras such that $\varphi(X) \subseteq kY$. The category of quadratic models over k will be written $\mathbf{quadmod}_k$.

Note that quadratic models are simply minimal semifree extensions of k . Soon we will also see that quadratic models are closely related to graded Lie algebras; in fact, we will see that they are equivalent to graded Lie algebras. Given a minimal semifree extensions of k , there is a functorial construction to produce a quadratic model:

Remark 2.3.5. Let $k[X]$ be a semifree extension of k . Any element in $k[X]$ can be written uniquely as a sum

$$z = c_0 + \sum_{n=1}^{\infty} c_n z_{[n]}$$

where $z_{[n]} \in X^n$ and $c_n \in k$. Thus we can decompose the differential as

$$\partial = \sum_{n=1}^{\infty} \partial_{[n]}$$

where $\partial_{[n]}: k[X] \rightarrow kX_n$ and for each element $z \in k[X]$, only $\partial_{[n]}(z) \neq 0$ for all but finitely many values of n .

Definition 2.3.6. Let $k[X]$ be a semifree extension of k . The **quadratic part** of the differential is the map $\partial_{[2]}: k[X] \rightarrow kX^2$ defined in [Remark 2.3.5](#). The **quadratic part** of $k[X]$ is the quadratic model $k[X]^q$ given by

$$\mathbf{quad}(k[X]) := k[X \mid \partial_q(x) = \partial^{[2]}(x)].$$

Exercise 2.3.7. Let $k[X]$ be a minimal semifree extension of k . Show that the quadratic part of $k[X]$ is a well-defined quadratic model.

Remark 2.3.8 (Quadratic part of a minimal model). Let R be a noetherian local ring and fix a minimal regular presentation $\widehat{R} \cong Q/I$ where $(Q, \mathfrak{m}, k)$ is a regular local ring and $I \subseteq \mathfrak{m}^2$. Given a minimal model $A := Q[X]$ for R over Q , set X_0 to denote a minimal set of generators for $\mathfrak{m}$. This notation allows us to write the maximal ideal of $Q[X]$ as (X) , where we now view $X = X_{\geq 0}$, and we can rewrite [Theorem 1.3.15](#) as

$$\partial(X) \subseteq (X)^2.$$

Set $k[X] = k[X_{\geq 0}] := Q[X_{\geq 0}] \otimes_Q k$, and note that as a k -vector space,

$$k[X]_0 = kX_0 \cong \mathfrak{m}/\mathfrak{m}^2.$$

The associated graded ring of $Q[X]$ with respect to (X) is

$$\mathrm{gr}_{(X)}(Q[X]) := \bigoplus_{n \geq 0} (X)^n / (X)^{n+1} \cong \mathfrak{m}/\mathfrak{m}^2 \oplus kX \oplus kX^2 \oplus \cdots.$$

The associated graded ring has its own internal grading: the degree n piece is given by

$$\mathrm{gr}_{(X)}(Q[X])^n = (X)^n / (X)^{n+1} \cong kX^n.$$

We claim that the differential on A induces a well-defined differential on $\mathrm{gr}_{(X)}(A)$. By [Exercise 2.3.3](#) any element $x \in (X)^n$ has $\partial(x) \in (X)^{n+1}$, and thus $[x] \in \mathrm{gr}_{(X)}(A)^n$ is taken to an element $[\partial(x)] \in \mathrm{gr}_{(X)}(A)^{n+1}$. We leave well-definedness as an exercise.

Most importantly, this construction is equivalent to taking the quadratic part of the differential in $k[X_{\geq 0}]$ as above: for any variable $x \in X$, by [Exercise 2.3.3](#) its differential $\partial(x)$ gives us a well-defined element in $\mathrm{gr}_{(X)}(A)^1$, and note that

$$[\partial(x)] = [\partial_q(x)] \pmod{(X)^3}.$$

We will denote the resulting quadratic model by $\mathrm{quad}(Q[X])$.

Definition 2.3.9. A **map of quadratic models** $\varphi: k[X] \rightarrow k[Y]$ is a map of dg algebras such that $\varphi(X) \subseteq kY$. The category of quadratic models (over k) will be written **quadmod**.

Notation 2.3.10. Fix a field k . We write $(-)^{\vee} = \mathrm{Hom}_k(-, k)$ to indicate taking k -duals.

Remark 2.3.11. In the case when X is a set, we should interpret $(X)^{\vee}$ as the set of all functions from X to k . Equivalently, $(X)^{\vee}$ is the product of $|X|$ many copies of k .

2.4 The homotopy Lie algebra of a quadratic model

To import ideas from rational homotopy theory to local commutative algebra, Avramov defined a graded Lie algebra from any semifree quadratic dg algebra A , building on a construction due to Sullivan. This is called the homotopy Lie algebra of A , and we will explain how it is built in this section.

We start with some generalities about pairings which will make our definitions more convenient.

Exercise 2.4.1. Find an example of a vector space V , elements $x, y \in V$, and functionals $u, v \in \text{Hom}_k(V, k)$ such that $u(x)v(y) \neq u(y)v(x)$.

Remark 2.4.2. Let V be a graded vector space over k . There is a natural **pairing**

$$\langle -, - \rangle: V^\vee \otimes_k V \longrightarrow k$$

given by evaluation $\langle f, v \rangle_V = f(v)$. This is a **perfect pairing**, meaning that it identifies the two sides as each other's dual.

We will need a quadratic version of this pairing, for elements of

$$\text{Sym}^2(V) = k[V]^2 = \text{span}\{xy : x, y \in V\}.$$

If you try to write down a pairing like this

$$\langle -, - \rangle: (V^\vee \otimes V^\vee) \otimes_k k[V]^2 \longrightarrow k$$

by matching up elements (and following the Koszul sign rule) like this

$$\langle u \otimes v, xy \rangle = (-1)^{|v||x|} u(x)v(y)$$

then you will have problems (no matter how clever with signs you are). For example, if $xy = yx$ but $u(x)v(y) \neq u(y)v(x)$, which we saw in [Exercise 2.4.1](#) can happen, then this pairing certainly is not well-defined.

We can fix this problem by working with symmetric elements of $V^\vee \otimes V^\vee$. Write

$$D_2(V^\vee) = \left\{ \sum u_i \otimes v_i \in V^\vee \otimes V^\vee \mid \sum u_i \otimes v_i = \sum (-1)^{|u_i||v_i|} v_i \otimes u_i \right\}$$

for the subspace of symmetric elements in $V^\vee \otimes V^\vee$. The pairing above is well-defined when we restrict it to $D_2(V^\vee)$: we get a pairing

$$\langle -, - \rangle: D_2(V^\vee) \otimes_k k[V]^2 \longrightarrow k$$

given by

$$\left\langle \sum u_i \otimes v_i, xy \right\rangle = \sum (-1)^{|v_i||x|} u_i(x)v_i(y).$$

There is one more complication: we will need to consider the shifts of these graded vector spaces, which will introduce additional signs. Given a graded vector space W , we write

$$D_2^{\text{anti}}(W) = \left\{ \sum u_i \otimes v_i \in W \otimes W \mid \sum u_i \otimes v_i = - \sum (-1)^{|u_i||v_i|} v_i \otimes u_i \right\}$$

for the space of antisymmetric tensors in $W \otimes W$. Finally, with this we write down one last pairing

$$\langle -, - \rangle : D_2^{\text{anti}}((\Sigma V)^\vee) \otimes_k k[V]^2 \longrightarrow k$$

given by

$$\langle \sum u'_i \otimes v'_i, xy \rangle = \sum (-1)^{|u_i|} u_i(x) v_i(y)$$

for $u_i, v_i \in V^\vee$ and $x, y \in V$. Here we have written $u' \in (\Sigma V)^\vee$ for the degree $|u| - 1$ element corresponding to $u \in V^\vee$.

Exercise 2.4.3. Let V be a graded vector space. Prove that $D_2(V^\vee)$ is isomorphic to the second divided power of $V^\vee$, and that it is spanned by elements of the form

$$u \otimes v + (-1)^{|u||v|} v \otimes u \quad \text{and} \quad w \otimes w \text{ when } |w| \text{ is even.}$$

Exercise 2.4.4. Prove that the pairing

$$\langle -, - \rangle : D_2(V^\vee) \otimes_k k[V]^2 \longrightarrow k$$

given by

$$\langle \sum u_i \otimes v_i, xy \rangle = \sum (-1)^{|v_i||x|} u_i(x) v_i(y)$$

is well-defined and perfect.

There is an adjunction formula relating the pairing between $V^\vee$ and V with the pairing between $D_2(V^\vee)$ and $k[V]^2$, which you will prove in the following exercise:

Exercise 2.4.5. Let $u, v \in V^\vee$ and let $x, y \in V$. The element v induces a unique derivation

$$\frac{d}{dv} : k[V]^i \rightarrow k[V]^{i-1}$$

such that $\frac{d}{dv}|_V = v$. Prove that

$$\langle u, \frac{d}{dv}(xy) \rangle = \langle u \otimes v + (-1)^{|u||v|} v \otimes u, xy \rangle.$$

Remark 2.4.6. We can think of this derivation $\frac{d}{dv}$ as partial differentiation with respect to v .

This idea will come back and take on an important role later.

Exercise 2.4.7. Prove that the map

$$D_2^{\text{anti}}((\Sigma V)^\vee) \longrightarrow D_2(V^\vee) \quad \sum u'_i \otimes v'_i \mapsto \sum (-1)^{|u_i|(|v_i|+1)} u_i \otimes v_i$$

is an isomorphism. Deduce from this that the pairing

$$\langle -, - \rangle : D_2^{\text{anti}}((\Sigma V)^\vee) \otimes_k k[V]^2 \longrightarrow k$$

defined in [Remark 2.4.2](#) is well-defined and perfect. Also deduce from this, along with [Exercise 2.4.5](#), that we have the following adjunction formula:

$$\langle u, \frac{d}{dv}(xy) \rangle = (-1)^{|v|} \langle u' \otimes v' - (-1)^{|u'||v'|} v' \otimes u', xy \rangle$$

for $u, v \in V^\vee$ and $x, y \in V$.

Definition 2.4.8. Let $A = k[X]$ be a quadratic semifree dg algebra. The **homotopy Lie algebra** of A is the graded vector space

$$\pi^*(A) = (\Sigma kX)^\vee,$$

and we equip this with the bracket and reduced square operation given by

$$[u, v](x) := \langle u \otimes v - (-1)^{|u||v|} v \otimes u, \partial(x) \rangle \quad \text{and} \quad w^{[2]}(x) := \langle w \otimes w, \partial(x) \rangle$$

for all $u, v, w \in \pi^*(A)$ with $|w|$ odd, and all $x \in X$. Here we have used the natural pairing $\langle u, x \rangle = u(x)$ on $\pi^*(A)$, and the pairing from [Remark 2.4.2](#) on $D_2^{\text{anti}}(\pi^*(A)) = D_2^{\text{anti}}((\Sigma kX)^\vee)$.

Note the similarity with [Example 2.1.6](#) in this definition: the Lie bracket is defined in terms of the (tensor) commutator, and the reduced square is defined in terms of the (tensor) square; then we extract the information we want from A by pairing with the quadratic differential.

Theorem 2.4.9 (Quillen, Sullivan, Avramov, Chevalley–Eilenberg). *If A is a quadratic model then $\pi^*(A)$ is a graded Lie algebra with the structure defined in [Definition 2.4.8](#). This assignment yields a contravariant equivalence of categories*

$$\mathbf{quadmod}_k \xrightarrow{\pi^*} \mathbf{Lie}_k.$$

Proof. Let $L = \pi^*(A)$. Let us check that L really is a graded Lie algebra by checking the axioms from [Definition 2.1.1](#). Note that

$$(u + v) \otimes (u + v) = u \otimes u + (u \otimes v + v \otimes u) + v \otimes v$$

for any odd elements $u, v \in \pi^*(A)$. The compatibility axiom between $[-, -]$ and $(-)^{[2]}$ follows by applying $\langle -, \partial(x) \rangle$ to this relation.

Similarly, the antisymmetry axioms follow from the fact that $u \otimes v - (-1)^{|u||v|} v \otimes u$ and $w \otimes w$ already satisfy the same antisymmetry properties, and then we apply $\langle -, \partial(x) \rangle$.

The hard part is to verify the Jacobi identities. We claim that in $A = k[X]$, the equation $\partial^2 = 0$ is equivalent to the three Jacobi identities on L .

To do this we will first need some observations about antisymmetrisation. For elements $u, v, w \in L$ we will write

$$\begin{aligned} (u \otimes v \otimes w)^{\text{anti}} &= u \otimes v \otimes w + (-1)^{|w|(|u|+|v|)} w \otimes u \otimes v + (-1)^{|u|(|v|+|w|)} v \otimes w \otimes u \\ &\quad - (-1)^{|u||v|} v \otimes u \otimes w - (-1)^{|v||w|} u \otimes w \otimes v - (-1)^{|u||v|+|u||v|+|v||w|} w \otimes v \otimes u. \end{aligned}$$

And we will write

$$(u \otimes v)^{\text{anti}} = [u, v]^\otimes = u \otimes v - (-1)^{|u||v|} v \otimes u.$$

By expanding everything out we obtain the identity

$$(u \otimes v \otimes w)^{\text{anti}} = (u \otimes v)^{\text{anti}} \otimes w + (-1)^{|w|(|u|+|v|)} (w \otimes u)^{\text{anti}} \otimes v + (-1)^{|u|(|v|+|w|)} (v \otimes w)^{\text{anti}} \otimes u. \quad (2.4.1)$$

There is a cubic version of the pairing from [Exercise 2.4.7](#):

$$\langle -, - \rangle : D_3^{\text{anti}}(L) \otimes_k k[X]^3 \longrightarrow k, \quad \langle \sum u_i \otimes v_i \otimes w_i, xyz \rangle = \sum (-1)^{|y|} u_i(x) v_i(y) w_i(z)$$

One needs to check that this is well-defined and perfect, just as for the quadratic version (the reader is no doubt suspecting that there is an n -tic version for every n , and the reader is correct).

Note that this satisfies

$$\langle \sum u_{ij} \otimes v_{ij} \otimes w_j, xyz \rangle = (-1)^{|x|+|y|} \sum_j \langle \sum_i u_{ij} \otimes v_{ij}, xy \rangle \langle w_j, z \rangle$$

whenever the sums $\sum_{i,j} u_{ij} \otimes v_{ij} \otimes w_j$ and $\sum_i u_{ij} \otimes v_{ij}$ are antisymmetric for all j .

In particular, if we have an element of the form $\sum_j (u_j \otimes v_j)^{\text{anti}} \otimes w_j$ which is in $D_3^{\text{anti}}(L)$, then using the Liebniz rule and the definition of the bracket on L , we have

$$\begin{aligned} & \langle \sum (u_j \otimes v_j)^{\text{anti}} \otimes w_j, \partial(xy) \rangle \\ &= \langle \sum (u_j \otimes v_j)^{\text{anti}} \otimes w_j, \partial(x)y \rangle + (-1)^{|x|} \langle \sum (u_j \otimes v_j)^{\text{anti}} \otimes w_j, x\partial(y) \rangle \\ &= \langle \sum (u_j \otimes v_j)^{\text{anti}} \otimes w_j, \partial(x)y \rangle + (-1)^{|x|+|x|(|y|-1)} \langle \sum (u_j \otimes v_j)^{\text{anti}} \otimes w_j, \partial(y)x \rangle \\ &= (-1)^{|x|-1} \sum \langle (u_j \otimes v_j)^{\text{anti}}, \partial(x) \rangle \langle w_j, y \rangle + (-1)^{|x|+|x|(|y|-1)+|y|-1} \sum \langle (u_j \otimes v_j)^{\text{anti}}, \partial(y) \rangle \langle w_j, x \rangle \\ &= (-1)^{|x|-1} \sum \langle [u_j, v_j], x \rangle \langle w_j, y \rangle + (-1)^{|x||y|+|y|-1} \sum \langle [u_j, v_j], y \rangle \langle w_j, x \rangle \\ &= (-1)^{|x|-1} \left[\sum \langle [u_j, v_j], x \rangle \langle w_j, y \rangle - (-1)^{(|y|+1)(|x|+1)} \sum \langle [u_j, v_j], y \rangle \langle w_j, x \rangle \right] \\ &= -\langle \sum [u_j, v_j] \otimes w_j - (-1)^{(|y|+1)(|x|+1)} w_j \otimes [u_j, v_j], xy \rangle \\ &= -\langle \sum ([u_j, v_j] \otimes w_j)^{\text{anti}}, xy \rangle \end{aligned}$$

Replacing xy with $\partial(x)$ in this equation yields

$$\langle \sum (u_i \otimes v_i)^{\text{anti}} \otimes w_j, \partial^2(x) \rangle = -\langle \sum ([u_i, v_i] \otimes w_j)^{\text{anti}}, \partial(x) \rangle = -\langle \sum [[u_i, v_i], w_j], x \rangle.$$

However, $\partial^2(x)$ is always zero. Since x is arbitrary, the vanishing of this pairing implies that $\sum [[u_i, v_i], w_j] = 0$. We have proved the following:

$$\text{If } \sum_{i,j} (u_i \otimes v_i)^{\text{anti}} \otimes w_j \text{ is antisymmetric, then } \sum_{i,j} [[u_i, v_i], w_j] = 0 \text{ in } L.$$

Applying this to Equation [2.4.1](#) immediately yields the classical Jacobi identity

$$[[u, v], w] + (-1)^{|u|(|v|+|w|)} [[v, w], u] + (-1)^{|v||w|+1} [[u, w], v] = 0.$$

Applying the same argument to $(u \otimes u)^{\text{anti}} \otimes u$ yields the Jacobi identity

$$[[u, u], u] = 0$$

as long as u is odd.

A similar argument applied to the antisymmetric element

$$u \otimes u \otimes v - (-1)^{|v|} u \otimes v \otimes u + v \otimes u \otimes u$$

shows that

$$[u^{[2]}, v] - [u, [u, v]] = 0,$$

when u is odd. This completes the proof that the Jacobi identities hold.

We have now shown that $\pi^*(A)$ is a well-defined graded Lie algebra. The rest of the proof, that this yields an equivalence of categories, will be completed as an exercise. $\square$

The homotopy Lie algebra construction has an inverse, taking a graded Lie algebra L to a quadratic semifree dg algebra $\mathcal{C}(L)$. This was originally introduced by Chevalley and Eilenberg in order to define Lie algebra cohomology (for ungraded Lie algebras). It was extended by Quillen and to graded Lie algebras of characteristic zero. Avramov adapted this construction to work over a field of any characteristic [Avr84, theorem 4.2], ultimately arriving at an equivalence of categories between $\mathbf{quadmod}_k$ and $\mathbf{Lie}_k$.

Definition 2.4.10. Let L be a finite type graded Lie algebra over k . The Chevalley–Eilenberg dg algebra of L is the free strictly graded commutative k -algebra

$$\mathcal{C}(L) = k[(\Sigma L)^\vee].$$

By the Leibniz rule, the differential is uniquely determined by its values on the generators of $(\Sigma L)^\vee$, and thus it is completely determined by the values of $\partial(x)$ for $x \in L$. We want this to be quadratic, i.e. $\partial(x) \in \mathrm{Sym}_k^2((\Sigma L)^\vee) = k[(\Sigma L)^\vee]^2$. By Exercise 2.4.7 we can define $\partial(x)$ by specifying the pairing $\langle -, \partial(x) \rangle$, and for this we can reverse Definition 2.4.8:

$$\langle u \otimes v - (-1)^{|u||v|} v \otimes u, \partial(x) \rangle := [u, v](x) \quad \text{and} \quad \langle w \otimes w, \partial(x) \rangle := w^{[2]}(x)$$

for all $u, v, w \in L$ with $|w|$ odd, and all $x \in (\Sigma L)^\vee$.

Exercise 2.4.11.

- 1) Verify that $\mathcal{C}(L)$ is a quadratic model for any graded Lie algebra L (the key point is to show that $\partial^2 = 0$, and the proof of this is the same as for Theorem 2.4.9 in reverse).
- 2) Prove that for any graded Lie algebra L and any quadratic model A we have $\mathcal{C}(\pi^*(A)) \cong A$ and $\pi^*(\mathcal{C}(L)) \cong L$.
- 3) Extend the definition of the homotopy Lie algebra and the Chevalley–Eilenberg dg algebra to morphisms. That is, if $A \rightarrow A'$ is a map of quadratic models, and $L \rightarrow L'$ is a map of graded Lie algebras, then define maps

$$\pi^*(A') \rightarrow \pi^*(A) \quad \text{and} \quad \mathcal{C}(L') \rightarrow \mathcal{C}(L).$$

Convince yourself that the make π^* and $\mathcal{C}$ into contravariant functors.

- 4) Deduce from all of this that there is a contravariant equivalence of categories

$$\mathbf{quadmod}_k \xrightleftharpoons[\mathcal{C}]{\pi^*} \mathbf{Lie}_k,$$

as stated in Theorem 2.4.9.

2.5 The homotopy Lie algebra of a local ring

Definition 2.5.1. Let R be a noetherian local ring. Fix a minimal regular presentation $\widehat{R} \cong Q/I$ and a minimal model $A = Q[X]$ for R over Q . The **homotopy Lie algebra** of R , denoted $\pi^*(R)$, is the graded Lie algebra

$$\pi^*(R) := \pi^*(\text{quad}(A)).$$

Remark 2.5.2. By [Theorem 1.3.29](#), any two minimal models for R over Q are isomorphic. Thus given any two minimal models $Q[X]$ and $Q[Y]$ for R , by [Remark 2.5.3](#) we have

$$(kY_{n-1})^\vee \cong (kX_{n-1})^\vee = \pi^n(R).$$

Thus the vector space $\pi^n(R)$ is independent of our choice of minimal model for R over Q .

In fact, more is true: using the functoriality of quadratic models, one can show that the bracket is independent of the choice of minimal model. Moreover, the construction of $\pi^*(R)$ is also independent on the choice of minimal regular presentation, a fact which follows from [Theorem 2.2.12](#).

Remark 2.5.3. Fix a minimal model $Q[X]$ for R over $(Q, \mathfrak{m}, k)$. Let us examine the graded k -vector space $\pi^*(R)$. In degree 1, we have

$$\pi^1(R) = (kX_0)^\vee \cong \mathfrak{m}/\mathfrak{m}^2,$$

which is a vector space of dimension $\text{embdim}(R) = \dim(Q)$. For each $n \geq 2$, we have the k -vector space

$$\pi^n(R) = (kX_{n-1})^\vee$$

of dimension $|X_{n-1}|$. By [Theorem 1.5.19](#),

$$\dim_k(\pi^n(R)) = |X_{n-1}| = \varepsilon_n(R).$$

For each $x \in X = X_{\geq 0}$ we set $x^\vee$ to be the k -linear map $kX_{|x|} \rightarrow k$ defined on basis elements by

$$x^\vee(y) = \begin{cases} 1 & \text{if } y = x \\ 0 & \text{otherwise.} \end{cases}$$

Note that $\{x^\vee \mid x \in X_{n-1}\}$ forms a basis for $\pi^n(R)$.

For example, if $\{x_1, \dots, x_e\}$ is a minimal generating set for $\mathfrak{m}$, or equivalently their images in $\mathfrak{m}/\mathfrak{m}^2$ form a basis, then we take $\{x_1^\vee, \dots, x_e^\vee\}$ as a basis for $\pi^1(R)$, where $x_i^\vee: \mathfrak{m}/\mathfrak{m}^2 \rightarrow k$ is the functional defined by

$$x_i^\vee(a_1x_1 + \dots + a_ex_e) = a_i.$$

Similarly, suppose that $\{f_1, \dots, f_n\}$ is a minimal generating set for I , so that $Q[X]$ is constructed with $X_1 = \{e_1, \dots, e_n\}$ and $\partial(e_i) = f_i$. Note that X_1 is in bijection with $\{f_1, \dots, f_n\}$. It is sometimes helpful to then identify kX_1 with $I/\mathfrak{m}I$ and the basis elements of $\pi^2(R)$ with our chosen minimal generators for I : we write $\{f_1^\vee, \dots, f_n^\vee\}$ for the basis of $\pi^2(R)$ where $f_i^\vee: kX_1 \rightarrow k$ is given by

$$f_i^\vee(a_1f_1 + \dots + a_nf_n).$$

Remark 2.5.4 (Calculating brackets). Fix a minimal regular presentation $\widehat{R} \cong Q/I$ and a minimal model $Q[X]$ for R over Q . Let us explicitly calculate the brackets on $\pi^*(R)$ in terms of the quadratic part of the differential on $k[X]$.

We start by endowing X with a total order, as follows: we order the variables in each X_n , and set $x < y$ whenever $|x| < |y|$. Fix $x, y, z \in X$, and consider the elements $y^\vee, z^\vee \in \pi^*(R)$. We include the shift in this notation, so $|y^\vee| = |y| + 1$ and $|z^\vee| = |z| + 1$.

Consider the quadratic part of the differential of x in $k[X]$, which can be written uniquely as

$$\partial_q(x) = \sum_{\substack{u \leq v \in X \\ |u|+|v|+1=|x|}} c_{x,u,v} uv$$

for some coefficients $c_{x,y,z} \in k$. By definition,

$$\begin{aligned} [z^\vee, y^\vee](x) &= \langle z^\vee y^\vee - (-1)^{|y^\vee| \cdot |z^\vee|} y^\vee z^\vee, \sum_{\substack{u < v \in X \\ |u|+|v|+1=|x|}} c_{x,u,v} uv \rangle \\ &= \sum_{\substack{u \leq v \in X \\ |u|+|v|+1=|x|}} (-1)^{|z|} c_{x,u,v} z^\vee(u) y^\vee(v) - \sum_{\substack{u \leq v \in X \\ |u|+|v|+1=|x|}} (-1)^{|y|+|y^\vee| \cdot |z^\vee|} y^\vee(u) z^\vee(v) \end{aligned}$$

Note that $y^\vee(u)$ is nonzero if and only if $y = u$, in which case $y^\vee(u) = 1$, and similarly for $y^\vee(v)$, $z^\vee(u)$, and $z^\vee(v)$. Therefore, whenever $y < z$ we have

$$[z^\vee, y^\vee](x) = (-1)^{|y|+|y^\vee| \cdot |z^\vee|+1} c_{x,y,z}$$

and so

$$[y^\vee, z^\vee](x) = -(-1)^{|y^\vee| \cdot |z^\vee|} [z^\vee, y^\vee](x) = (-1)^{|y|} c_{x,y,z}.$$

Similarly, we can compute the reduced square: when $|y^\vee| = |y| + 1$ is odd,

$$\begin{aligned} (y^\vee)^{[2]}(x) &= \langle y \otimes y, \sum_{\substack{u < v \in X \\ |y|+|z|+1=|x|}} c_{x,u,v} uv \rangle \\ &= (-1)^{|y|} c_{x,y,y} \\ &= c_{x,y,y}. \end{aligned}$$

Thus

$$[y^\vee, z^\vee] = (-1)^{|y|} \sum_{x \in X} c_{x,y,z} x^\vee \quad \text{and} \quad (y^\vee)^{[2]} = \sum_{x \in X} c_{x,y,y} x^\vee \text{ if } |y| \text{ is odd.}$$

We note that this is induced by the dual of the quadratic part of ∂ . More precisely, ∂_q gives us a map $\partial_q: kX \rightarrow kX^2$, and we claim that its dual $(\partial_q)^\vee: (kX^2)^\vee \rightarrow (kX)^\vee$ determines the bracket. Given $y, z \in X$, we consider $yz \in X^2$, and $(yz)^\vee \in (kX)^\vee$. Note that yz lives in degree $|y| + |z|$, so $(yz)^\vee \in (kX_{|y|+|z|+1})^\vee$, which indeed corresponds to an element of

$$\pi^{|y|+|z|+2}(R) = \pi^{|y^\vee|+|z^\vee|}(R).$$

One can now see that up to signs, the map $(yz)^\vee$ matches $[y^\vee, z^\vee]$ as above.

Remark 2.5.5 (Sjöden, 1976 [Sjö76]). Let us examine the recipe for taking brackets of elements in $\pi^1(R)$ a bit closer. Once we fix a minimal generating set $X_0 = \{x_1, \dots, x_e\}$ for the maximal ideal $\mathfrak{m}$ of Q , an element in kX_0^2 simply corresponds to an element in $\mathfrak{m}^2$ of $\mathfrak{m}$ -adic order exactly 2. To determine $[x_i^\vee, x_j^\vee]$, we need to look at the differential in $Q[X]$ of the variables in X_1 . But note that $X = \{y_1, \dots, y_n\}$ with each $\partial(y_\ell) = f_\ell$ a minimal generator of the defining ideal I of R . Since we have taken a minimal regular presentation, $f_\ell \in \mathfrak{m}^2$. We now write each f_ℓ as

$$f_\ell = \sum_{i \leq j} c_{\ell, i, j} x_i x_j + w,$$

for some $w \in \mathfrak{m}^3$, and whenever $j < i$, we get

$$[x_i^\vee, x_j^\vee] = \sum_{\ell=1}^n c_{\ell, i, j} f_\ell^\vee \quad \text{and} \quad (x_i^\vee)^{[2]} = \sum_{\ell=1}^n c_{\ell, i, i} f_\ell^\vee$$

While this recipe for $[\pi^1(R), \pi^1(R)]$ was easy to extract from our definition of $\pi^*(R)$, this recipe was first found by Sjöden using a very different definition of the homotopy Lie algebra which makes this a lot less obvious.

Remark 2.5.6. Since X_1 corresponds to a chosen minimal generating set for the defining ideal I of R over Q , with each variable corresponding to one of the chosen minimal generators, we will sometimes identify the corresponding element of $\pi^2(R)$ with the minimal generator $f \in I$, by writing $f^\vee$. Note, however, that this only makes sense once we choose a complete minimal generating set for I , giving us a basis for kX_1 .

Let us see this in practice.

Example 2.5.7. Let $Q = k[[t]]$ and $R = Q/(x^2)$. Since $f = x^2$ is a regular element in Q , the Koszul complex on x^2 is a minimal model for R over Q :

$$Q[X] = 0 \longrightarrow Q \xrightarrow{x^2} Q \longrightarrow 0.$$

As a graded vector space, the homotopy Lie algebra $\pi^*(R)$ looks like

$$\begin{array}{ccc} k \cdot x^\vee & & k \cdot f^\vee \\ 1 & & 2 \end{array}$$

By degree reasons, the only potentially nontrivial brackets are those involving two elements of $\pi^1(R)$. Thus it suffices to compute $(x^\vee)^{[2]}$. To do that, we note that

$$f = 1 \cdot x^2,$$

and thus $(x^\vee)^{[2]} = f^\vee$. Whenever $\text{char}(k) \neq 2$, we have $[x^\vee, x^\vee] = 2f^\vee$.

Example 2.5.8. Let $Q = k[[x, y]]$ and $R = Q/(f)$ with $f = xy$. The homotopy Lie algebra $\pi^*(R)$ looks like

$$\begin{array}{ccc} k \cdot x^\vee \oplus k \cdot y^\vee & & k \cdot f^\vee \\ 1 & & 2 \end{array}$$

We have

$$f = 1 \cdot xy,$$

and thus $[x^\vee, y^\vee] = [y^\vee, x^\vee] = f^\vee$.

Example 2.5.9. Let $Q = k[[x]]$ and $R = Q/(f)$ with $f = x^3$. As in [Example 2.5.7](#), the homotopy Lie algebra $\pi^*(R)$ looks like

$$\begin{array}{ccc} k \cdot x^\vee & & k \cdot f^\vee \\ 1 & & 2 \end{array}$$

For degree reasons, the only potentially nontrivial brackets are those involving two elements of $\pi^1(R)$. However, note that $f = x^3 \in \mathfrak{m}^3$, so

$$f = 0 \cdot x^2 + x^3,$$

and thus $[x^\vee, x^\vee] = 0$. We conclude that $\pi^*(R)$ has trivial bracket.

One can compute information about $\pi^*(R)$ using Macaulay2.

Macaulay2. In Macaulay2, we can extract information about $\pi^*(R)$ by computing a minimal model via the `DGAlgebras` package, or we can use the `HomotopyLieAlgebra` package. The method `bracketMatrix` in the `HomotopyLieAlgebra` package allows us to calculate $[\pi^a(R), \pi^b(R)]$ on bases via `bracketMatrix(A, a, b)` with $A = Q[X]$ a minimal model for R . The entries in the matrix are elements of QX_{a+b+1} .

```
i1 : needsPackage "HomotopyLieAlgebra";
```

```
i2 : Q = QQ[x,y];
```

```
i3 : I = ideal"xy";
```

```
i4 : A = minimalModel I
```

```
o4 = {Ring => Q
      Underlying algebra => Q[T
      1,1
      Differential => {x*y}
```

```
o4 : DGAlgebra
```

We are now ready to ask for bracket matrices:

```
i5 : bracketMatrix(A, 1, 1)
```

```
o5 = | 0      T_(1,1) |
      | T_(1,1) 0      |
```

```
o5 : Matrix (Q[T_ ] ) <-- (Q[T_ ] )
      2      2
      1,1      1,1
```

Note that here $T_{(1,1)}$ is the variable we adjoined in degree 1 with $\partial(T_{(1,1)}) = xy$. So the matrix tells us

$$[x^\vee, y^\vee] = T_{(1,1)},$$

which matches our calculation in [Example 2.5.8](#), where we used the notation $f^\vee$ for $T_{1,1}$.

The fact that all of the examples above lead to finite dimensional graded Lie algebras is simply a consequence of all rings being complete intersections. In fact, using only what we know about minimal models we can already prove the following theorem:

Theorem 2.5.10. *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring.*

- 1) *The ring R is a regular ring if and only if $\pi^*(R) = \pi^1(R)$.*
- 2) *The ring R is a complete intersection if and only if $\pi^*(R) = \pi^1(R) \oplus \pi^2(R)$.*
- 3) *If R is not a complete intersection, then $\pi^*(R)$ is an infinite dimensional graded Lie algebra, and $\pi^n(R)$ is nonzero for all $n \geq 1$.*

Proof. Fix a minimal regular presentation $\widehat{R} \cong Q/I$ and a minimal model $Q[X]$ for R over Q . For $n \geq 1$, $\pi^n(R) = 0$ if and only if $X_{n-1} = \emptyset$, or equivalently $\varepsilon_n(R) = 0$, as we noted in [Remark 2.5.3](#).

The ring R is regular if and only if $\widehat{R} = Q$, and thus 1) is immediate. Moreover, 2) is just a restating of [Lemma 1.3.31](#), while 3) is a consequence of Halperin's Rigidity Theorem, which says that if $\varepsilon_n(R) = 0$ for some n , then R must be a complete intersection (see [Theorem 1.5.23](#)). $\square$

Let us try some examples that are *not* complete intersections.

Example 2.5.11. In [Example 1.3.11](#), we computed a partial minimal model $Q[X_{\leq 3}]$ for $k[x, y, z]/(xy, xz, yz)$. We saw that

$$X_1 = \{T_{1,1}, T_{1,2}, T_{1,3}\} \quad X_2 = \{T_{2,1}, T_{2,2}\} \quad X_3 = \{T_{3,1}, T_{3,2}, T_{3,3}\}$$

with

$$\begin{aligned} \partial(T_{1,1}) &= xy & \partial(T_{1,2}) &= xz & \partial(T_{1,3}) &= yz \\ \partial(T_{2,1}) &= zT_1 - yT_2 & \text{and} & & \partial(T_{2,2}) &= zT_1 - xT_3 \end{aligned}$$

$$\partial(T_{3,1}) = T_{1,1}T_{1,2} - xT_{2,1} \quad \partial(T_{3,2}) = T_{1,2}T_{1,3} + zT_{2,1} - zT_{2,2} \quad \partial(T_{3,2}) = T_{1,1}T_{1,3} - yT_{2,2}.$$

From this, we immediately obtain that

$$\pi^1(R) \cong k^3 \cong kx^\vee \oplus ky^\vee \oplus kz^\vee \quad \pi^2(R) \cong k^3 \cong kT_{1,1}^\vee \oplus kT_{1,2}^\vee \oplus kT_{1,3}^\vee$$

$$\pi^3(R) \cong k^2 \cong kT_{2,1}^\vee \oplus kT_{2,1}^\vee \quad \pi^4(R) \cong k^3 \cong kT_{3,1}^\vee \oplus kT_{3,2}^\vee \oplus kT_{3,3}^\vee.$$

We can also read off the brackets up to $[\pi^2(R), \pi^2(R)] \subseteq \pi^4(R)$. For example,

$$[T_{1,1}^\vee, T_{1,2}^\vee] = (-1)^{|T_{1,2}|}(T_{3,1}) = -T_{3,1} \quad \text{and} \quad [x^\vee, T_{2,1}^\vee] = (-1)^{|T_{2,1}|}(-T_{1,3}) = T_{1,3}.$$

Exercise 2.5.12. Let $Q = k[[x, y]]$ and $R = Q/(x^2, xy)$.

- 1) Compute $\pi^1(R)$, $\pi^2(R)$, and $\pi^3(R)$.
- 2) Compute $[\pi^1(R), \pi^1(R)]$ and $[\pi^1(R), \pi^2(R)]$.
- 3) Check your work with Macaulay2.
- 4) Let $Q[X]$ be a minimal model for R over Q , where we adjoined $u, v \in X_1$ so that $\partial(u) = x^2$ and $\partial(v) = xy$. Use Macaulay2 to find $[u^\vee, v^\vee] \in \pi^4(R)$.

Chapter 3

Differential graded modules

3.1 Differential graded module resolutions

Fix a commutative ring Q , and throughout this section let A denote a nonnegatively graded dg Q -algebra, that is, $A_i = 0$ for all $i < 0$. We will be particularly interested in the situation when $A = Q[X]$ is a semifree extension of Q or when $A = R = Q/I$ is a quotient ring of Q . Recall from [Definition 3.1.1](#) that a dg A -module will mean a *left* dg A -module, unless specified otherwise. In fact, here is the definition again:

Definition 3.1.1. A (left) **dg module** M over a dg R -algebra A is a complex of R -modules with the structure of a graded module over the underlying ring A that satisfies the Leibniz rule, that is, for all $a \in A$ and $m \in M$,

$$\partial_M(am) = \partial_A(a)m + (-1)^{|a|}a\partial_M(m).$$

Remark 3.1.2. Equivalently, note that a dg A -module M is a complex equipped with a map of complexes

$$\begin{aligned} \mu_M: A \otimes_R M &\longrightarrow M \\ a \otimes m &\longmapsto am \end{aligned}$$

that is associative and unital, so that the following diagrams commute:

$$\begin{array}{ccc} A \otimes_Q A \otimes_Q M & \xrightarrow{\mu_A \otimes \text{id}} & A \otimes_Q M \\ \downarrow \text{id} \otimes \mu_M & & \downarrow \mu_M \\ A \otimes_Q M & \xrightarrow{\mu_M} & M \end{array} \qquad \begin{array}{ccc} A \otimes_Q M & \xrightarrow{\mu_M} & M \\ \eta_A \otimes \text{id} \uparrow & \nearrow \cong & \\ Q \otimes_Q M & & \text{natural} \end{array}$$

Remark 3.1.3. A **right dg A-module** M is a graded right A -module that satisfies

$$\partial_M(ma) = \partial_M(m)a + (-1)^{|m|}m\partial_A(a).$$

Exercise 3.1.4. Show that if M is a dg A -module, then $H(M)$ is a graded $H(A)$ -module.

Notation 3.1.5. Let A be a dg Q -algebra and M be a dg A -module. We let $A^\natural$ and $M^\natural$ denote the underlying graded Q -modules.

Remark 3.1.6. Note that $A^\natural$ is a graded Q -algebra and $M^\natural$ is a graded $A^\natural$ -module.

Remark 3.1.7 (dg E -modules). Consider a Koszul complex $E = \text{Kos}^R(f_1, \dots, f_n)$ and a complex of R -modules F . A dg E -module structure on F amounts to R -linear maps $\sigma_1, \dots, \sigma_n \in \text{Hom}_R(F, F)$ of degree 1 such that

$$\sigma_i \sigma_j = -\sigma_j \sigma_i, \quad \sigma_i^2 = 0, \quad \text{and} \quad \partial \sigma_i + \sigma_i \partial = f_i \cdot \text{id}_F.$$

If $E = Q[e_1, \dots, e_n \mid \partial(e_i) = f_i]$, then the action of e_i on F corresponds to applying σ_i .

Let us see a concrete example.

Example 3.1.8. Let k be a field and consider $Q = k[[x, y]]$ and $f = x^3 + y^4$. Let $E = Q[e \mid \partial(e) = f]$ be the Koszul complex on f , and let F be the Q -complex

$$0 \longrightarrow Q \xrightarrow{\begin{bmatrix} -y \\ x \end{bmatrix}} Q^2 \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} Q \longrightarrow 0.$$

In order to give an E -module structure on F , we need to give nullhomotopy for multiplication by f on F . One can check that

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q^2 & \xrightarrow{\begin{bmatrix} -y \\ x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} & Q \longrightarrow 0 \\ & & & \swarrow & \downarrow \begin{bmatrix} x^2 \\ y^3 \end{bmatrix} & \swarrow & \\ 0 & \longrightarrow & Q^2 & \xrightarrow{\begin{bmatrix} -y \\ x \end{bmatrix}} & Q^3 & \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} & Q \longrightarrow 0 \end{array}$$

is indeed a nullhomotopy for multiplication by f .

Definition 3.1.9. Let A be a dg algebra and M and N be dg A -modules. The **tensor product** of M and N over A , written $M \otimes_A N$, is the dg A -module

$$M \otimes_Q N / (am \otimes n - (-1)^{|a| \cdot |m|} m \otimes an \mid a \in A, m \in M, n \in N).$$

Remark 3.1.10. To define the tensor product $M \otimes_A N$, one takes the graded A -module

$$M \otimes_Q N / (am \otimes n - (-1)^{|a| \cdot |m|} m \otimes an \mid a \in A, m \in M, n \in N),$$

and then checks that the differential on the complex $M \otimes_Q N$ induces a well-defined map on the quotient, making it into a complex.

Example 3.1.11. Let $A = Q[X]$ be a minimal model for R over Q . Then $R = Q[X]/(I + (X))$, so

$$M \otimes_{Q[X]} R \cong \frac{M}{(I + (X))M}.$$

Definition 3.1.12. For dg A -modules M and N , we define $\text{Hom}_A(M, N)$ to be Q -subcomplex of $\text{Hom}_Q(M, N)$ consisting of all A -linear maps from M to N . That is to say,

$$\text{Hom}_A(M, N) := \{\alpha \in \text{Hom}_Q(M, N) \mid \alpha(am) = (-1)^{|a||\alpha|} a\alpha(m) \text{ for all } a \in A, m \in M\}$$

with differential

$$\partial_{\text{Hom}_A(M, N)}(\alpha) = \partial^N \alpha - (-1)^{|\alpha|} \alpha \partial^M.$$

Exercise 3.1.13. Show that for all dg A -modules M and N , the following hold:

- 1) The degree zero cycles of $\text{Hom}_A(M, N)$ are exactly the morphisms of dg A -module from M to N . Furthermore, elements of $H_0(\text{Hom}_A(M, N))$ are equivalences of morphisms of dg A -modules from M to N identified up to homotopy.
- 2) $\text{End}_A(M) = \text{Hom}_A(M, M)$ is a dg subalgebra of $\text{End}_Q(M)$.
- 3) $\text{Hom}_A(M, N)$ is a dg $(\text{End}_A(N), \text{End}_A(M))$ -bimodule.

Definition 3.1.14. Let M and N be dg A -modules. A map of complexes $\varphi: M \rightarrow N$ is a **map of dg A -modules** provided φ is a morphism of the underlying complexes of Q -modules such that

$$\varphi(am) = a\varphi(m)$$

for all $a \in A$ and $m \in M$.

Notation 3.1.15. As we did with maps of dg algebras, we will write $\varphi: M \xrightarrow{\simeq} N$ when φ is a quasiisomorphism.

Definition 3.1.16. Maps of dg A -modules $\varphi, \psi: M \rightarrow N$ are **homotopic**, denoted $\varphi \sim \psi$, if there exists an A -linear map $\sigma: M \rightarrow N$ of degree 1 such that

$$\partial_N \sigma + \sigma \partial_M = \varphi - \psi.$$

Here A -linear means that for all $a \in A$ and all $m \in M$ we have

$$\sigma(am) = (-1)^{|a|} a\sigma(m).$$

Definition 3.1.17. Let A be a dg Q -algebra and M be a dg A -module. For each integer ℓ , the **ℓ th suspension of M** is the dg A -module $\Sigma^\ell M$ given by

$$(\Sigma^\ell M)_n := M_{n-i}, \quad \partial^{\Sigma^\ell M} := (-1)^\ell \partial^M, \quad \text{and} \quad a \cdot \Sigma^\ell m := (-1)^{|a| \cdot \ell} \Sigma^\ell(am).$$

Remark 3.1.18. We can view a noetherian local ring as a dg algebra A , where $A_0 = R$ and $A_n = 0$ for all $n \geq 1$. In this case, a dg module over A is simply an A -complex, and all of the above notions are the familiar ones.

We will see more examples later, when we discuss dg modules in more detail.

Definition 3.1.19. Given a map of dg A -modules $\varphi: M \rightarrow N$, the **(mapping) cone** or **cone** of φ is the dg A -module with underlying graded A -module $N \oplus \Sigma M$ and differential

$$\begin{array}{ccc} N_n & \xrightarrow{\partial^N} & N_{n-1} \\ \oplus & \nearrow \varphi & \oplus \\ \left[\begin{array}{cc} \partial^n & \varphi \\ 0 & -\partial^M \end{array} \right] : & & \\ M_{n-1} & \xrightarrow{\partial^M} & M_{n-2} \end{array}$$

Remark 3.1.20. Given any map of dg modules $\varphi: M \rightarrow N$, there is a short exact sequence of dg modules

$$0 \rightarrow N \rightarrow \text{cone}(\varphi) \rightarrow \Sigma M \rightarrow 0.$$

Exercise 3.1.21. Show that φ is a quasiisomorphism if and only if $\text{cone}(\varphi)$ is exact.

Exercise 3.1.22. Let $\varphi: A \rightarrow B$ be a map of dg modules. Show that φ is nullhomotopic if and only if φ factors through the canonical map $A \rightarrow \text{cone}(\text{id}_A)$.

Definition 3.1.23. For dg A -modules M and N , we define $\text{Hom}_A(M, N)$ to be Q -subcomplex of $\text{Hom}_Q(M, N)$ consisting of all A -linear maps from M to N . That is to say,

$$\text{Hom}_A(M, N) := \{ \alpha \in \text{Hom}_Q(M, N) \mid \alpha(am) = (-1)^{|a||\alpha|} a\alpha(m) \text{ for all } a \in A, m \in M \}$$

with differential

$$\partial_{\text{Hom}_A(M, N)}(\alpha) = \partial^N \alpha - (-1)^{|\alpha|} \alpha \partial^M.$$

Definition 3.1.24. A dg A -module P is **semiprojective** if for every map of dg A -modules $\alpha: P \rightarrow N$ and each surjective quasiisomorphism of dg A -modules $\gamma: M \rightarrow N$ there exists a map of dg A -module $\beta: P \rightarrow M$, unique up to homotopy, such that $\alpha = \gamma\beta$. That is, P has the following lifting property:

$$\begin{array}{ccc} & & M \\ & \nearrow \beta & \downarrow \gamma \simeq \\ P & \xrightarrow{\alpha} & N. \end{array}$$

Exercise 3.1.25. Let P be a dg A -module. Show that the following are equivalent:

- 1) The dg module P is semiprojective.

2) For any surjective quasiisomorphism $M \rightarrow N$, the induced map

$$\beta_* = \text{Hom}_A(P, \beta): \text{Hom}_A(P, M) \rightarrow \text{Hom}_A(P, N)$$

given by $\gamma \mapsto \beta\gamma$ is a surjective quasiisomorphism.

3) The graded A -module $P^\natural$ is a projective and $\text{Hom}_A(P, C) \simeq 0$ whenever $C \simeq 0$.

4) As a dg A -module, P is a direct summand of a semifree dg A -module.

Definition 3.1.26. A **semiprojective resolution** of a dg A -module M is a surjective quasiisomorphism of dg A -modules $\varepsilon: P \rightarrow M$ where P is a semiprojective dg A -module.

Remark 3.1.27. We will soon show that semiprojective resolutions exist. Given that, it follows from the lifting property in [Definition 3.1.24](#) that any two semiprojective resolutions of M are homotopy equivalent dg A -modules.

Remark 3.1.28. For dg A -modules M and N , define

$$\text{Ext}_A(M, N) := H(\text{Hom}_A(P, N))$$

where P is a semiprojective resolution of M over A . Since by [Remark 3.1.27](#) any two semiprojective resolutions of M are homotopy equivalent, $\text{Ext}_A(M, N)$ is independent of the choice of M .

We claim that an element $[\alpha]$ of $\text{Ext}_A(M, N)$ is the class of a morphism of dg A -modules

$$\alpha: P \rightarrow \Sigma^{|\alpha|} N.$$

Indeed,

$$\begin{aligned} \text{Ext}_A^i(M, N) &= H^i(\text{Hom}_A(P, N)) \\ &= H_{-i}(\text{Hom}_A(P, N)) \\ &\cong H_0(\Sigma^i \text{Hom}_A(P, N)) \\ &\cong H_0(\text{Hom}_A(P, \Sigma^i N)) \end{aligned}$$

and now use [Exercise 3.1.25](#).

Remark 3.1.29. Given a morphism of dg algebras $\varphi: A \rightarrow B$, any dg B -module M can be regarded as a dg A -module by restricting scalars on φ . Explicitly, M will be considered as a dg A -module with underlying graded Q -complex (M, ∂^M) and the A -action is determined by $a \cdot m := \varphi(a)m$.

Lemma 3.1.30. *Let $\varphi: A \rightarrow B$ be a map of dg algebras. Then for dg B -modules M and N , this induces a natural map*

$$\text{Ext}_\varphi(M, N): \text{Ext}_B(M, N) \rightarrow \text{Ext}_A(M, N)$$

that is an isomorphism whenever φ is a quasiisomorphism.

Proof. The map is the induced map in homology of the map of complexes

$$\mathrm{Hom}_B(P, N) \xrightarrow{\mathrm{Hom}_B(\mu, N)} \mathrm{Hom}_B(B \otimes_A P', N) \cong \mathrm{Hom}_A(P', N)$$

where $P' \rightarrow M$ is a semiprojective resolution over A , and $P \rightarrow M$ is a semiprojective resolution over B . The map μ is a lift of the multiplication map $B \otimes_A P' \rightarrow B \otimes_A M \rightarrow M$ in the sense that the following commutes:

$$\begin{array}{ccc} & & P \\ & \nearrow \mu & \downarrow \simeq \\ B \otimes_A P' & \longrightarrow & M. \end{array}$$

Such a lift exists by [Definition 3.1.24](#), and any choice of lift is unique up to homotopy. In particular, the map on Ext modules is well-defined.

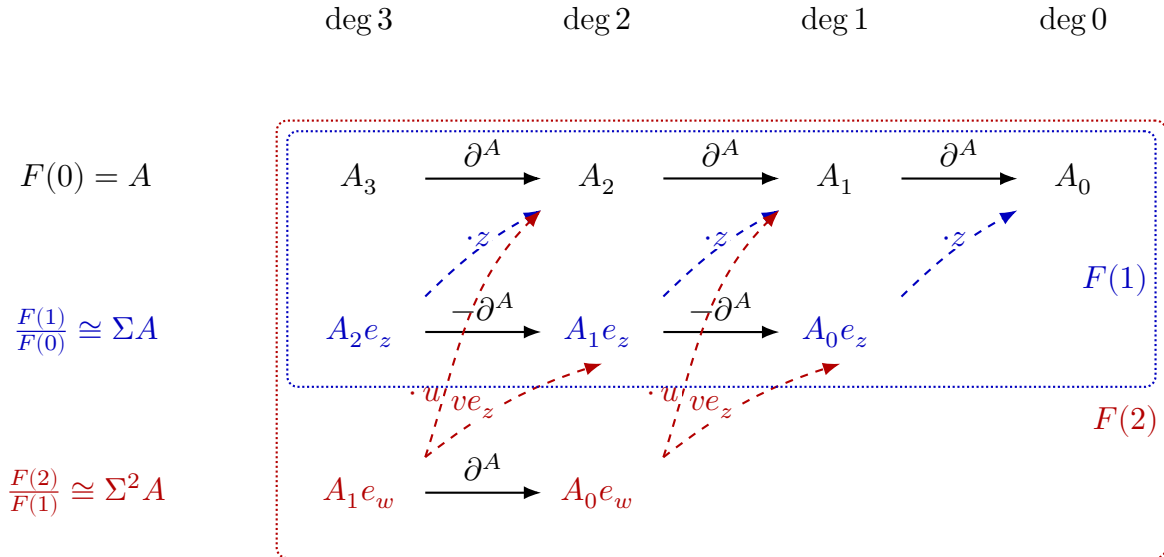
When φ is a quasiisomorphism, it follows that $B \otimes_A P'$ is again a semiprojective resolution of M over B , and hence the map μ above is a homotopy equivalence. In particular, the induced map $\mathrm{Hom}_B(\mu, N)$ is a homotopy equivalence, and so the map on Ext modules is an isomorphism. $\square$

Definition 3.1.31. A dg A -module F is **semifree** if it admits a filtration by dg A -submodules

$$0 = F(-1) \subseteq F(0) \subseteq F(1) \subseteq F(2) \subseteq \dots \subseteq F$$

where $\cup F(i) = F$ and each subquotient $F(i)/F(i-1)$ is a sum of shifts of A .

Remark 3.1.32. Here is a schematic of a semifree dg A -module with $F(2) = F$ and $F(i)/F(i-1) \cong \Sigma^i A$ for $i = 0, 1, 2$ and zero otherwise.



$$\begin{aligned} \partial(ae_z) &= \partial^A(a)e_z + (-1)^{|a|}az, \\ \partial(be_w) &= \partial^A(b)e_w + (-1)^{|b|}(bu + bve_z), \quad w = u + ve_z \in A_1 \oplus A_0e_z. \end{aligned}$$

Exercise 3.1.33. Prove the following facts about semifree dg modules.

- 1) If F is a dg A -module that is bounded below and $F^\natural$ is free over $A^\natural$, then F is semifree over A .
- 2) If F is semifree, the $F^\natural$ is a free graded $A^\natural$ -module.
- 3) If F is semifree, then F is semiprojective.

Hint: Mimic the proof of [Exercise 1.3.21](#) to show that $\mathrm{Hom}_A(F, \beta)$ is a surjective quasiisomorphism for any surjective quasiisomorphism of dg A -modules β .

Theorem 3.1.34. *Every dg A -module M admits a semifree resolution over A , which is unique up to homotopy equivalence.*

Proof. Fix a generating set $Z(0)$ for $Z(M)$. Define

$$F(0) = \bigoplus_{z \in Z(0)} \Sigma^{|z|} A$$

and $X(0) \subseteq F(0)$ in bijection with $Z(0)$. Set

$$\begin{aligned} F(0) &\xrightarrow{\varepsilon(0)} M \\ \Sigma^{|z|} 1 &\longmapsto z \end{aligned}$$

Note that $H(\varepsilon(0)): H(F(0)) \rightarrow H(M)$ is surjective, since the map on cycles is surjective.

Assuming we have constructed

$$F(0) \subseteq F(1) \subseteq \dots \subseteq F(i)$$

with $F(j)/F(j-1)$ a direct sum of shifts of A and compatible maps $\varepsilon(j): F(j) \rightarrow M$ extending $\varepsilon(0)$, let us outline the construction of the next step of the filtration $F(i+1)$. Fix a set of cycles $Z(i)$ in $F(i)$ whose homology classes form a generating set for $\ker(H(\varepsilon(i)))$, and note that for each $z \in Z(i)$ there exists $m_z \in M$ such that

$$\partial^M(m_z) = \varepsilon(i)(z). \tag{3.1.1}$$

Now define

$$F(i+1) := \mathrm{cone} \left(\bigoplus_{z \in Z(i)} \Sigma^{|z|} A \xrightarrow{\oplus \Sigma^{|z|} 1 \mapsto z} F(i-1) \right)$$

and $\varepsilon(i+1): F(i+1) \rightarrow M$ given by

$$\left(f, \sum_{\text{finite}} (\Sigma^{|z|+1} r) \right) \mapsto \varepsilon(i)(f) + \sum_{\text{finite}} r m_z.$$

Moreover, we take $X(i+1)$ to be a set of variables. It follows from $\partial^M(m_z) = \varepsilon(i)(z)$ that $\varepsilon(i+1)$ is map of dg A -modules extending $\varepsilon(i)$. Furthermore, as $F(i+1)$ is a mapping cone, the quotient $F(i+1)/F(i)$ is a direct sum of shifts of A , namely, it is

$$\bigoplus_{z \in Z(i)} \Sigma^{|z|+1} A.$$

This completes the inductive step of the construction.

Now set

$$F := \varinjlim F(i) \quad \text{and} \quad \varepsilon := \varinjlim \varepsilon(i): F \rightarrow M.$$

By construction, F is semifree, $H(\varepsilon)$ is surjective (since it extends $\varepsilon(0)$), and $\ker(H(\varepsilon))$ is zero by the choice of cycles in the inductive step. Therefore, ε is a quasiisomorphism, and so it remains to use the exact sequences involving a dg A -module, its boundaries, its cycles, and its homology, together with the fact that ε is surjective on cycles and homology to conclude that ε is a surjective. $\square$

Remark 3.1.35. By analyzing the proof of [Theorem 3.1.34](#), it follows that if $H_0(A)$ is a noetherian ring and $H(M)$ is finitely generated over $H_0(A)$, then there exists a semifree resolution $F \rightarrow M$ such that

$$F^\natural \cong \bigoplus_{i=n}^{\infty} \Sigma^i (A^\natural)^{\oplus b_i}$$

for some n and nonnegative integers b_i . Equivalently, there is an isomorphism of graded $A^\natural$ -modules

$$F^\natural \cong A^\natural \otimes_Q G$$

where G is a bounded below graded Q -module such that each G_i is a finite rank free Q -module.

Remark 3.1.36. When A is a ring and M is a module, the proof of [Theorem 3.1.34](#) produces a free resolution of M over A .

Example 3.1.37. Assume A is a ring and M is an R -module and consider a free resolution

$$F = \cdots \longrightarrow F_3 \xrightarrow{\partial_3} F_2 \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \longrightarrow 0$$

of M over A . Note that $F(i) = F_{\leq i}$ for each i determines a filtration showing that F is semifree. Furthermore, for each i we claim that

$$F(i+1) = \text{cone} \left(\Sigma^i F_{i+1} \xrightarrow{\partial_{i+1}} F(i) \right).$$

Indeed, consider the map of complexes $\Sigma^i F_{i+1} \xrightarrow{\partial_{i+1}} F(i)$ depicted below:

degree	$i+1$	i	$i-1$	$i-2$	$\cdots$
$\Sigma^i F_{i+1}:$	0	$\longrightarrow F_{i+1}$	$\longrightarrow 0$	$\longrightarrow 0$	$\longrightarrow \cdots$
$\downarrow$	$\downarrow$	$\downarrow \partial_{i+1}$	$\downarrow$	$\downarrow$	$\downarrow$
$F_{\leq i}:$	0	$\longrightarrow F_i$	$\xrightarrow{\partial_i} F_{i-1}$	$\xrightarrow{\partial_{i-1}} F_{i-2}$	$\longrightarrow \cdots$

Its mapping cone is exactly the A -complex $F_{\leq i+1} = F(i+1)$, as claimed. In particular, we have a short exact sequence of complexes

$$0 \longrightarrow F(i) \longrightarrow F(i+1) \longrightarrow \Sigma^{i+1} F_{i+1} \longrightarrow 0.$$

Theorem 3.1.38. *Let $A \rightarrow B$ be a quasiisomorphism of dg algebras. Let M be a dg B -module and let $\varepsilon: F \rightarrow M$ be a map of dg A -modules, where we view M as a dg A -module via restriction of scalars. Let F be a semifree (respectively, semiprojective) dg A -module. Then $\varepsilon: F \rightarrow M$ is a semifree (respectively, semiprojective) resolution over A if and only if*

$$B \otimes_A F \xrightarrow{\text{id}_B \otimes \varepsilon} B \otimes_A F \xrightarrow{\mu} M$$

is a semifree (respectively, semiprojective) resolution over B .

Theorem 3.1.39. *Let $A \rightarrow B$ be a quasiisomorphism of local dg algebras with common residue field k . Let M be a dg B -module with $H(M)$ finitely generated over $H(B)$. Hence, F is a minimal semifree A -resolution if and only if $B \otimes_A F$ is a minimal semifree B -resolution.*

Corollary 3.1.40. *Let R be a noetherian local ring with minimal regular presentation $\widehat{R} \cong Q/I$, and let $Q[X]$ be the minimal model of R over Q . For any finitely generated R -module M , $\text{pdim}_R(M) < \infty$ if and only if the underlying graded A -module to a minimal semifree resolution $F \xrightarrow{\sim} M$ of M over A has*

$$F \cong \bigoplus_{i=0}^n \Sigma^i A^{\beta_i}.$$

Proof. The minimal semifree resolution for M over $Q[X]$ has the form

$$F \cong \bigoplus_{i \geq 0} \Sigma^i A^{\beta_i}.$$

By [Theorem 3.1.39](#), the minimal free resolution for M over R has the form

$$R \otimes_{Q[X]} F \cong \bigoplus_{i \geq 0} \Sigma^i R^{\beta_i}.$$

Thus

$$\text{pdim}_R(M) < \infty \iff \beta_i = 0 \text{ for all } i \gg 0.$$

□

Remark 3.1.41. In [Theorem 3.1.38](#), M need not be an R -module. When M is an R -complex, one takes F to be a semifree resolution; the conclusions of the lemma, suitably modified for the in particular statement, still hold.

Using NAK (both the graded and non-graded versions), one can prove the following result, in a way analogous to how one shows the existence of minimal free resolutions:

Theorem 3.1.42. *Let A be a local dg algebra, nonnegatively graded, and let M be a dg A -module with $H(M)$ finitely generated over $H(A)$. There exists a minimal semifree resolution $F \xrightarrow{\simeq} M$ with*

$$\partial(F) \subseteq \mathfrak{m}_A F.$$

Example 3.1.43. Let $Q = k[[x, y]]$ and $R = Q/(xy)$. Consider

$$A = Q[e \mid \partial(e) = xy] = \text{Kos}^Q(xy) \xrightarrow{\simeq} R = Q/(f, e)$$

and F given by the following two pictures:

$$\begin{array}{ccccccc} \cdots & & Q^2 & \xrightarrow{\begin{bmatrix} y & 0 \\ 0 & x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} & Q \longrightarrow 0 \\ & \nearrow xy & \oplus & \nearrow -xy & \oplus & \nearrow xy & \\ (Qe)^2 & \xrightarrow{\begin{bmatrix} y & 0 \\ 0 & x \end{bmatrix}} & (Qe)^2 & \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} & Qe & & \end{array}$$

and

$$\begin{array}{ccccccc} Q^2 & \xrightarrow{\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} y & 0 \\ 0 & x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} & Q \longrightarrow 0 \\ \searrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & & \oplus & \begin{bmatrix} -1 & -1 \end{bmatrix} & \oplus & & \\ \cdots & & (Qe)^2 & \xrightarrow{\begin{bmatrix} y & 0 \\ 0 & x \end{bmatrix}} & (Qe)^2 & \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} & Qe \end{array}$$

One can check that F is a semifree resolution for k over A .

3.2 Algebra and module structures on Ext

Let A be a (nonnegatively graded) dg Q -algebra.

Remark 3.2.1. Fix dg A -modules M, N , and let $F \xrightarrow{\simeq} M$ and $G \xrightarrow{\simeq} N$ be semiprojective resolutions over A . Recall that by [Exercise 3.1.13](#), $\text{Hom}_A(F, G)$ is a dg $\text{End}_A(G)\text{-End}_A(F)$ -bimodule acting on $\text{Hom}_A(F, G)$ where the actions are given by post- and pre-composition of A -linear maps on G, F , respectively. Given other semiprojective dg A -modules $F' \xrightarrow{\simeq} M$ and $G' \xrightarrow{\simeq} N$, we have that the pairs F and F' and G and G' are each homotopy equivalent, respectively, and so these define quasiisomorphisms

$$\text{End}_A(F) \simeq \text{End}_A(F'), \quad \text{End}_A(G) \simeq \text{End}_A(G') \quad \text{and} \quad \text{Hom}_A(F, G) \simeq \text{Hom}_A(F', G').$$

Furthermore, the bimodule actions are compatible in homology in the sense that the following diagram commutes:

$$\begin{array}{ccccc}
\mathrm{H}(\mathrm{End}_A(G)) \otimes_Q \mathrm{H}(\mathrm{Hom}_A(F, G)) & \longrightarrow & \mathrm{H}(\mathrm{Hom}_A(F, G)) & \longleftarrow & \mathrm{H}(\mathrm{Hom}_A(F, G)) \otimes_Q \mathrm{H}(\mathrm{End}_A(F)) \\
\downarrow \cong & & \downarrow \cong & & \downarrow \cong \\
\mathrm{H}(\mathrm{End}_A(G')) \otimes_Q \mathrm{H}(\mathrm{Hom}_A(F', G')) & \rightarrow & \mathrm{H}(\mathrm{Hom}_A(F', G')) & \leftarrow & \mathrm{H}(\mathrm{Hom}_A(F', G')) \otimes_Q \mathrm{H}(\mathrm{End}_A(F')).
\end{array}$$

In summary, we have proved the following:

Theorem 3.2.2. *For any dg A -module M , the graded Q -module $\mathrm{Ext}_A(M, M)$ is a graded Q -algebra with the composition product given by*

$$[\alpha] \cdot [\beta] := [\alpha \circ \beta]$$

where $[\alpha], [\beta] \in \mathrm{H}(\mathrm{End}_A(F))$ and $F \xrightarrow{\cong} M$ is a semiprojective resolution of M over A . Furthermore, for any dg A -module N , $\mathrm{Ext}_A(M, N)$ is a graded $\mathrm{Ext}_A(N, N)$ - $\mathrm{Ext}_A(M, M)$ bimodule with composition actions given by

$$[\alpha] \cdot [\beta] := [\alpha \circ \beta] \quad \text{and} \quad [\beta] \cdot [\gamma] := [\beta \circ \gamma]$$

where $[\alpha] \in \mathrm{H}(\mathrm{End}_A(G))$, $[\beta] \in \mathrm{H}(\mathrm{Hom}_A(F, G))$, and $[\gamma] \in \mathrm{H}(\mathrm{End}_A(F))$ with $F \xrightarrow{\cong} M$ and $G \xrightarrow{\cong} N$ semiprojective resolutions over A . Moreover, these structures are independent of the choices of semiprojective resolutions.

Remark 3.2.3. Given a map of dg A -modules $M \rightarrow N$, then the induced map

$$\mathrm{Ext}_A(N, k) \rightarrow \mathrm{Ext}_A(M, k)$$

is $\mathrm{Ext}_A(k, k)$ -linear.

Corollary 3.2.4. *Let $A \twoheadrightarrow k$ be a map of dg algebras with k a field. Then*

$$\mathrm{Ext}_A(k, k) = \bigoplus_{i=0}^{\infty} \mathrm{Ext}_A^i(k, k)$$

is a nonnegatively graded k -algebra with k in degree 0. Furthermore, for any dg A -module M , $\mathrm{Ext}_A(M, k)$ is a graded left $\mathrm{Ext}_A(k, k)$ -module.

Proof. The fact $\mathrm{Ext}_A^0(k, k) = k$ follows from the map $A \rightarrow k$ being surjective. Also, since A is nonnegatively graded, it follows that k has a semifree resolution over A that is concentrated in nonnegative homological degrees. In particular, it follows that $\mathrm{Ext}_A^i(k, k) = 0$ for $i < 0$. The remaining claims follow immediately from the previous result. $\square$

Lemma 3.2.5 (Serre). *Let A be a local dg algebra with $A_0 = R$ a noetherian local ring $(R, \mathfrak{m}, k)$. Let $x \in \mathfrak{m} \setminus \mathfrak{m}^2$, and set $B = A[e \mid \partial(e) = x]$. If F is the minimal semifree resolution of a dg B -module M , then it is the minimal semifree A -resolution.*

3.3 Application: Gulliksen–Schoeller

We can finally prove Gulliksen and Schoeller’s theorem [Theorem 1.5.25](#):

Theorem 3.3.1 (Gulliksen, 1968 [[Gul68](#)], Schoeller, 1967 [[Sch67](#)]). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. An acyclic closure for k is a minimal free resolution for k .*

Sketch of proof. Fix a minimal regular presentation $\widehat{R} \cong Q/I$ and let $Q[X]$ be a minimal model for R over Q . Consider the zigzag from [Remark 1.4.24](#):

$$K^R \xleftarrow{\simeq} k^Q \otimes_Q Q[X] \xrightarrow{\simeq} k \otimes_Q Q[X] =: k[X],$$

and recall that $\partial(X) \subseteq (X)^2$, by [Exercise 1.3.16](#). In [Construction 1.5.16](#), we constructed

$$K^R \langle Y_{\geq 2} \rangle \xleftarrow{\simeq} k^Q \otimes_Q Q[X] \langle Y_{\geq 2} \rangle \xrightarrow{\simeq} k[X] \langle Y_{\geq 2} \rangle$$

with

$$Y_i = \{y_x \mid x \in X_{i-1}\} \quad \text{and} \quad \partial(k[X] \langle Y_{\geq 2} \rangle) \subseteq \mathfrak{m}_{k[X]} \cdot k[X] \langle Y_{\geq 2} \rangle.$$

By construction,

$$k[X] \langle Y_{\geq 2} \rangle \xrightarrow{\simeq} k,$$

so $K^R \langle Y_{\geq 2} \rangle$ is a resolution for k . Moreover, from the construction it follows that

$$\partial(K^R \langle Y_{\geq 2} \rangle) \subseteq \mathfrak{m}_{K^R} \cdot K^R \langle Y_{\geq 2} \rangle.$$

This says that $K^R \langle Y_{\geq 2} \rangle$ is the minimal semifree resolution of k over K^R . Now by [Lemma 3.2.5](#), it must also be the minimal semifree resolution over R ; but that is the same as saying it is a minimal free resolution over R . $\square$

Chapter 4

Complete intersections and the conormal module

4.1 The conormal module

We have seen that K^R is an R -module that detects the complete intersection property. We will now discuss another important module that also detects whether R is a complete intersection.

Definition 4.1.1. The **conormal module** of a surjective ring homomorphism $\varphi: Q \twoheadrightarrow R$ with kernel I is the module I/I^2 .

Remark 4.1.2. Note that I/I^2 is not only a Q -module but also an R -module.

Before we can talk more about conormal modules, we need a few auxiliary lemmas, which have several other applications.

Lemma 4.1.3. *Let R be any noetherian ring and I a nonzero ideal in R . If I is a free R -module, then either $I = R$ or $I = (a)$ for a regular element $a \in R$.*

Proof. Since I is finitely generated, we have $I \cong R^n$ for some $n \geq 1$. Any distinct nonzero $x, y \in I$ satisfy the relation $xy - yx = 0$, so there are no two linearly independent elements in I . We conclude that $n = 1$ and $I \cong R$. Let $\{a\}$ be a basis for I , and suppose that I is a proper ideal. The map

$$\begin{array}{ccc} R & \xrightarrow{\pi} & I \\ r & \longmapsto & ra \end{array}$$

is an isomorphism, so

$$\text{ann}_R(a) = \ker(\pi) = 0$$

and a is a regular element in R . □

Exercise 4.1.4. Let R be any noetherian local ring. Show that any finite exact sequence of finitely generated free R -modules

$$0 \longrightarrow R^{\beta_d} \longrightarrow \dots \longrightarrow R^{\beta_0} \longrightarrow 0$$

satisfies

$$\sum_{i=0}^d (-1)^i \beta_i = 0.$$

The following statement might appear simple, but it has very powerful consequences, as we will soon see.

Theorem 4.1.5. *Let I be a nonzero ideal in a noetherian local ring R . If I has finite projective dimension, then I contains a regular element.*

Proof. Consider any finite (augmented) free resolution of the nonzero ideal I by finitely generated R -modules

$$0 \longrightarrow R^{\beta_d} \longrightarrow R^{\beta_{d-1}} \longrightarrow \cdots \longrightarrow R^{\beta_1} \longrightarrow R^{\beta_0} \longrightarrow I \longrightarrow 0.$$

Since localization is exact, for all prime ideals P we get a finitely generated free resolution for I_P

$$0 \longrightarrow R_P^{\beta_d} \longrightarrow \cdots \longrightarrow R_P^{\beta_0} \longrightarrow I_P \longrightarrow 0.$$

In particular, I_P has finite projective dimension over R_P .

Let P be an associated prime of R . We saw above that I_P has finite projective dimension over R_P . Since P is an associated prime of R , it consists of only zerodivisors, so $\text{depth}(R_P) = 0$. By the Auslander–Buchsbaum formula, [Theorem 0.3.7](#), we conclude that I_P must be a free module over R_P . By [Exercise 4.1.4](#),

$$\text{rank}_{R_P}(I_P) = r := \sum_i (-1)^i \beta_i.$$

In particular, this rank r does not depend on P .

Take any associated prime P of I (as an R -module). The inclusion of R -modules $I \subseteq R$ gives us $\text{Ass}(I) \subseteq \text{Ass}(R)$, so P is an associated prime of R . Since $P \in \text{Ass}(I) \subseteq \text{Supp}(I)$, we conclude that $I_P \neq 0$. Therefore, there exists an associated prime P of R such that $I_P \neq 0$. We saw above that I_P is a free module of rank r , and since $I_P \neq 0$ we conclude that $r \geq 1$.

Now suppose that I contains no regular elements, so that I is contained in the union of the associated primes of R . By Prime Avoidance, I must be contained in some associated prime of R , say P . In particular, $I_P \neq R_P$. We have also shown that I_P must be a free module over R_P of rank $r \geq 1$, and thus $I_P \neq 0$. By [Lemma 4.1.3](#), we conclude that I_P is generated by a regular element in R_P . Since $P \in \text{Ass}(R)$, there are no regular elements in R_P , so we have reached a contradiction. Therefore, I contains a regular element. $\square$

Exercise 4.1.6. Let M be a finitely generated R -module.

- 1) Show that if $\underline{x}$ is a regular sequence on M , then $\mathrm{Tor}_1^R(M, R/(\underline{x})) = 0$.
- 2) Show that if $\underline{x}$ is a regular sequence on M and also on R , then $\mathrm{Tor}_i^R(M, R/(\underline{x})) = 0$ for all $i \geq 1$.
- 3) Give an example of a ring R , a finitely generated R -module M , and a regular sequence $\underline{x}$ on M such that $\mathrm{Tor}_i^R(M, R/(\underline{x})) \neq 0$ for some $i \geq 2$.

Exercise 4.1.7. Let I be a nonzero ideal in a noetherian local ring R . Show that if I is generated by a regular sequence, then I/I^2 is a free R -module.

More surprisingly, the converse also holds.

Theorem 4.1.8 (Vasconcelos, 1967 [Vas67], Ferrand, 1967 [Fer67]). *Let I be a nonzero ideal of finite projective dimension in a noetherian local ring $(Q, \mathfrak{m}, k)$. Set $R = Q/I$. If I/I^2 is a free R -module, then I is generated by a regular sequence.*

Proof. By Theorem 4.1.5, I must contain a regular element, and thus I is not contained in the union of the associated primes of R . Moreover, $I \not\subseteq \mathfrak{m}I$ by NAK. By Prime Avoidance,

$$I \not\subseteq \mathfrak{m}I \cup \bigcup_{P \in \mathrm{Ass}(R)} P.$$

Thus we can pick a minimal generator $x \in I$ that is also a regular element on R .

We proceed by induction on $\mu(I)$. If $\mu(I) = 1$, then $I = (x)$ is generated by a regular sequence. When $\mu(I) > 1$, note that $I/(x)$ is generated by $\mu(I) - 1$ elements, and since x is regular on R then it is also regular on I . We will prove that

$$\mathrm{pdim}_{Q/(x)}(I/(x)) < \infty \quad \text{and} \quad \frac{I/(x)}{(I/(x))^2} \text{ is a free } R\text{-module.}$$

Induction will then complete the proof: the induction hypothesis gives us that $I/(x)$ is generated by a regular sequence, which together with x we can lift to a regular sequence generating I .

Since x is regular on R and I , by Exercise 4.1.6 we know that

$$\mathrm{Tor}_i^Q(Q/(x), I) = 0 \text{ for all } i \geq 1.$$

Given a finite free resolution F for I over Q , note that

$$\mathrm{H}_i(F \otimes_Q Q/(x)) = \mathrm{Tor}_i^Q(Q/(x), I) = 0 \text{ for all } i \geq 1$$

and

$$\mathrm{H}_0(F \otimes_Q Q/(x)) = I \otimes_Q Q/(x) = I/(x)I.$$

We conclude that $F \otimes_Q Q/(x)$ is a finite free resolution for $I/(x)I$ over $Q/(x)$. We will use this to prove that $I/(x)$ also has finite projective dimension.

We have

$$\frac{I/I^2}{\mathfrak{m}I/I^2} \cong I/\mathfrak{m}I.$$

Since $x \notin \mathfrak{m}I$, its image in the free module I/I^2 is a basis element, and thus we can extend it to $x, x_2, \dots, x_n$ such that $x + I^2, x_2 + I^2, \dots, x_n + I^2$ is a basis for I/I^2 . Set $J = (x_2, \dots, x_n)$. Set $J = (x_2, \dots, x_n)$. Given any $z \in J \cap (x)$, note that

$$z = ax \text{ and } z = a_2x_2 + \dots + a_nx_n \text{ for some } a, a_2, \dots, a_n \in Q.$$

But in I/I^2 , the elements $x + I^2, x_2 + I^2, \dots, x_n + I^2$ are linearly independent, and thus all the coefficients $a, a_2, \dots, a_n$ are zero in R , so in particular $a \in I$ and $z = ax \in (x)I$. We conclude that $J \cap (x) \subseteq (x)I$. Therefore, by the Third Isomorphism Theorem

$$I/(x) = J + (x)/(x) \cong J/J \cap (x).$$

Moreover, the inclusion $J \subseteq I$ induces a map on the quotients

$$J/J \cap (x) \longrightarrow I/J \cap (x).$$

Composing with the canonical quotient maps $I/J \cap (x) \twoheadrightarrow I/(x)I$ and $I/(x)I \twoheadrightarrow (x)$, we get a homomorphism

$$\begin{aligned} I/(x) \cong J/J \cap (x) &\longrightarrow I/J \cap (x) \longrightarrow I/(x)I \longrightarrow I/(x) \\ x_i + (x) &\longmapsto x_i + (J \cap (x)) \longmapsto x_i + (x)I \longmapsto x_i + I. \end{aligned}$$

Since $I/(x)$ is generated by $x_i + (x)$, this composition is the identity map, giving us a splitting of the surjection $I/(x)I \twoheadrightarrow I/(x)$. We conclude that $I/(x)$ is a direct summand of $I/(x)I$. By [Exercise 0.2.13](#), $I/(x)$ also has finite projective dimension over $Q/(x)$.

It remains to show that $\frac{I/(x)}{(I/(x))^2}$ is a free R -module. This quotient

$$\frac{I/(x)}{(I/(x))^2} \cong \frac{I/(x)}{I^2 + (x)/(x)} \cong I/I^2 + (x)$$

is isomorphic to the quotient of I/I^2 by the free submodule generated by $x + I^2$, and thus it is free, with basis given by the images of $x_2, \dots, x_n$. $\square$

Corollary 4.1.9. *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. The following are equivalent:*

- 1) *The ring R is a complete intersection.*
- 2) *For some regular presentation $\widehat{R} \cong Q/I$, the conormal module I/I^2 is a free $\widehat{R}$ -module.*
- 3) *For every regular presentation $\widehat{R} \cong Q/I$, the conormal module I/I^2 is a free $\widehat{R}$ -module.*

Proof. If R is a complete intersection, then by [Corollary 1.4.23](#) every minimal regular presentation $\widehat{R} \cong Q/I$ is such that I is generated by a regular sequence. By [Exercise 4.1.7](#), I/I^2 is free.

Conversely, suppose that there exists a minimal regular presentation $\widehat{R} \cong Q/I$ such that I/I^2 is a free R -module. By [Theorem 4.1.8](#), I is generated by a regular sequence, and thus R is a complete intersection. $\square$

Vasconcelos conjectured that [Theorem 4.1.8](#) could be further extended: that in order for R to be a complete intersection, it suffices to assume that for some regular presentation $\widehat{R} \cong Q/I$, the conormal module I/I^2 has finite projective dimension. As we will see in the chapter, this is indeed true; however, we will need to develop a new tool in order to prove it: the homotopy Lie algebra of R .

4.2 Derivations

First, let us give a brief overview of the classical theory of derivations.

Definition 4.2.1. Let Q be a commutative ring. Let A be commutative Q -algebra and M an A -module. A Q -linear **derivation** from A to M is a Q -linear map $\theta: A \rightarrow M$ satisfying the Leibniz rule: for all $x \in A$ and $y \in M$,

$$\theta(xy) = \theta(x)y + x\theta(y).$$

Example 4.2.2. We saw in [Example 2.1.10](#) that $\text{Der}_Q(A, A)$ is a Lie algebra.

Example 4.2.3. If A is a graded Q -algebra (in the usual sense, not the graded commutative sense), then one can show that the map $\varepsilon: A \rightarrow A$

$$\varepsilon(a) = |a| \cdot a$$

is a derivation, called the Euler derivation.

Example 4.2.4. If $A = Q[x]$ is a polynomial ring over Q , then $\text{Der}_Q(A, M) \cong M$ via $\alpha \mapsto \alpha(x)$.

More generally:

Theorem 4.2.5. If $A = Q[x_1, \dots, x_n]$ is a polynomial ring over Q , then $\text{Der}_Q(A, M) \cong M^n$, via the isomorphism given by

$$\begin{aligned} \text{Der}_Q(A, M) &\longrightarrow M^n \\ \alpha &\longmapsto (\alpha(x_i)). \end{aligned}$$

Example 4.2.6. If $Q = k$ has prime characteristic p , then

$$\begin{aligned} \text{Der}_k(k[x]/(x^p), M) &\longrightarrow M \\ \theta &\longmapsto \theta(x). \end{aligned}$$

There is a universal derivation:

Theorem 4.2.7. Let A be commutative Q -algebra. There exists a Q -module $\Omega_{A|Q}$ and a derivation $d: A \rightarrow \Omega_{A|Q}$ such that for every derivation $A \rightarrow M$ there exists a module map $f: \Omega_{A|Q} \rightarrow M$ such that the following diagram commutes:

$$\begin{array}{ccc} & \Omega_{A|Q} & \\ \nearrow & & \searrow \exists f \\ A & \xrightarrow{\theta} & M. \end{array}$$

Moreover, if $A = Q[x_1, \dots, x_n]$ is a polynomial ring over Q , then $\Omega_{A|Q} \cong A^n$ is a free A -module. If A is a finitely generated Q -algebra, then $\Omega_{A|Q}$ is a finitely generated A -module.

Definition 4.2.8. The module $\Omega_{A|Q}$ with the property in [Theorem 4.2.7](#) is known as the **module of Kähler differentials**.

Construction 4.2.9. Here is an explicit construction of the module of Kähler differentials:

$$\Omega_{A|Q} := \frac{\bigoplus_{a \in A} A \cdot da}{(d(a+b) - d(a) - d(b), d(q), d(ab) - d(a)b - ad(b) \mid a, b \in A, q \in Q)}$$

and the universal differential is given by $a \mapsto da$.

Note that $d: A \rightarrow \Omega_{A|Q}$ is a Q -linear derivation by construction. Moreover, the universal property and the Yoneda Lemma give us a bijection

$$\begin{aligned} \text{Hom}_A(\Omega_{A|Q}, M) &\longleftrightarrow \text{Der}_Q(A, M) \\ f: (da \mapsto d(da)) &\qquad \theta: a \mapsto f(da). \end{aligned}$$

The fact that $\Omega_{A|Q}$ satisfies the necessary universal property follows by construction.

We are now ready for the dg version, which is what we will be using in the next section.

Definition 4.2.10. Let A be dg Q -algebra and B a dg A -module. A Q -linear **derivation** from A to B is a Q -linear map $\theta: A \rightarrow B$ of degree $|\theta|$ satisfying the following graded Leibniz rule: for all $x, y \in A$,

$$\theta(xy) = \theta(x)y + (-1)^{|\theta||x|}x\theta(y).$$

If θ is also a chain map $A \rightarrow \Sigma^{|\theta|} B$, then we say that θ is a **chain derivation**.

We denote that set of all Q -linear derivations from A to B by $\text{Der}_Q(A, B)$, and endow it with a complex structure as in [Definition 1.2.11](#), with

$$\text{Der}_Q(A, B)_n = \{\theta: A \rightarrow B \mid \theta \text{ is a degree } n \text{ } Q\text{-linear derivation}\}$$

and differential

$$\partial(\theta) = \partial_B \theta - (-1)^{|\theta|} \theta \partial_A.$$

In fact, $\text{Der}_Q(A, B)$ is a dg A -module via

$$a \cdot \theta \text{ is given by } (a \cdot \theta)(x) = a\theta(x),$$

and $\text{Der}_Q(A, B) \subseteq \text{Hom}_Q(A, B)$ is an inclusion of dg A -modules.

We will be primarily interested in derivations from a minimal model to itself.

Exercise 4.2.11. Let R be a noetherian local ring and fix a minimal regular presentation $\widehat{R} \cong Q/I$ and $Q[X]$ a minimal model for R over Q . Let $\theta: Q[X] \rightarrow Q[X]$ be a derivation of degree $d \leq 0$ such that $\theta((X_{\geq 0})) \subseteq (X_{\geq 0})$.

- 1) Show that $\theta((X_{\geq 0})^n) \subseteq (X_{\geq 0})^n$ for all $n \geq 1$.

2) Show that θ induces a well-defined derivation $\theta^{(n)}: k[X_{\geq n}] \rightarrow k[X_{\geq n}]$.

Discussion 4.2.12 (Kähler differentials). Let A be a dg Q -algebra. One can show that the functor defined on the category of dg A -modules by $\text{Der}_Q(A, -)$ is representable; its representative is known as the dg A -module of **Kähler differentials**, denoted $\Omega_{A|Q}$, and satisfying the natural isomorphism

$$\text{Der}_R(A, M) \cong \text{Hom}_A(\Omega_{A|Q}, M).$$

The module of Kähler differentials comes equipped with a universal map

$$d: A \rightarrow \text{Hom}_A(\Omega_{A|Q}, M)$$

such that for every Q -linear derivation $\theta: A \rightarrow B$ there is an A -linear map $d_\theta: \Omega_{A|Q} \rightarrow B$ such that

$$\begin{array}{ccc} A & \xrightarrow{\theta} & B \\ d \downarrow & \nearrow d_\theta & \\ \Omega_{A|Q} & & \end{array}$$

commutes. If θ is a chain derivation, then d_θ is a map of dg A -modules.

The long exact sequence on $\text{Hom}_A(\Omega_{A|Q}, -)$ gives us that any short exact sequence of dg A -modules

$$0 \longrightarrow U \longrightarrow V \longrightarrow W \longrightarrow 0$$

gives rise to a long exact sequence

$$\cdots \longrightarrow H_n(\text{Der}_R(A, U)) \longrightarrow H_n(\text{Der}_R(A, V)) \longrightarrow H_n(\text{Der}_R(A, W)) \longrightarrow \cdots.$$

We will be particularly interested in the case when $A = Q[X]$ is a minimal model for R over Q . In what follows, we will use some basic properties of $\Omega_{A|Q}$ without proof:

- The underlying A -module to $\Omega_{A|Q}$ is free on a basis dX_i in degree preserving bijection with X_i .
- The projection map $\Omega_{A|Q} \rightarrow R \otimes_A \Omega_{A|Q}$ is a quasiisomorphism. Moreover, $R \otimes_A \Omega_{A|Q}$ is a minimal complex of free R -modules, and as a graded R -module

$$R \otimes_A \Omega_{A|Q} \cong \bigoplus_{i \geq 1} R \cdot dX_i.$$

As an R -complex, $R \otimes_A \Omega_{A|Q}$ is isomorphic to

$$R[X]/((X^2) + R) \cong RX.$$

- There is a relationship between $\Omega_{A|Q}$ and the conormal module:

$$H_1(\Omega_{A|Q}) \cong H_1(R \otimes_A \Omega_{A|Q}) \cong I/I^2.$$

Notice that by Hom-tensor adjunction, restricting $\text{Der}_Q(Q[X], -)$ to Q -modules annihilated by I , that is, those with an R -module structure,

$$\text{Der}_Q(Q[X], -) \cong \text{Hom}_{Q[X]}(\Omega_{Q[X]|Q}, -) \cong \text{Hom}_R(R \otimes_{Q[X]} \Omega_{Q[X]|Q}, -).$$

Thus to give a (chain) derivation from $Q[X]$ to an R -module M is equivalent as giving a (chain) map $RX \rightarrow M$.

Remark 4.2.13. Let $Q[X]$ be any semifree extension of Q . By combining Q -linearity with the Leibniz rule, we see that any derivation $\theta: Q[X] \rightarrow M$ is completely determined by the values of the variables $\theta(x)$ for all $x \in X$: in general, given variables $x_1, \dots, x_n$,

$$\theta(x_1 \cdots x_n) = \sum_{j=1}^n (-1)^{|x_1| \cdots |x_{j-1}| \cdot |\theta|} x_1 \cdots x_{j-1} \theta(x_j) x_{j+1} \cdots x_n.$$

Theorem 4.2.14. *Given a dg A -module M , there is an isomorphism*

$$\begin{aligned} \text{Der}_Q(A, M) &\xleftarrow{\cong} \prod_{x \in X} M \\ \theta: x_i &\mapsto m_i & (m_i)_{x_i \in X} \end{aligned}$$

Example 4.2.15. Let us compute $R \otimes_{Q[X]} \Omega_{Q[X]|Q}$ for $R = Q/I$ with $Q = k[[x, y, z]]$ and $I = (xy, zx, yz)$. Back in [Example 1.3.11](#), we computed the first few steps in a minimal model $Q[X]$. Since $R \otimes_{Q[X]} \Omega_{Q[X]|Q} \cong RX$, the complex looks like

$$\cdots \longrightarrow RX_3 \longrightarrow RX_2 \longrightarrow RX_1 \longrightarrow 0$$

and the differential is induced by the differential on $Q[X]$. More precisely, we read off the *linear* part of $\partial_{Q[X]}$, as we have modded out by $(X)^2$, and set all elements in $IQ[X]$ to zero. Recall from [Example 1.3.11](#) that

$$X_1 = \{T_{1,1}, T_{1,2}, T_{1,3}\} \quad X_2 = \{T_{2,1}, T_{2,2}\} \quad X_3 = \{T_{3,1}, T_{3,2}, T_{3,3}\}$$

with

$$\begin{aligned} \partial(T_{2,1}) &= zT_{1,1} - yT_{1,2} & \text{and} & & \partial(T_{2,2}) &= zT_{1,1} - xT_{1,3} \\ \partial(T_{3,1}) &= T_{1,1}T_{1,2} - xT_{2,1} & \partial(T_{3,2}) &= T_{1,2}T_{1,3} + zT_{2,1} - zT_{2,2} & \partial(T_{3,3}) &= T_{1,1}T_{1,3} - yT_{2,2}. \end{aligned}$$

Note that none of the coefficients above are in I , though if we did indeed see any coefficients in I they would be set to zero. We conclude that in $R \otimes_{Q[X]} \Omega_{Q[X]|Q} \cong RX$,

$$\begin{aligned} \partial_{RX}(T_{2,1}) &= zT_1 - yT_2 & \text{and} & & \partial_{RX}(T_{2,2}) &= zT_1 - xT_3 \\ \partial_{RX}(T_{3,1}) &= -xT_{2,1} & \partial_{RX}(T_{3,2}) &= zT_{2,1} - zT_{2,2} & \partial_{RX}(T_{3,3}) &= -yT_{2,2}. \end{aligned}$$

Thus $R \otimes_{Q[X]} \Omega_{Q[X]|Q} \cong RX$ looks like

$$\begin{array}{c} \dots \longrightarrow \begin{array}{c} RT_{3,1} \\ RT_{3,2} \\ RT_{3,3} \end{array} \xrightarrow{\begin{pmatrix} -x & z & 0 \\ 0 & -z & -y \end{pmatrix}} \begin{array}{c} RT_{2,1} \\ \oplus \\ RT_{2,2} \end{array} \xrightarrow{\begin{pmatrix} z & z \\ -y & 0 \\ 0 & -x \end{pmatrix}} \begin{array}{c} RT_{1,1} \\ \oplus \\ RT_{1,2} \\ \oplus \\ RT_{1,3} \end{array} \end{array}$$

Exercise 4.2.16. Let $Q = k[[x, y, z]]$, $I = (x^2, xy)$, and $R = Q/I$. Compute $R \otimes_{Q[X]} \Omega_{Q[X]|Q}$ up to degree 3.

Theorem 4.2.17. Let R be a noetherian local ring, $\widehat{R} \cong Q/I$ a minimal regular presentation, and $Q[X]$ a minimal model for R over Q . There is a canonical isomorphism of graded vector spaces

$$\mathrm{Der}_Q(Q[X], k) \cong \pi^*(R).$$

Proof. Here we view k as the quotient of $Q[X]$ by $(X_{\geq 0})$.

First, note that by degree reasons there are no positive degree derivations from $Q[X]$ to k . Let $\theta: Q[X] \rightarrow k$ be a derivation of degree $-d$, with $d \geq 0$. As noted in [Remark 4.2.13](#), θ is completely determined by the values of the variables. Again by degree reasons, $\theta(X_n) = 0$ for all $n \neq d$, so θ is completely determined by its $Q \cdot X_d \rightarrow k$. Equivalently, θ determines a k -linear map $kX_d \rightarrow k$, which is by definition an element of $\pi^d(R)$. In fact, $\pi^d(R) = (kX_d)^\vee$ by definition, and thus the map that assigns to each θ the corresponding element of $(kX_d)^\vee$ is an isomorphism. $\square$

While not all derivations are chain derivations, the following construction allows us to build a chain derivation out of any derivation.

Construction 4.2.18. Let R be a noetherian local ring and fix a minimal regular presentation $\widehat{R} \cong Q/I$ for R with $(Q, \mathfrak{m}, k)$. Let $A = Q[X]$ be a minimal model for R over Q .

As we saw in [Remark 4.2.13](#), an element $z \in \pi^n(R)$ can be identified with a Q -linear derivation $z: Q[X] \rightarrow k$, which is the same data as a Q -linear map $z: QX_{n-1} \rightarrow k$. Fix a lift $\tilde{z}: QX_{n-1} \rightarrow Q$. From this data, we take the unique derivation

$$\frac{d}{dz}: Q[X] \rightarrow Q[X]$$

that satisfies

$$\frac{d}{dz}(x) = \begin{cases} \tilde{z}(x) & \text{if } x \in X_{n-1} \\ 0 & \text{if } x \in X_\ell \text{ for any } \ell \neq n-1. \end{cases}$$

By Q -linearity and the Leibniz rule, there exists a unique derivation of degree $-(n-1)$ with these properties. However, $\frac{d}{dz}$ might not be a chain derivation. We take

$$\theta_z := \left[\partial, \frac{d}{dz} \right]$$

which is now a map of degree $-n$. We claim θ_z is a chain derivation, and indeed this is the next exercise.

Exercise 4.2.19. Consider the map θ_z defined in [Construction 4.2.18](#).

- 1) Show that $\theta_z: Q[X] \rightarrow Q[X]$ is a chain derivation.
- 2) Show that $\text{im}(\theta_z) \subseteq (X)$.

Remark 4.2.20. In the construction of $\frac{d}{dz}$, suppose that $z = y^\vee$. Then for $x \in X$,

$$\frac{d}{dz}(x) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise.} \end{cases}$$

Exercise 4.2.21. Let $u, z \in X$ with $u \neq z$.

- 1) Show that

$$\frac{d}{dz}(zu) = u \quad \text{and} \quad \frac{d}{dz}(uz) = (-1)^{|z|}u.$$

- 2) Show that if $|z|$ is even, then

$$\frac{d}{dz}(z^2) = 2z.$$

Exercise 4.2.22. Let $Q[X]$ be a minimal model for R over Q . Consider the short exact sequence

$$0 \longrightarrow (X) \longrightarrow Q[X] \longrightarrow k \longrightarrow 0$$

and the corresponding long exact sequence

$$\cdots \longrightarrow \text{Der}_R(Q[X], (X))_n \longrightarrow \text{Der}_R(Q[X], Q[X])_n \longrightarrow \text{Der}_R(Q[X], k)_n \longrightarrow \cdots.$$

- 1) Show that the connecting homomorphism

$$\text{Der}_R(Q[X], k) \longrightarrow \Sigma \text{Der}_R(Q[X], (X))$$

is given by $z \mapsto \theta_z$ as in [Construction 4.2.18](#).

- 2) Show that up to homotopy, θ_z does not depend on the choice of lift $\tilde{z}$ of z .

4.3 Vasconcelos' conjecture

Our goal in this section is to prove a conjecture of Vasconcelos we alluded to in [Section 4.1](#):

Conjecture 4.3.1 (Vasconcelos, 1978 [[Vas78](#)]). *Let $(Q, \mathfrak{m}, k)$ be a regular local ring and consider $R = Q/I$ with $I \subseteq \mathfrak{m}^2$. If I/I^2 is an R -module of finite projective dimension, then I must be generated by a regular sequence.*

To prove this theorem, we need to better understand the role of nilpotent elements in the homotopy Lie algebra $\pi^*(R)$.

Theorem 4.3.2 (Avramov–Halperin, 1987 [[AH87](#)]). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. The following are equivalent:*

- 1) *The ring R is a complete intersection.*
- 2) *Every element in $\pi^2(R)$ is central, that is, $\zeta^2(\pi(R)) = \pi^2(R)$.*
- 3) *Every element in $\pi^2(R)$ is nilpotent.*

Note that if R is a complete intersection, then $\pi^*(R) = \pi^1(R) \oplus \pi^2(R)$ by [Theorem 2.5.10](#), and thus every element in $\pi^2(R)$ is central by degree reasons. Moreover, central elements are always nilpotent. Thus the interesting direction is the fact that if all elements of $\pi^2(R)$ are nilpotent, then R must be a complete intersection. In the next chapter, we will give an alternative proof of this theorem, which can be found in [[BGP24](#), Theorem 3.1].

Later, in 2001, Iyengar proved that free summands of I/I^2 give rise to central elements in $\pi^2(R)$. By [Theorem 4.1.8](#), if I/I^2 is free, then R is a complete intersection, and by [Theorem 4.3.2](#), every element in $\pi^2(R)$ is central. Thus it is natural to ask whether central elements of $\pi^2(R)$ always come from free summands of the conormal module I/I^2 . The answer, surprisingly, is no: in upcoming work of these authors and Mark Waker, we construct counterexamples.

Vasconcelos' [Conjecture 4.3.1](#) was solved by Briggs (the first author of these notes) following a similar strategy that constructs radical elements in $\pi^2(R)$. The key idea is to show that if θ_z factors through a finitely generated R -module of finite projective dimension, then z is nilpotent. We present his proof below.

The proof will use derivations on a minimal model $A = Q[X]$ for R over Q . First, let us give a brief roadmap for what the proof will look like:

- 1) Each derivation $\theta: Q[X] \rightarrow Q[X]$ gives rise to a commutative diagram

$$\begin{array}{ccc} \Omega_{Q[X]|Q} & \xrightarrow{f} & A \\ \downarrow \simeq & & \downarrow \simeq \\ \Omega_{Q[X]|Q} \otimes_Q R & \xrightarrow{f \otimes \text{id}_R} & R \end{array}$$

- 2) Any derivation $\theta: Q[X] \rightarrow Q[X]$ with $\theta(X) \subseteq (X)$ induces a well-defined derivation

$$\theta^{(n)}: k[X_{\geq n}] \rightarrow k[X_{\geq n}]$$

- 3) If θ is a derivation such that the corresponding map $\Omega_{Q[X]|Q} \otimes_Q R \rightarrow R$ factors through an R -module M with $\text{pdim}_R(M) < \infty$, meaning that

$$\begin{array}{ccc} \Omega_{Q[X]|Q} \otimes_Q R & \xrightarrow{f \otimes \text{id}_R} & R \\ & \searrow & \nearrow \\ & M & \end{array}$$

commutes, then there exists n such that the map $\theta^{(n)}$ induced by θ on

$$k[X_{\geq n}] \rightarrow k[X_{\geq n}]$$

has image in $(X_{\geq n})^2$.

- 4) Now take $M = \text{Syz}_1^R(I/I^2)$, and note that

$$\text{pdim}_R(M) < \infty \iff \text{pdim}_R(I/I^2) < \infty.$$

Moreover, we apply the previous steps to $\theta = \theta_z$, for each $z \in \pi^2$. We prove that θ_z factors through $M = \text{Syz}_1^R(I/I^2)$, and therefore for large enough n , the map induced by θ_z on $k[X_{\geq n}]$ has image in $(X_{\geq n})^2$. We then show that this conditions implies that $\text{ad}^n(z) = 0$.

- 5) The final step is applying Avramov–Halperin’s [Theorem 4.3.2](#): the previous step shows every element in $\pi^2(R)$ is nilpotent, which finishes the proof.

Lemma 4.3.3. *Let $Q[X]$ be a minimal model for R over Q . Let $\theta: Q[X] \rightarrow Q[X]$ be a Q -linear derivation of degree $-d$ for some $d \geq 0$. If $\text{im}(\theta) \subseteq (X)$, then θ induces a degree $-d$ map $\tilde{\theta}: \pi^*(R) \rightarrow \pi^*(R)$. Moreover, when $\theta = \theta_\alpha$ for some $\alpha \in \pi^*(R)$, then $\tilde{\theta}_\alpha = -\text{ad}(\alpha)$.*

Proof. Since $\text{im}(\theta) \subseteq (X)$, the Leibniz rule gives us

$$\theta((X)^2) \subseteq (X)^2.$$

Therefore, θ induces a well-defined map

$$kX \cong k[X]/(X)^2 \rightarrow k[X]/(X)^2 \cong kX.$$

Taking the dual, we get a well defined map $\tilde{\theta}: \pi^*(R) \rightarrow \pi^*(R)$, which is the dual of θ .

To show that $\tilde{\theta}_\alpha = -\text{ad}(\alpha)$, the bilinearity of the Lie bracket allows us to reduce to the case when $\alpha = z^\vee$ for some $z \in X$. Thus

$$\theta = \theta_z = \left[\partial, \frac{d}{dz} \right].$$

Note that we want to compute the induced map on $\text{quad}(k[X])$, so we will take the quadratic part of the differential, ∂_q . Note moreover that in order to check that $\tilde{\theta}_z = -\text{ad}(z^\vee)$, it suffices to check that for all $y \in X$,

$$\tilde{\theta}_z(y^\vee) = -[z^\vee, y^\vee].$$

Let $x \in X$. We leave it as an exercise to check that if $x \leq z$, then $\theta_z(x) = 0$, and that moreover $[z^\vee, y^\vee](x) = 0$ for all y . This shows that $\tilde{\theta}_z(y^\vee)(x) = -[z^\vee, y^\vee](x)$ for all $x \leq z$.

Let $x \in X$ such that $x > z$, and write

$$\partial_q(x) = \sum_{\substack{u \leq v \in X \\ |u|+|v|+1=|x|}} C_{x,u,v} uv.$$

Our assumption that $x > z$ leaves z a chance of appearing in the expansion of $\partial_q(x)$. Then $\frac{d}{dz}(x) = 0$, so that

$$\begin{aligned} \theta_z(x) &= -(-1)^{|z|} \frac{d}{dz} \partial_q(x) \\ &= -(-1)^{|z|} \frac{d}{dz} \left(\sum_{\substack{u \leq v \in X \\ |u|+|v|+1=|x|}} C_{x,u,v} uv \right) \\ &= -(-1)^{|z|} \left(\sum_{u < z} (-1)^{|u| \cdot \frac{d}{dz}|x|} C_{x,u,z} u + \frac{d}{dz} (C_{x,z,z} z^2) + \sum_{z < v} C_{x,z,v} v \right) \quad \text{by Exercise 4.2.21} \\ &= -(-1)^{|z|} \left(\sum_{u < z} (-1)^{|u| \cdot |z|} C_{x,u,z} u + \frac{d}{dz} (C_{x,z,z} z^2) + \sum_{z < v} C_{x,z,v} v \right) \end{aligned}$$

Note that by Exercise 4.2.21,

$$\frac{d}{dz} (C_{x,z,z} z^2) = \begin{cases} 0 & \text{if } |z| \text{ is odd} \\ 2C_{x,z,z} z & \text{if } |z| \text{ is even.} \end{cases}$$

We can now write the dual map. If $z < y$, then recall that

$$[z^\vee, y^\vee] = (-1)^{|z|} \sum_{x \in X} c_{x,z,y} x^\vee.$$

Using the formula above we obtain

$$(\theta_z)^\vee(y^\vee) = -(-1)^{|z|} \sum_{|y|+|z|+1=|x|} C_{x,z,y} x^\vee = -[z^\vee, y^\vee].$$

Now we still need to check that $\tilde{\theta}_z(y^\vee) = -[z^\vee, y^\vee]$ when $z = y$ and $z > y$, which we leave as an exercise for the reader. $\square$

Lemma 4.3.4. *Let $\alpha, \beta: Q[X] \rightarrow (X)$ be Q -linear chain derivations which are cohomologous in $\text{Der}_{Q[X]}(Q[X], (X))$, meaning that there exists a derivation $\sigma: Q[X] \rightarrow Q[X]$ with $\text{im}(\sigma) \subseteq (X)$ such that*

$$\alpha - \beta = \partial\sigma - \sigma\partial.$$

Then α and β induce the same map $\tilde{\alpha} = \tilde{\beta}$ on $\pi^(R)$.*

In particular, if $z \in \pi^*(R)$ is such that θ_z is cohomologous in $\text{Der}_Q(Q[X], Q[X])$ to some derivation γ with $\gamma(Q[X]) \subseteq (X)$ and such that for all $n \gg 0$, the map

$$\gamma_n: k[X_{\geq n}] \longrightarrow k[X_{\geq n}]$$

is the zero map, then z is nilpotent.

Proof. Suppose that $\sigma: Q[X] \longrightarrow Q[X]$ is a derivation with $\text{im}(\sigma) \subseteq (X)$ such that

$$\alpha - \beta = \partial\sigma - \sigma\partial.$$

Since $\sigma((X)) \subseteq (X)$, the Leibniz rule gives us that $\sigma((X)^2) \subseteq (X)^2$. Moreover, recall that $\partial((X)) \subseteq (X)^2$, by [Theorem 1.3.15](#). Therefore,

$$(\partial\sigma - \sigma\partial)(X) \subseteq \partial((X)) + \sigma((X)^2) \subseteq (X)^2.$$

Therefore, $\alpha - \beta$ induces the zero map on $kX \cong k[X]/(X)^2$, and thus also on $\pi^*(R)$. We conclude that α and β induce the same map on $\pi^*(R)$.

Now suppose that $z \in \pi^*(R)$ is such that θ_z and γ induce the same map on $\pi^*(R)$, where γ_n is the zero map. By [Lemma 4.3.3](#), the map on $\pi^*(R)$ induced by θ_z (and therefore also γ) is $-\text{ad}(z)$, so that the map induced by γ_n is

$$-\text{ad}(z): \pi^{>n}(R) \longrightarrow \pi^{>n}(R).$$

We conclude that $\text{ad}(z)^n = 0$, and thus z is nilpotent. $\square$

Theorem 4.3.5. *Let $(R, \mathfrak{m})$ be a noetherian local ring and fix a minimal regular presentation for R , $\widehat{R} \cong Q/I$. Let $Q[X]$ be a minimal model for R over Q , and consider the canonical quasiisomorphism $\pi: Q[X] \longrightarrow R$. Fix $z \in \pi^*(R)$. Consider the composition $\overline{\theta_z}$*

$$Q[X] \xrightarrow{\theta_z} Q[X] \xrightarrow{\pi} R.$$

If there exists a finitely generated R -module M of finite projective dimension $\text{pdim}_R(M)$ and a chain derivation $Q[X] \longrightarrow M$ such that $\overline{\theta_z}$ factors through M , that is, if there exists a chain derivation $\varphi: Q[X] \longrightarrow M$ and a map of R -modules $\psi: M \longrightarrow R$ such that

$$\begin{array}{ccc} Q[X] & \xrightarrow{\theta_z} & R \\ & \searrow \varphi & \nearrow \psi \\ & M & \end{array}$$

commutes, then z is a nilpotent element in $\pi^(R)$.*

Remark 4.3.6. By [Exercise 4.2.19](#), $\text{im}(\theta_z) \subseteq (X_{\geq 0})$, and thus the image of $\overline{\theta_z}$ is contained in the maximal ideal $\mathfrak{m}$ of R . Thus we may rewrite the statement as asking for a factorization

$$\begin{array}{ccc} Q[X] & \xrightarrow{\theta_z} & \mathfrak{m} \\ & \searrow \varphi & \nearrow \psi \\ & M & \end{array}$$

Proof of Theorem 4.3.5. By Theorem 3.1.34, there exists a semifree dg $Q[X]$ -module resolution P for M . Since M has finite projective dimension over R , then P can be chosen to be a finitely generated free module over $Q[X]$, by Corollary 3.1.40.

By Theorem 4.2.12, to give a chain derivation $Q[X] \rightarrow M$ is equivalent to giving an R -linear chain map $R \otimes_Q \Omega_{Q[X]|Q} \rightarrow M$. Similarly, the chain derivation θ_z corresponds to a Q -linear chain map $\Omega_{Q[X]|Q} \rightarrow Q[X]$. These give us a commutative diagram

$$\begin{array}{ccccc}
 \Omega_{Q[X]|Q} & \xrightarrow{\theta_z} & (X) & & \\
 \downarrow \simeq & \searrow \beta & \nearrow \alpha & & \downarrow \simeq \\
 & & P & & \\
 R \otimes_{Q[X]} \Omega_{Q[X]|Q} & \longrightarrow & M & \longrightarrow & \mathfrak{m}.
 \end{array}$$

Moreover, since $\Omega_{Q[X]|Q}$ and P are both complexes of free Q -modules, we can construct lifts α and β as above, so that the trapezia commute. Note that we are not claiming the triangle above commutes. However, by the commutativity of the trapezia and the outer square, θ_z and $\alpha\beta$ commute after composing with the quasiisomorphism $(X) \twoheadrightarrow \mathfrak{m}$. By Exercise 1.3.22, θ_z and $\alpha\beta$ are homotopic. By Lemma 4.3.4, θ_z and $\alpha\beta$ induced the same map on $\pi^*(R)$.

Fix a finite generating set S for P over $Q[X]$, and consider $\alpha(S)$. Since S is finite, there exists n such that $\alpha(S) \subseteq (X_{\leq n})$. Moreover, since the derivation $\gamma: Q[X] \rightarrow Q[X]$ corresponding to $\alpha\beta$ is obtained by precomposing $\alpha\beta$ with the universal derivation $d: Q[X] \rightarrow \Omega_{Q[X]|Q}$ (see Theorem 4.2.12), we conclude that

$$\text{im}(\gamma) \subseteq \text{im}(\alpha) \subseteq (X_{\leq n}).$$

Therefore, the map $k[X_{\leq n}] \rightarrow k[X_{\leq n}]$ induced by γ is the zero map. Thus the map on $\pi^{>n}(R) \rightarrow \pi^{>n}(R)$ induced by γ is also the zero map. But γ and θ_z induce the same map on $\pi^*(R)$, and that map is $-\text{ad}(z)$. We conclude that

$$\text{ad}(z): \pi^{>n}(R) \rightarrow \pi^{>n}(R)$$

is the zero map. Therefore, z must be a nilpotent element, as $\text{ad}^p(z)$ lands in higher and higher degrees as p grows. $\square$

Exercise 4.3.7. Consider the diagram of R -module homomorphisms

$$\begin{array}{ccccc}
 F & & G & & H \\
 \downarrow f & & \downarrow g & & \downarrow h \\
 A & \xrightarrow{a} & B & \xrightarrow{b} & C
 \end{array}$$

where the vertical maps are surjective, the top modules $(F, G, \text{ and } H)$ are free R -modules, and the bottom row is exact. Show that there exists lifts α and β of a and b making the

following diagram

$$\begin{array}{ccccc}
\ker(f) & \xrightarrow{\alpha} & \ker(g) & \xrightarrow{\beta} & \ker(h) \\
\downarrow & & \downarrow & & \downarrow \\
F & \xrightarrow{\alpha} & G & \xrightarrow{\beta} & H \\
\downarrow f & & \downarrow g & & \downarrow h \\
A & \xrightarrow{a} & B & \xrightarrow{b} & C
\end{array}$$

commute.

Remark 4.3.8. Let R be a noetherian local ring and fix a minimal regular presentation for R , $\widehat{R} \cong Q/I$. Note that

$$\pi^2(R) \cong \operatorname{Hom}_k(kX_1, k) \cong \operatorname{Hom}_Q(QX_1, k) \cong \operatorname{Hom}_Q(I, k) \cong \operatorname{Hom}_R(I/I^2, k).$$

Thus given an element $z \in \pi^2(R)$, we can identify it with an R -module map $z: I/I^2 \rightarrow k$.

Exercise 4.3.9. Let M be a finitely generated module over a noetherian local ring R . Show that if $\operatorname{pdim}_R(M) < \infty$, then $\operatorname{pdim}_R(\operatorname{Syz}^n(M)) < \infty$ for all $n \geq 1$.

Theorem 4.3.10. Let $(R, \mathfrak{m})$ be a noetherian local ring and fix a minimal regular presentation for R , $\widehat{R} \cong Q/I$. Let $z \in \pi^2(R)$, and consider the corresponding map of k -vector spaces $\tilde{z}: I/I^2 \rightarrow k$. If $\tilde{z}$ factors through a finitely generated R -module N of finite projective dimension, then z is nilpotent.

Proof. Let F be a free R -module surjecting onto N minimally, and consider the corresponding short exact sequence

$$0 \longrightarrow M \longrightarrow F \longrightarrow N \longrightarrow 0.$$

The minimality assumption gives us $M \cong \operatorname{Syz}^1(M)$, so $\operatorname{pdim}_R(M) < \infty$ by [Exercise 4.3.9](#).

Fix a minimal model $Q[X]$ for R over Q . Since QX_1 surjects onto I , we get an exact sequence

$$QX_2 \oplus QX_1^2 \longrightarrow QX_1 \longrightarrow I \longrightarrow 0.$$

Note that $\partial(X_1^2) \subseteq IX_1$. Tensoring with R , we obtain an exact sequence

$$RX_2 \longrightarrow RX_1 \longrightarrow I/I^2 \longrightarrow 0.$$

Finally, consider the short exact sequence

$$0 \longrightarrow \mathfrak{m} \longrightarrow R \longrightarrow k \longrightarrow 0.$$

By [Exercise 4.3.7](#), we get a commutative diagram

$$\begin{array}{ccccc}
RX_3 & & 0 & & 0 \\
\downarrow & & \downarrow & & \downarrow \\
RX_2 & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & \mathfrak{m} \\
\downarrow & & \downarrow & & \downarrow \\
RX_1 & \xrightarrow{\alpha} & F & \xrightarrow{\beta} & R \\
\downarrow & & \downarrow & & \downarrow \\
I/I^2 & \xrightarrow{a} & N & \xrightarrow{b} & k \\
\downarrow & & \downarrow & & \downarrow \\
0 & & 0 & & 0
\end{array}$$

where we replaced $\ker(RX_1 \rightarrow I/I^2)$ by its submodule RX_2 . We can extend $\alpha: RX_1 \rightarrow M$ trivially into a map $\alpha: RX \rightarrow M$. We claim this is in fact a map of complexes, and to show that we need only to prove that

$$\begin{array}{ccc}
RX_3 & \longrightarrow & 0 \\
\downarrow & & \downarrow \\
RX_2 & \xrightarrow{\alpha} & M
\end{array}$$

commutes.

To show that, note that the map $RX_2 \rightarrow M$ is the restriction to RX_2 of the map

$$\ker(RX_1 \rightarrow I/I^2) \rightarrow \ker(F \rightarrow N) = M$$

induced by $RX_1 \rightarrow F$. Thus

$$RX_3 \xrightarrow{\partial} RX_2 \rightarrow M = RX_3 \xrightarrow{\partial} RX_2 \xrightarrow{\partial} \ker(RX_1 \rightarrow I/I^2) \rightarrow M$$

is the zero map, and we have constructed a map of complexes $RX \rightarrow M$.

By [Theorem 4.2.12](#), we have constructed a chain map

$$R \otimes_{Q[X]} \Omega_{Q[X]|Q} \rightarrow M,$$

which is equivalent to giving a chain derivation $\gamma: Q[X] \rightarrow M$.

Now back to the original diagram, the composition $RX_1 \xrightarrow{\alpha} F \xrightarrow{\beta} R \twoheadrightarrow k$ corresponds to the element $z \in \pi^2(R)$ we started with, while the lift $RX_2 \xrightarrow{\alpha} M \xrightarrow{\beta} \mathfrak{m}$ corresponds to $\widetilde{\theta}_z \otimes_Q R$, where $\widetilde{\theta}_z$ is the map on $\pi^*(R)$ induced by z , as in [Lemma 4.3.3](#).

Therefore, we have shown that $\widetilde{\theta}_z \otimes_Q R$ factors through M , an R -module of finite projective dimension. By [Theorem 4.3.5](#), z is a nilpotent element. $\square$

We are now ready to prove Vasconcelos' [Conjecture 4.3.1](#):

Theorem 4.3.11 (Briggs, 2022 [[Bri22](#)]). *Let R be a noetherian local ring, and fix a minimal regular presentation $\widehat{R} \cong Q/I$ for R . If I/I^2 has finite projective dimension, then I is generated by a regular sequence.*

Proof. Since I/I^2 has finite projective dimension, then any R -module map $I/I^2 \rightarrow k$ trivially factors through a module of finite projective dimension (I/I^2 itself). By [Theorem 4.3.10](#), every element in $\pi^2(R)$ is nilpotent, so by [Theorem 4.3.2](#) we conclude that I is generated by a regular sequence. $\square$

Chapter 5

Cohomological support varieties

Fix a noetherian local ring R . We will associate to each finitely generated R -module a variety, called the cohomological support variety of M , that encodes homological information about M . Cohomological support varieties were first defined and studied by Avramov in the 1980s, inspired by work of Quillen [Qui71]. The definition first appeared in print in full generality in work of Jorgensen [Jor02], while the general theory was developed by Pollitz [Pol19, Pol21].

In this chapter, we will introduce support varieties and use them to characterize complete intersections. We will finish the chapter with a connection to the homotopy Lie algebra, and prove Avramov–Halperin’s characterization of complete intersections via the nilpotent elements of $\pi^2(R)$.

5.1 Resolutions over complete intersections

Let us go back to thinking about free resolutions, with no requirement of a dg algebra structure. While resolutions of modules over complete intersections can be infinite, we can nevertheless construct them using only a finite set of data.

Definition 5.1.1 (Systems of higher homotopies). Let $(Q, \mathfrak{m}, k)$ be a noetherian local ring and let F be a free resolution (not necessarily finite) for the finitely generated Q -module M . Let $f \in \text{ann}_Q(M)$. A **system of higher homotopies** for f on F , sometimes shortened to **sohh**, is a collection of Q -linear maps σ_i of degree $2i - 1$

$$\sigma_i : F_\bullet \longrightarrow F_{\bullet+2i-1}$$

for each $i \geq 0$ satisfying the following conditions:

$$\sigma_0 = \partial_F \qquad \sigma_1 \sigma_0 + \sigma_0 \sigma_1 = f \cdot \text{id}_F \qquad \sum_{i=0}^n \sigma_i \sigma_{n-i} = 0 \text{ for all } n \geq 2.$$

For example, here are depictions of σ_0 and σ_1

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & F_3 & \xrightarrow{\sigma_0} & F_2 & \xrightarrow{\sigma_0} & F_1 & \xrightarrow{\sigma_0} & F_0 & \longrightarrow & 0 \\
& & & & \swarrow \sigma_1 & & \swarrow \sigma_1 & & \swarrow \sigma_1 & & \\
\cdots & \longrightarrow & F_3 & \xrightarrow{\sigma_0} & F_2 & \xrightarrow{\sigma_0} & F_1 & \xrightarrow{\sigma_0} & F_0 & \longrightarrow & 0
\end{array}$$

and also σ_2 :

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & F_4 & \longrightarrow & F_3 & \longrightarrow & F_2 & \longrightarrow & F_1 & \longrightarrow & F_0 \\
& & & & & & \searrow \sigma_2 & & \searrow \sigma_2 & & \\
\cdots & \longrightarrow & F_4 & \xleftarrow{\sigma_2} & F_3 & \xleftarrow{\sigma_2} & F_2 & \longrightarrow & F_1 & \longrightarrow & F_0
\end{array}$$

Remark 5.1.2. Let $R = Q/(f)$. The condition $f \in \text{ann}_Q(M)$ is equivalent to saying that M is an R -module.

Remark 5.1.3. The map $f \cdot \text{id}_F$ is a lift of the map on M given by multiplication by f . If $f \in \text{ann}_Q(M)$, then multiplication by f is zero on M . Thus $f \cdot \text{id}_F$ is a lift of the zero map $M \rightarrow M$ to F . Note that the zero map on F is also a lift of the zero map of M to F .

Since any two lifts to F of the same map on M must be homotopic, we conclude that $f \cdot \text{id}_F$ must be nullhomotopic. The condition

$$\sigma_1 \sigma_0 + \sigma_0 \sigma_1 = f \cdot \text{id}_F$$

says precisely that σ_1 is a nullhomotopy for $f \cdot \text{id}_F$. In particular, we can always choose such a map σ_1 . More generally, for all $n \geq 2$,

$$\sum_{i=0}^n \sigma_i \sigma_{n-i} = 0 \iff \sigma_n \partial + \partial \sigma_n = - \sum_{i=1}^{n-1} \sigma_i \sigma_{n-i} =: \tau_n$$

says that σ_n is a nullhomotopy for τ_n .

One can then show that a system of higher homotopies always exists.

Lemma 5.1.4 (Systems of higher homotopies exist). *Let $(R, \mathfrak{m}, k)$ be a noetherian local ring. Let M be an R -module and $f \in \text{ann}(M)$, and let F be a free resolution for M over R , not necessarily finite. Then there exists a system of higher homotopies for f on F .*

Proof. We will construct the σ_n inductively. By definition, σ_0 is the differential. Moreover, we saw in Remark 5.1.3 that we can always find σ_1 . So we need only to construct σ_n for $n \geq 2$. Given $\sigma_0, \dots, \sigma_{n-1}$ satisfying our desired properties, note that for each i , we have

$$\partial \sigma_i = \tau_i - \sigma_i \partial.$$

Therefore,

$$\partial \sigma_i \sigma_{n-i} = \tau_i \sigma_{n-i} - \sigma_i \partial \sigma_{n-i} = \tau_i \sigma_{n-i} - \sigma_i (\tau_{n-i} - \sigma_{n-i} \partial) = \tau_i \sigma_{n-i} - \sigma_i \tau_{n-i} + \sigma_i \sigma_{n-i} \partial$$

and

$$\partial\tau_n = -\sum_{i=1}^{n-1} \partial\sigma_i\sigma_{n-i} = \sum_{i=1}^{n-1} (\sigma_i\tau_{n-i} - \tau_i\sigma_{n-i} - \sigma_i\sigma_{n-i}\partial) = \sum_{i=1}^{n-1} (\sigma_i\tau_{n-i} - \tau_i\sigma_{n-i}) + \tau_n\partial.$$

Now note that

$$\sum_{i=1}^{n-1} (\sigma_i\tau_{n-i} - \tau_i\sigma_{n-i}) = -\sum_{i=1}^{n-1} \sum_{j=1}^{n-i-1} \sigma_i\sigma_j\sigma_{n-i-j} + \sum_{i=1}^{n-1} \sum_{j=1}^{i-1} \sigma_j\sigma_{i-j}\sigma_{n-i}$$

and both sums above can be rewritten as

$$\sum_{i=1}^{n-1} \sum_{j=1}^{n-i-1} \sigma_i\sigma_j\sigma_{n-i-j} = \sum_{\substack{s+t+u=n \\ s,t,u \geq 1}} \sigma_s\sigma_t\sigma_u = \sum_{i=1}^{n-1} \sum_{j=1}^{i-1} \sigma_j\sigma_{i-j}\sigma_{n-i}.$$

Therefore, $\partial\tau_n = \tau_n\partial$.

Note that τ_n is a map of degree $2n - 2 \geq 2$ on F , and that $2n - 2$ is even. The condition $\partial\tau_n = \tau_n\partial$ says that τ_n is a cycle in $\text{Hom}_Q(F, F)$. On the other hand, the quasi-isomorphism $F \rightarrow M$ induces a quasiisomorphism $\text{Hom}_Q(F, F) \rightarrow \text{Hom}_Q(F, M)$, so τ_n corresponds to a cycle of degree $2n - 2$ in $\text{Hom}_Q(F, M)$. But $F_{<0} = 0$, so $\text{Hom}_Q(F, M)$ is concentrated in negative degrees, and thus τ_n must be a boundary in $\text{Hom}_Q(F, F)$. Thus there is a nullhomotopy σ_n of degree $2n - 1$ for τ_n . $\square$

In order to construct a system of higher homotopies for f on F , we set $\sigma_0 = \partial$ and construct σ_1, σ_2 , and so on inductively. In order to do this, we start by looking at the image of σ_1 applied to F_0 , then F_1 , and so on, and find each of these maps inductively. Note, however, that there are many choices along the way, so while this proves that a system of higher homotopies exists, it is not unique.

Example 5.1.5. Let $Q = k[[x, y]]$ and $M = Q/(x^2, xy)$, and note that x^2 annihilates M . Let us construct a system of higher homotopies for x^2 on the minimal free resolution F for M over Q , which is given in the diagram below. By definition, σ_0 is the differential on F , so there is nothing to do at this stage. Next, we construct σ_1 , which has degree 1. Then σ_1 has the following shape, where a, b, c, d are elements of Q :

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q & \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x^2 & xy \end{bmatrix}} & Q \longrightarrow 0 \\ & & \downarrow \cdot x^2 & \swarrow \begin{bmatrix} a & b \end{bmatrix} & \downarrow \cdot x^2 & \swarrow \begin{bmatrix} c & d \end{bmatrix} & \downarrow \cdot x^2 \\ 0 & \longrightarrow & Q & \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x^2 & xy \end{bmatrix}} & Q \longrightarrow 0 \end{array}$$

Consider $1 \in F_0 = Q$, and note that

$$x^2 = \partial\sigma_1(1) = \begin{bmatrix} x^2 & xy \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = cx^2 + dxy.$$

Therefore, we must have $cx^2 + dxy = x^2$, so we can take $c = 1$ and $d = 0$. Note, however, that there are other possible choices: for example, we could have instead taken $c = 1 + y$ and $d = -x$. We can now update our diagram to

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q & \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x^2 & xy \end{bmatrix}} & Q & \longrightarrow & 0 \\ & & \downarrow \cdot x^2 & & \downarrow \cdot x^2 & & \downarrow \cdot x^2 & & \\ 0 & \longrightarrow & Q & \xrightarrow{\begin{bmatrix} a & b \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} & Q & \longrightarrow & 0 \\ & & \uparrow \cdot x^2 & & \uparrow \cdot x^2 & & \uparrow \cdot x^2 & & \end{array}$$

Now looking at the image of $(1, 0), (0, 1) \in F_1 = Q^2$ under σ_1 , we have

$$\begin{aligned} \begin{bmatrix} x^2 \\ 0 \end{bmatrix} &= x^2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ x^2 \end{bmatrix} &= x^2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= (\sigma_1\partial + \partial\sigma_1) \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) & &= (\sigma_1\partial + \partial\sigma_1) \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\ &= \begin{bmatrix} x^2 \\ 0 \end{bmatrix} + \begin{bmatrix} ay \\ -ax \end{bmatrix} & &= \begin{bmatrix} xy \\ 0 \end{bmatrix} + \begin{bmatrix} by \\ -bx \end{bmatrix} \\ &= \begin{bmatrix} x^2 + ay \\ -ax \end{bmatrix}. & &= \begin{bmatrix} by + xy \\ -bx \end{bmatrix}. \end{aligned}$$

Therefore, we can take $a = 0$ and $b = -x$, giving us the following σ_1 :

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q & \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x^2 & xy \end{bmatrix}} & Q & \longrightarrow & 0 \\ & & & & \swarrow \begin{bmatrix} 0 & -x \end{bmatrix} & & \swarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} & & \\ 0 & \longrightarrow & Q & \longrightarrow & Q^2 & \longrightarrow & Q & \longrightarrow & 0 \end{array}$$

The next step would be to find σ_2 . But σ_2 would be a map of degree 3, while F only has length 2, so all the higher homotopies vanish.

Exercise 5.1.6. Find a system of higher homotopies for multiplication by xy on the minimal free resolution F in [Example 5.1.5](#).

Remark 5.1.7. Let $R = Q/I$ and let M be an R -module. Fix $f \in I$ and consider the koszul complex

$$E = \text{Kos}^Q(f) = Q[e \mid \partial(e) = f].$$

Suppose that F is a free resolution for M over Q , and that F has a dg E -module structure.

As we discussed in [Remark 3.1.7](#), if we set $\sigma := e \cdot (-) : F \rightarrow F$ to be given by multiplication by e , then σ is a nullhomotopy for $f \cdot \text{id}_F$, and $\sigma^2 = 0$. Thus

$$\sigma_0 = \partial_F \quad \sigma_1 = \sigma \quad \sigma_n = 0 \text{ for } n \geq 2$$

determines a system of higher homotopies for f on F .

Conversely, any system of higher homotopies $\{\sigma_i\}$ for f on F with no honest higher homotopies, meaning $\sigma_n = 0$ for all $n \geq 2$, corresponds to a dg E -module structure on F , where we set

$$e \cdot x := \sigma_1(x) \quad \text{for all } x \in F.$$

The key point is that the definition of a system of higher homotopies together with the fact that $\sigma_2 = 0$ forces

$$\sigma_1^2 = -(\sigma_2 \partial + \partial \sigma_2) = 0.$$

We can use systems of higher homotopies to compute free resolutions for modules over complete intersections.

Construction 5.1.8 (Shamash Construction). Let $(Q, \mathfrak{m}, k)$ be a noetherian local ring and consider $R = Q/(f)$ for some $f \in \mathfrak{m}$. Let M be an R -module, and F be a free resolution for M over Q . Let $\{\sigma_i\}$ be a system of higher homotopies for f on F . Let $\mathcal{S} = Q[\chi]$ be a graded polynomial ring over Q , where χ has degree -2 , which is known as the **ring of cohomological operators**, and let Γ be the graded dual of $\mathcal{S}$, which is a divided power algebra with basis $\{t^{(a)}\}$ for $a \geq 0$, with $\chi_i^{(0)} = 1$. By convention, we write $\chi_i^{(j)} = 0$ whenever $j < 0$. Then Γ is an $\mathcal{S}$ -module via

$$\chi \cdot t^{(a)} := \begin{cases} t^{(a-1)} & \text{if } a \geq 1 \\ 0 & \text{otherwise.} \end{cases}.$$

$\text{Sh}(F, \sigma)$ The **Shamash construction** or **Shamash resolution** on F and $\sigma = \{\sigma_i\}$ is the complex of R -modules $\text{Sh}(F, \sigma) = R \otimes_Q \Gamma \otimes_Q F$ given by

$$\cdots \longrightarrow \bigoplus_{i \geq 0} R t^{(i)} \otimes_Q F_{n-2i} \xrightarrow{\partial} \bigoplus_{i \geq 0} R t^{(i)} \otimes_Q F_{n-1-2i}$$

$n \qquad \qquad \qquad n-1$

with differential

$$\partial(t^{(i)} \otimes u) = \sum_{j=0}^i t^{(i-j)} \otimes \sigma_j(u).$$

Theorem 5.1.9 (Shamash, 1969 [Sha69]). *Let $(Q, \mathfrak{m}, k)$ be a noetherian local ring and consider $R = Q/(f)$ for some regular element $f \in \mathfrak{m}$. Let M be an R -module, and F be a free resolution for M over Q . Let $\{\sigma_i\}$ be a system of higher homotopies for f on F . Then $\text{Sh}(F, \sigma)$ is a free resolution for M over R .*

We will primarily use the Shamash construction when Q is a regular local ring, so that any nonzero nonunit f is a regular element, and the construction applies to give a free resolution over the hypersurface $Q/(f)$.

Remark 5.1.10. Let Q be a regular local ring of dimension d and take any nonzero element $f \neq 0$. Let M be a finitely generated R -module, where $R = Q/(f)$. Since Q is regular, $\text{pdim}_Q(M) \leq d$. Let F be the minimal free resolution of M over Q :

$$0 \longrightarrow F_d \longrightarrow \cdots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow 0.$$

Let us apply the Shamash construction to F . Note that in even degrees, we only use F_i with i even, and in odd degrees with we will only use F_i with i odd. The Shamash resolution $\text{Sh}(F, \sigma)$ starts with

$$\begin{array}{ccccccc} \cdots & \longrightarrow & \begin{array}{c} Rt^{(0)} \otimes_Q F_3 \\ \oplus \\ Rt^{(1)} \otimes_Q F_1 \end{array} & \longrightarrow & \begin{array}{c} Rt^{(0)} \otimes_Q F_2 \\ \oplus \\ Rt^{(1)} \otimes_Q F_0 \end{array} & \longrightarrow & Rt^{(0)} \otimes_Q F_1 \xrightarrow{\partial} Rt^{(0)} \otimes_Q F_0 \longrightarrow 0 \\ & & 3 & & 2 & & 1 & & 0 \end{array}$$

But in high enough degrees (meaning, above degree d), the resolution starts repeating itself. Let us see this in the case when $d = 2a$ is even:

$$\begin{array}{ccccccc} & & Rt^{(b+1)} \otimes_Q F_{2a} & & Rt^{(b+1)} \otimes_Q F_{2a-1} & & Rt^{(b)} \otimes_Q F_{2a} \\ & & \oplus & & \oplus & & \oplus \\ \cdots & \longrightarrow & Rt^{(b+2)} \otimes_Q F_{2a-2} & \longrightarrow & Rt^{(b+2)} \otimes_Q F_{2a-3} & \longrightarrow & Rt^{(b+1)} \otimes_Q F_{2a-2} \longrightarrow \cdots \\ & & \oplus & & \oplus & & \oplus \\ & & \vdots & & \vdots & & \vdots \\ & & \oplus & & \oplus & & \oplus \\ & & Rt^{(b+1+a)} \otimes_Q F_0 & & Rt^{(b+a)} \otimes_Q F_1 & & Rt^{(b+a)} \otimes_Q F_0 \\ & & 2a+2b+2 & & 2a+2b+1 & & 2a+2b \end{array}$$

Note that for any $b \geq 1$ and $0 \leq i \leq a$, and any $u \in F_{2a-1-i}$,

$$\partial(t^{(b+i)} \otimes u) = \sum_{j=0}^{b+i} t^{(b+i-j)} \otimes \sigma_j(u).$$

Similarly, for any $b \geq 1$ and $0 \leq i \leq a$, and any $u \in F_{2a-i}$,

$$\partial(t^{(b+i)} \otimes u) = \sum_{j=0}^{b+i} t^{(b+i-j)} \otimes \sigma_j(u).$$

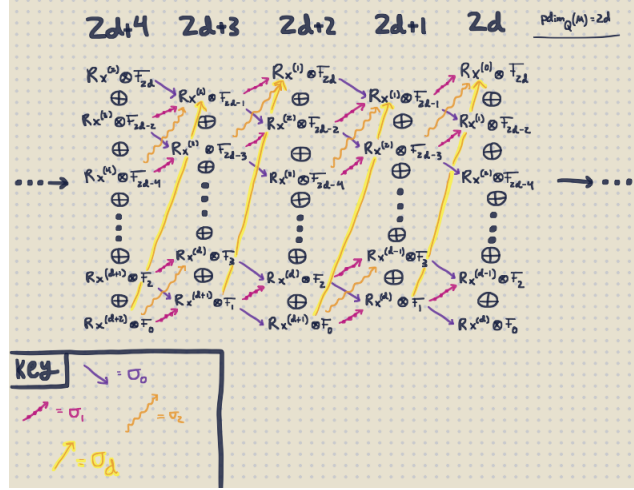


Figure 5.1: The differential in $\text{Sh}(F, \sigma)_{\geq 0}$. Diagram drawn by Fletcher Fletcher.

Therefore, the resolution becomes 2-periodic above degree d . A very similar calculation shows that the same holds when $d = 2a + 1$, though we leave the details as an exercise. In fact, by [Exercise 4.1.4](#), when $\ell \gg 0$ we have

$$\text{rank}(\text{Sh}(F, \sigma)_\ell) = \sum_{i \geq 0} \text{rank}(F_{2i}) = \sum_{i \geq 0} \text{rank}(F_{2i+1}),$$

so the ranks of the free modules in $\text{Sh}(F, \sigma)$ eventually become constant.

Note that this is *not* necessarily a minimal free resolution for M over R , but the minimal free resolution for M does split off as a free summand of $\text{Sh}(F, \sigma)$. Since $\text{Sh}(F, \sigma)$ is eventually 2-periodic, then so is the minimal resolution: one can see more precisely that there is a repeating splitting in all large even degrees, and another repeating splitting in all large odd degrees. The key point is that a splitting corresponds to contracting away copies of the trivial complex

$$\cdots 0 \longrightarrow R \xrightarrow{1} R \longrightarrow 0 \longrightarrow \cdots$$

which can be seen directly from the maps in $\text{Sh}(F, \sigma)$ with no knowledge of the minimal free resolution for M over R .

We have thus proved the following theorem:

Theorem 5.1.11 (Eisenbud, 1980 [[Eis80](#)]). *Let $(Q, \mathfrak{m})$ be a regular local ring, $f \in \mathfrak{m} \setminus \mathfrak{m}^2$, and consider the hypersurface ring $R = Q/(f)$. Every minimal free resolution of a finitely generated module over R is eventually 2-periodic: for all finitely generated R -module M , there exist an integer n and maps A and B such that the tail of the resolution looks like*

$$\cdots \longrightarrow R^n \xrightarrow{A} R^n \xrightarrow{B} R^n \xrightarrow{A} \cdots.$$

This is what leads to the concept of a *matrix factorization*.

We can use this idea to determine finite projective dimension over a hypersurface.

Definition 5.1.12. A $\mathbb{Z}/2$ complex F over R consists of the following data: two R -modules E and O , with E sitting in even degree (or 0 modulo 2) and O sitting in odd degree (or 1 modulo 2), and two maps

$$O \xrightarrow{f} E \quad \quad E \xrightarrow{g} O$$

such that $fg = 0$ and $gf = 0$. The **homology** of F is given by

$$H_{\text{even}}(F) = \ker(g)/\text{im}(f) \quad \text{and} \quad H_{\text{even}}(F) = \ker(f)/\text{im}(g).$$

Remark 5.1.13. Any 2-periodic complex can be reinterpreted as a $\mathbb{Z}/2$ complex. The advantage of thinking of $\mathbb{Z}/2$ complexes is only evident when dealing with graded modules and graded maps.

Lemma 5.1.14. Let Q be a regular local ring and $R = Q/(f)$ with $f \neq 0$, and let M be a finitely generated R -module. Let F be a free resolution of M over Q , and $\{\sigma_i\}$ be a system of higher homotopies for f on F . Consider the $\mathbb{Z}/2$ complexes

$$\mathcal{P} := \cdots \longrightarrow \bigoplus_{i \geq 0} F_{2i} \otimes_Q k \xrightarrow{\partial} \bigoplus_{i \geq 0} F_{2i+1} \otimes_Q k \xrightarrow{\partial} \bigoplus_{i \geq 0} F_{2i} \otimes_Q k \longrightarrow \cdots$$

with

$$\partial(u \otimes 1) = \sum_j \sigma_j(u) \otimes 1$$

and

$$\mathcal{P}^* := \cdots \longrightarrow \bigoplus_{i \geq 0} \text{Hom}_Q(F_{2i}, k) \xrightarrow{\partial} \bigoplus_{i \geq 0} \text{Hom}_Q(F_{2i+1}, k) \xrightarrow{\partial} \bigoplus_{i \geq 0} \text{Hom}_Q(F_{2i}, k) \longrightarrow \cdots$$

with

$$\partial = \text{Hom}_Q \left(\sum_j \sigma_j, k \right).$$

Then $\text{pdim}_R(M) < \infty \iff P \text{ is exact} \iff P^* \text{ is exact}.$

Proof. By [Exercise 0.2.6](#),

$$\beta_i^R(M) = \dim_k \text{Tor}_i^R(M, k),$$

so M has finite projective dimension if and only if $\text{Tor}_i^R(M, k) = 0$ for $i \gg 0$. Let F be a finite resolution for M over Q , and σ a system of higher homotopies for f on F . Then

$$\text{Tor}_i^R(M, k) = H_i(\text{Sh}(M, \sigma) \otimes_R k).$$

We conclude that M has finite projective dimension if and only if $\text{Sh}(M, \sigma) \otimes_R k$ is eventually zero. Thus we need only to consider the tail of the complex $\text{Sh}(F, \sigma) \otimes_R k$.

We described the tail of $\text{Sh}(F, \sigma)$ in [Remark 5.1.10](#), and it follows immediately from that description that the tail of $\text{Sh}(F, \sigma) \otimes_R k$ matches the complex $\mathcal{P}$ above. Finally, we conclude that M has finite projective dimension if and only if P is exact.

Similarly, [Exercise 0.2.6](#) also says

$$\beta_i^R(M) = \dim_k \text{Ext}_R^i(M, k),$$

and the tail of $\text{Hom}_Q(\text{Sh}(F, \sigma), k)$ is the complex P^* above. Alternatively, note that $\mathcal{P}^*$ is the k -linear dual of $\mathcal{P}$, so as complexes of k -vector spaces, $\mathcal{P}$ is exact if and only if $\mathcal{P}^*$ is exact. $\square$

Remark 5.1.15. Fix any R -module N . Since the resolution of any finitely generated module M over a hypersurface is eventually 2-periodic, for large enough values of i there are only two modules $\text{Tor}_i(M, N)$ and $\text{Ext}_R^i(M, N)$: the even and odd ones. Indeed,

$$\text{Ext}_R^i(M, N) \cong \text{Ext}_R^{i+2}(M, N) \quad \text{and} \quad \text{Tor}_i^R(M, N) \cong \text{Tor}_{i+2}^R(M, N) \quad \text{for all } i \gg 0.$$

The proof of [Lemma 5.1.14](#) says that the homology of the complex $\mathcal{P}$ computes the stable values for Tor :

$$H_{\text{even}}(\mathcal{P}) = \text{Tor}_{2i}^R(M, k) \text{ for } i \gg 0 \quad \text{and} \quad H_{\text{odd}}(\mathcal{P}) = \text{Tor}_{2i+1}^R(M, k) \text{ for } i \gg 0.$$

Similarly, the complex $\mathcal{P}^*$ computes the stable values for Ext :

$$H_{\text{even}}(\mathcal{P}^*) = \text{Ext}_R^{2i}(M, k) \text{ for } i \gg 0 \quad \text{and} \quad H_{\text{odd}}(\mathcal{P}^*) = \text{Ext}_R^{2i+1}(M, k) \text{ for } i \gg 0.$$

Remark 5.1.16. Since Q is regular, we can always choose a finite resolution F or M when constructing the complex $\mathcal{P}$ in [Lemma 5.1.14](#). Suppose that F has length d . After fixing basis for $F_1, \dots, F_d$, our complex $\mathcal{P}$ looks like

$$\cdots \longrightarrow \mathcal{P}_{\text{even}} \xrightarrow{A} \mathcal{P}_{\text{odd}} \xrightarrow{B} \mathcal{P}_{\text{even}} \xrightarrow{A} \cdots$$

Since $\mathcal{P}$ is a complex,

$$\text{rank } A \leq \text{rank}(\ker B) \quad \text{and} \quad \text{rank } B \leq \text{rank}(\ker A).$$

Moreover, $\mathcal{P}$ is exact if and only if equality holds in both inequalities above. Note however that by the Rank–Nullity Theorem, we only need to check equality for one: if $\mathcal{P}_{\text{odd}}$ has rank N , then

$$\text{rank } A = \text{rank}(\ker B) \iff N - \text{rank}(\ker A) = N - \text{rank } B \iff \text{rank } B = \text{rank}(\ker A).$$

These ideas were extended by Eisenbud to any complete intersection.

Definition 5.1.17 (Systems of higher homotopies: general definition). Let $(R, \mathfrak{m}, k)$ be a noetherian local ring and let F be a free resolution for the finitely generated R -module M , not necessarily finite. Let $\underline{f} = f_1, \dots, f_n \in \text{ann}_R(M)$. Given an n -tuple $\omega = (\omega_1, \dots, \omega_n) \in \mathbb{Z}^n$, set $|\omega| := \omega_1 + \dots + \omega_n$. A **system of higher homotopies** for $\underline{f}$ on F is a collection of R -linear maps

$$\sigma_\omega : F_\bullet \longrightarrow F_{\bullet+2|\omega|-1}$$

where $\omega = (\omega_1, \dots, \omega_n) \in \mathbb{Z}^n$ with $\omega_i \geq 0$ for all i and such that σ_ω has degree $2|\omega| - 1$, satisfying the following conditions:

$$\sigma_{\mathbf{0}} = \partial_F \quad \sigma_{\mathbf{e}_i} \sigma_{\mathbf{0}} + \sigma_{\mathbf{0}} \sigma_{\mathbf{e}_i} = f_i \cdot \text{id}_F \quad \sum_{u+v=\omega} \sigma_u \sigma_v = 0 \quad \text{for all } |\omega| \geq 2.$$

Here $\mathbf{0}$ denotes the n -vector with all entries 0, and $\mathbf{e}_i$ the vector with i th coordinate 1 and all other coordinates 0.

Exercise 5.1.18. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring and let F be a free resolution for the finitely generated R -module M , not necessarily finite. Let $f_1, \dots, f_n \in \text{ann}_R(M)$. Show that there exists a system of higher homotopies for $f_1, \dots, f_n$ on F .

If $\{\sigma_\omega\}$ is a system of higher homotopies for $\underline{f}$ on F , then the collection $\{\sigma_{n\mathbf{e}_i}\}$ is a system of higher homotopies for f_i on F . In fact, one can do more:

Exercise 5.1.19. Let $(R, \mathfrak{m}, k)$ be a noetherian local ring and let F be a free resolution for the finitely generated R -module M , not necessarily finite. Let $\{\sigma_\omega\}$ is a system of higher homotopies for $\underline{f} = f_1, \dots, f_n$ on F . Show that for all $a_1, \dots, a_n \in R$ not all zero, the maps

$$\sigma_i := \sum_{|\omega|=i} a_1^{\omega_1} \dots a_n^{\omega_n} \sigma_\omega$$

form a system of higher homotopies for $a_1 f_1 + \dots + a_n f_n$ on F .

Remark 5.1.20. Consider the system of higher homotopies for f_a from [Exercise 5.1.19](#). The differential of the 2-periodic complex $\mathcal{P}$ from [Lemma 5.1.14](#) becomes

$$\partial(u \otimes 1) = \sum_{\omega} \sigma_\omega(u) \otimes a_1^{\omega_1} \dots a_n^{\omega_n}.$$

Construction 5.1.21 (Ring of cohomological operators). Let $(Q, \mathfrak{m}, k)$ be a noetherian local ring and consider $R = Q/(f_1, \dots, f_n)$ for some $f_1, \dots, f_n \in \mathfrak{m}$. The **ring of cohomological operators** for R over Q is the graded polynomial $\mathcal{S} = \mathcal{S}_Q = Q[\chi_1, \dots, \chi_n]$ where each χ_i has cohomological degree 2.¹ Let $\Gamma := \text{Hom}_Q(\mathcal{S}, Q)$ be the graded dual of $\mathcal{S}$, and note that Γ is a divided power algebra with Q -basis

$$\{t^{(a)} = t_1^{(a_1)} \dots t_n^{(a_n)} \mid |a| = a_1 + \dots + a_n \geq 0\}.$$

¹As Avramov wisely pointed out in [\[Avr10\]](#), “This will not be surprising once the χ_i ’s reveal their cohomological nature.”

We set $t_i^{(0)} = 1$ and $\chi_i^{(j)} = 0$ whenever $j < 0$. Then Γ is an $\mathcal{S}$ -module via

$$\chi_i \cdot t_1^{(a_1)} \dots t_n^{(a_n)} := \begin{cases} t_1^{(a_1)} \dots t_i^{(a_i-1)} \dots t_n^{(a_n)} & \text{if } a_i \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$

We will also consider the ring of cohomological operators $\mathcal{S}_R = \mathcal{S} \otimes_Q R$ of R over R and the ring of cohomological operators $\mathcal{S}_k = \mathcal{S} \otimes_Q k$ of R over k .

Construction 5.1.22 (Eisenbud–Shamash Construction). Let $(Q, \mathfrak{m}, k)$ be a noetherian local ring and consider $R = Q/(f_1, \dots, f_n)$ for some $f_1, \dots, f_n \in \mathfrak{m}$. Let M be an R -module, and F be a free resolution for M over Q . Let $\sigma = \{\sigma_\omega\}$ be a system of higher homotopies for $f_1, \dots, f_n$ on F . Let $\mathcal{S} = Q[\chi_1, \dots, \chi_n]$ be the ring of cohomological operators over R , and let Γ be the graded dual of $\mathcal{S}$. The **Shamash construction** or **Shamash resolution** on F and $\sigma = \{\sigma_i\}$ is the complex of R -modules $\text{Sh}(F, \sigma)$ given by

$$\text{Sh}(F, \sigma)_\ell := \bigoplus_{\substack{i_1 + \dots + i_n = d \\ d \geq 0}} Rt_1^{(i_1)} \dots t_n^{(i_n)} \otimes_Q F_{\ell-2d}$$

and differential

$$\partial(t_1^{(i_1)} \dots t_c^{(i_c)} \otimes u) = \sum_\omega t_1^{(i_1 - \omega_1)} \dots t_c^{(i_c - \omega_c)} \otimes \sigma_\omega(u).$$

Theorem 5.1.23 (Eisenbud, 1980 [Eis80]). Let Q be a regular local ring, $\underline{f} = f_1, \dots, f_n$ a regular sequence on Q , and $R = Q/(\underline{f})$. Let M be an R -module. Given a free resolution F for M over Q and a system of higher homotopies $\sigma = \{\sigma_\omega\}$ for $\underline{f}$ on F , the Eisenbud–Shamash construction $\text{Sh}(F, \sigma)$ is a free resolution for M over R .

However, resolutions over complete intersections are no longer eventually 2-periodic.

Remark 5.1.24. Let $R = Q/(\underline{f})$ be a complete intersection of codimension c , that is, assume that $(Q, \mathfrak{m})$ is a regular local ring and that $\underline{f} = f_1, \dots, f_c \in \mathfrak{m}^2$. Consider a finite free resolution F for M over Q , a system of higher homotopies σ for $\underline{f}$ on F , and the free resolution $G = \text{Sh}(F, \sigma)$ for M over R built by [Theorem 5.1.23](#). Let us try to repeat the idea of [Remark 5.1.10](#). In homological degrees above $\text{pdim}_Q(R)$, we now have

$$G_{2n} = \sum_{\substack{2a+2|i|=2n \\ a, i_j \geq 0}} Rt_1^{(i_1)} \dots t_c^{(i_c)} \otimes_Q F_{2a} \quad \text{and} \quad G_{2n+1} = \sum_{\substack{2a+2|i|=2n \\ a, i_j \geq 0}} Rt_1^{(i_1)} \dots t_c^{(i_c)} \otimes_Q F_{2a+1}.$$

Each F_{2j} appears repeatedly in each homological degree:

- In homological degree $2n$, F_{2j} appears with multiplicity the number of monomials in c variables of degree $n - j$.
- In homological degree $2n+1$, F_{2j+1} appears with multiplicity the number of monomials in c variables of degree $n - j$.

The number of monomials in c variables of degree d is

$$\binom{d+c-1}{d},$$

which is a polynomial in d of degree $c-1$. Since the ranks of the F_i are constant, we may consider them as coefficients, which leads us to conclude that $\text{rank}(G_n)$ grows at most like a polynomial f of degree $c-1$ in n . Therefore

$$\beta_n^R(M) \leq \text{rank}(G_n) \leq f(n).$$

In fact, this discussion shows that all finitely generated modules over a complete intersection have finite complexity:

Definition 5.1.25. We say that a finitely generated R -module M has **finite complexity** if there is a polynomial $f \in \mathbb{Z}[t]$ such that

$$\beta_i(M) \leq f(i)$$

for all i . If no such polynomial exists, we say M has **infinite complexity**.

We can even give a value to the complexity of a module.

Definition 5.1.26. Let M be a finitely generated R -module of finite complexity. If M has finite projective dimension, we say M has complexity 0, and write $\text{cx}(M) = 0$. Otherwise, we say M has complexity d , and write $\text{cx}(M) = d$, if $d-1$ is the smallest degree of a polynomial f with $\beta_i(M) \leq f(i)$ for all i .

Remark 5.1.27. Some authors define the complexity of M to be the least integer d such that $\beta_n(M) \leq C \cdot n^{d-1}$ for some constant C and all $n \gg 0$. This has the advantage that it includes complexity 0 as part of the definition. Also, note that this other definition is equivalent to our definition, as only the largest power term of the polynomial really matters, and when $\text{cx}(M) > 0$ we can always change the constant C so that the inequality applies to all (not just large) n .

Example 5.1.28. Complexity 1 means that M has infinite projective dimension but the Betti numbers are bounded above by a constant. For example, over $R = k[[x]]/(x^2)$, the module k has complexity 1, since the Betti numbers of k are constant: $\beta_i(k) = 1$ for all i .

In fact, we can do more: we can show that for every complete intersection R of codimension c , there exists a finitely generated R -module M with complexity exactly c .

Exercise 5.1.29. Let $(Q, \mathfrak{m}, k)$ be a regular local ring and $R = Q/(f_1, \dots, f_c)$ for some $f_1, \dots, f_c \in \mathfrak{m}^2$. Show that under an appropriate choice of resolution and system of higher homotopies, the Shamash construction leads to the minimal free resolution for k over R .

Exercise 5.1.30. Let $(Q, \mathfrak{m}, k)$ be a regular local ring and $R = Q/(f_1, \dots, f_c)$ for some $f_1, \dots, f_c \in \mathfrak{m}^2$. Show that exists a finitely generated R -module M with complexity exactly c .

Exercise 5.1.31. Let Q be a regular local ring and $R = Q/I$ with I minimally generated by $f = f_1, \dots, f_n$. Let F be a free resolution of R over Q that has a structure of a DG Q -algebra. Fix $e_1, \dots, e_n \in F_1$ with $\partial(e_i) = f_i$. Show that we get a system of higher homotopies $\{\sigma_\omega\}$ for $\underline{f}$ on F by setting

$$\sigma_{e_i}(-) = e_i \cdot - \quad \text{and} \quad \sigma_\omega(u) = 0 \text{ for all } |\omega| \geq 2.$$

When $R = Q/I$ is not a complete intersection, we often can (and should!) replace R by $E = \text{Kos}^Q(f_1, \dots, f_n)$, where I is minimally generated by $f_1, \dots, f_n$. The rough idea is that E behaves like a complete intersection for many purposes, since we can also obtain semifree E -resolutions following the Eisenbud–Shamash construction.

Remark 5.1.32. Given any R -module M , we can give M an E -module structure by taking the action of the generators $e_1, \dots, e_n$ to be zero. In fact, viewing M as a complex concentrated in degree zero, one can check that this gives M the structure of a dg E -module.

The idea for the following theorem already appears in work of Avramov–Buchweitz [AB00a]; a version of this can also be found in Pollitz’s work [Pol21] and in upcoming work of Kalir–Pollicove–Watson.

Theorem 5.1.33. *Let $R = Q/I$ be a noetherian local ring with $I = (f_1, \dots, f_n)$, and set $E = \text{Kos}^Q(f_1, \dots, f_n)$. Let M be a finitely generated R -module, and let F be a free resolution for M over Q . Let $\sigma = \{\sigma_\omega\}$ be a system of higher homotopies for $f_1, \dots, f_n$ over F . Then $\text{Sh}(F, \sigma, E) = E \otimes_Q \Gamma \otimes_Q F$ is a semifree resolution for M over E , where*

$$\text{Sh}(F, \sigma, E)_\ell := \bigoplus_{\substack{i_0 + i_1 + \dots + i_n = d \\ d \geq 0}} E_{i_0} \otimes t_1^{(i_1)} \dots t_n^{(i_n)} \otimes_Q F_{\ell-2d}$$

and the differential is given by

$$\partial = \sum_{|\omega| \geq 0} (-1)^{|\omega|} \text{id}_E \otimes \chi^\omega \otimes \sigma_\omega + \sum_{i=1}^n e_i \otimes \chi_i \otimes 1 + \partial_E \otimes \text{id}_\Gamma \otimes \text{id}_F$$

that is,

$$\begin{aligned} \partial(e \otimes t_1^{(i_1)} \dots t_n^{(i_n)} \otimes u) &= \sum_{\omega} e \otimes t_1^{(i_1 - \omega_1)} \dots t_n^{(i_n - \omega_n)} \otimes \sigma_\omega(u) \\ &+ \sum_{j=1}^n e \otimes t_1^{(i_1)} \dots t_j^{(i_j - 1)} \dots t_n^{(i_n)} \otimes e_j u \\ &+ \partial_E(e) \otimes t_1^{(i_1)} \dots t_n^{(i_n)} \otimes u. \end{aligned}$$

Remark 5.1.34. The $\mathcal{S}$ -module structure on Γ we defined before, given by

$$\chi_i \cdot t_1^{(a_1)} \cdots t_n^{(a_n)} := \begin{cases} t_1^{(a_1)} \cdots t_i^{(a_i-1)} \cdots t_n^{(a_n)} & \text{if } a_i \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

gives $\text{Sh}(F, \sigma, E)$ the structure of a graded module over $\mathcal{S}$.

In recent years, Pollitz has exploited this idea of using E as a replacement for R in various directions; we will see some of them in the rest of the chapter.

Exercise 5.1.35. Let $(Q, \mathfrak{m}, k)$ be a noetherian local ring, and let N be a Q -module. Show that if F is a bounded below complex of free Q -modules, then

$$\text{Hom}_Q(F, N) \cong \text{Hom}_Q(F, Q) \otimes_Q N.$$

Theorem 5.1.36. Let $R = Q/I$ be a noetherian local ring with $I = (f_1, \dots, f_n)$, and set $E = \text{Kos}^Q(f_1, \dots, f_n)$. Let M be a finitely generated R -module, and let F be a free resolution for M over Q . Let $\sigma = \{\sigma_\omega\}$ be a system of higher homotopies for $f_1, \dots, f_n$ over F . Consider the ring of cohomological operators $\mathcal{S} = \mathcal{S}_k = k[\chi_1, \dots, \chi_n]$. Then there is an isomorphism of $\mathcal{S}$ -modules

$$\text{Hom}_E(\text{Sh}(F, \sigma, E)) \cong \mathcal{S} \otimes_Q \text{Hom}_Q(F, k)$$

Moreover, $\text{Ext}_E(M, k)$ is a finitely generated graded $\mathcal{S}$ -module.

Proof. By [Theorem 5.1.33](#), $\text{Sh}(F, \sigma, E)$ is a free resolution for M over E . Therefore,

$$\text{Hom}_E(M, k) \cong H(\text{Hom}_E(\text{Sh}(F, \sigma, E), k).$$

Besides $\mathcal{S} = \mathcal{S}_k$, consider also $\mathcal{S}_Q$ and its dual $\Gamma = \Gamma_Q$. First we do a calculation that will soon reveal itself to be useful:

$$\begin{aligned} \text{Hom}_Q(\Gamma, k) &\cong \text{Hom}_Q(\text{Hom}_Q(\mathcal{S}_Q, Q), k) && \text{by definition of } \Gamma \\ &\cong \text{Hom}_Q(\text{Hom}_Q(\mathcal{S}_Q, Q), Q) \otimes_Q k \\ &\cong \mathcal{S}_Q \otimes_Q k \\ &\cong \mathcal{S}_k. \end{aligned}$$

Then

$$\begin{aligned} \text{Hom}_E(E \otimes_Q \Gamma \otimes_Q F, k) &\cong \text{Hom}_Q(\Gamma \otimes_Q F, \text{Hom}_E(E, k)) && \text{by Hom-tensor adjunction} \\ &\cong \text{Hom}_Q(\Gamma \otimes_Q F, k) \\ &\cong \text{Hom}_Q(F, \text{Hom}_Q(\Gamma, k)) && \text{by Hom-tensor adjunction} \\ &\cong \text{Hom}_Q(F, \mathcal{S}_k) && \text{by the calculation above} \\ &\cong \text{Hom}_Q(F, Q) \otimes_Q \mathcal{S}_k && \text{by Exercise 5.1.35} \\ &\cong \text{Hom}_Q(F, Q) \otimes_Q k \otimes_Q \mathcal{S}_k && \text{since } k \otimes_Q \mathcal{S}_k \otimes_Q \mathcal{S}_k \\ &\cong \text{Hom}_Q(F, k) \cong \mathcal{S}_k && \text{by Exercise 5.1.35.} \end{aligned}$$

Therefore,

$$\mathrm{Hom}_E(M, k) \cong \mathrm{H}(\mathrm{Hom}_E(\mathrm{Sh}(F, \sigma, E), k) \cong \mathrm{H}(\mathrm{Hom}_Q(F, k) \cong \mathcal{S}_k).$$

On the other hand, now back to $\mathcal{S} = \mathcal{S}_k$, we get an induced graded $\mathcal{S}$ -module structure on $\mathrm{Ext}_E(M, k)$ with

$$\chi_i : \mathrm{Ext}_R^i(M, k) \longrightarrow \mathrm{Ext}_R^{i+2}(M, k).$$

Note that the graded module underlying $\mathcal{S} \otimes_Q \mathrm{Hom}_E(\mathcal{T}, k)$ is a finitely generated graded $\mathcal{S}$ -module, and thus $\mathrm{Ext}_E(M, k)$ is a finitely generated graded $\mathcal{S}$ -module. $\square$

5.2 Cohomological support varieties

Remark 5.2.1. Let R be a noetherian local ring with minimal regular presentation $\widehat{R} \cong Q/I$ for some regular local ring $(Q, \mathfrak{m}, k)$. Note that $I/\mathfrak{m}I$ is a k -vector space of dimension $n = \mu(I)$, which we will identify with affine n -space $\mathbb{A}_k^n$. A choice of coordinates for $\mathbb{A}_k^n$ corresponds to a choice of basis for k^n , and thus to a choice of minimal generating set $\underline{f} = f_1, \dots, f_n$ for I . We will write $[f] := f + \mathfrak{m}I$ for the class of $f \in I$ in $I/\mathfrak{m}I$.

Any R -module M is also a module over $Q/(f)$ for any $f \in I$, since $fM = 0$.

Definition 5.2.2 (Cohomological support varieties). Let R be a noetherian local ring and let M be a finitely generated R -module. Let $\widehat{R} \cong Q/I$ be a minimal regular presentation with $(Q, \mathfrak{m}, k)$. The **cohomological support variety** of M is given by

$$V_R(M) := \{[f] \in I/\mathfrak{m}I \mid [f] = 0 \text{ or } \text{pdim}_{Q/(f)}(\widehat{M}) = \infty\}.$$

We will informally refer to the hypersurfaces $Q/(f)$, where f ranges over the minimal generators of I , as **intermediate hypersurfaces**.

It is not clear from the definition above that this is in fact a variety; we will prove this later via an alternative definition. It is also not clear that this is well-defined, meaning that it does not depend on the choice of representative f for $[f]$, nor that the definition does not depend on the choice of minimal regular presentation, but we will skip such details.

Remark 5.2.3. Suppose that $[f]$ and $[g]$ are two points on the same line through the origin, but not the origin, so we can assume that $g = \lambda f$ for some unit $\lambda \in Q$. Then $(f) = (g)$, so $[f] \in V_R(M)$ if and only if $[g] \in V_R(M)$. Note moreover that $[0] \in V_R(M)$ by definition. This shows that $V_R(M)$ is a union of lines through the origin. Adding to this the fact that $V_R(M)$ is an affine variety, we conclude that it is in fact the affine cone of a projective variety. However, there are advantages to considering $V_R(M)$ as an affine variety instead of the appropriate projective version, which will unfortunately not be evident in these lectures.

Remark 5.2.4. Fix a minimal generating set $f_1, \dots, f_n$ for I . For each $a \in k$, write $\tilde{a}$ for a lift of a to Q . For each $a \in \mathbb{A}_k^n$, let $f_a := \tilde{a}_1 f_1 + \dots + \tilde{a}_n f_n$, where $\tilde{a}_i$ for a lift of a_i to Q . Note that any two choices of lift $\tilde{a}_i$ for a_i differ by an element in $\mathfrak{m}$. Therefore, given two lifts f and g of a , note that $f - g \in \mathfrak{m}I$. Conversely, given a and b in $\mathbb{A}_k^n$,

$$f_a - f_b \in \mathfrak{m}I \iff (a_1 - b_1)f_1 + \dots + (a_n - b_n)f_n \in \mathfrak{m}I.$$

By NAK, the last condition is equivalent to asking for $a_i - b_i \in \mathfrak{m}$ for all i . Thus if $f - g \in \mathfrak{m}I$ then $a = b$. We conclude that the assignment $a \mapsto f_a$ induces a bijection between $\mathbb{A}_k^n$ and $I/\mathfrak{m}I$.

We can now rewrite the definition of $V_R(M)$ as

$$V_R(M) = \{a \in \mathbb{A}_k^n \mid a = \mathbf{0} \text{ or } \text{pdim}_{Q/(f_a)}(\widehat{M}) = \infty\}.$$

However, we have not yet proved well-definedness of $V_R(M)$: we still need to check that if $f - g \in \mathfrak{m}I$, then

$$\text{pdim}_{Q/(f)}(\widehat{M}) = \infty \iff \text{pdim}_{Q/(g)}(\widehat{M}) = \infty.$$

We will prove this in [Lemma 5.2.18](#).

Example 5.2.5. Let R be a noetherian local ring with minimal regular presentation $\widehat{R} \cong Q/I$, where $(Q, \mathfrak{m}, k)$ is a regular local ring and $I \subseteq \mathfrak{m}^2$. Assume $n := \mu(I) \geq 1$. For every minimal generator $f \in I \subseteq \mathfrak{m}^2$, by [Exercise 0.2.17](#) the ring $Q/(f)$ is not regular, and thus $\text{pdim}_{Q/(f)} k = \infty$ by Auslander–Buchsbaum–Serre ([Theorem 0.2.20](#)). We conclude that

$$V_R(k) = \mathbb{A}_k^n.$$

Example 5.2.6. Suppose that R is a complete intersection, and fix a minimal regular presentation $\widehat{R} \cong Q/I$. Any minimal generator $f \in I \setminus \mathfrak{m}I$ can be completed to a minimal generating set $\underline{f} = f, f_2, \dots, f_n$ for I , and $\underline{f}$ is necessarily a regular sequence. In particular, the sequence $f_2, \dots, f_n$ is regular on $Q/(f)$, so the Koszul complex $\text{Kos}^{Q/(f)}(f_2, \dots, f_n)$ is a minimal free resolution for $\widehat{R}$ over $Q/(f)$. In particular, $\text{pdim}_{Q/(f)} \widehat{R} < \infty$, and we conclude that $V_R(R) = \{\mathbf{0}\}$.

Pollitz showed that the converse holds, that is, that the fact that $V_R(R) = \{\mathbf{0}\}$ characterizes complete intersections; we will discuss his proof in the next section.

Above we saw two extreme examples where we were able to compute $V_R(M)$ completely. But in general, how can we decide whether M has finite projective dimension over each intermediate hypersurface? We can do this taking advantage of the Eisenbud–Shamash construction we saw above.

Construction 5.2.7 (How to compute cohomological support varieties). Let R be a noetherian local ring and let $\widehat{R} \cong Q/I$ be a minimal regular presentation for R with $(Q, \mathfrak{m}, k)$. Suppose I is minimally generated by $\underline{f} = f_1, \dots, f_n$. Fix a finite free resolution F for $\widehat{M}$ over Q with $F_i = 0$ for all $i \geq 2d + 1$, and a system of higher homotopies $\{\sigma_\omega\}$ for $\underline{f}$ on F . Consider the ring of cohomological operators $\mathcal{S} = \mathcal{S}_k = k[\chi_1, \dots, \chi_n]$ in n variables. Fix bases for each k -vector space $F_i^* \text{Hom}_Q(F_i, k)$, and let $\sigma_\omega^{(i)}$ be the matrix that represents the map

$$\text{id}_{\mathcal{S}} \otimes_Q \text{Hom}_Q(\sigma_\omega, \text{id}_k) : \mathcal{S} \otimes_Q F_{i+2|\omega|-1}^* \longrightarrow \mathcal{S} \otimes_Q F_i^*$$

in the chosen bases. For all appropriate i and j , set

$$\sigma_i^j := \sum_{2|\omega|-1=j-i} \chi_1^{\omega_1} \cdots \chi_n^{\omega_n} \sigma_\omega^{(i)} : \mathcal{S} \otimes_Q F_j^* \longrightarrow F_i^*.$$

Note that σ_i^j is a matrix representing the map from degree i to degree j .

Let $N := \sum_{i=0}^d \text{rank } F_{2i}$, and consider the following two $N \times N$ matrices:

	F_0^*	F_2^*	F_4^*	$\cdots$	F_{2d}^*
F_1^*	σ_1^0	σ_1^2	0	$\cdots$	0
F_3^*	σ_3^0	σ_3^2	σ_3^4		0
$\cdots$	$\vdots$		$\ddots$	$\vdots$	
F_{2d-3}^*	σ_{2d-3}^0	σ_{2d-3}^2	σ_{2d-3}^4	$\cdots$	0
F_{2d-1}^*	σ_{2d-1}^0	σ_{2d-1}^2	σ_{2d-1}^4	$\cdots$	σ_{2d-1}^{2d}

$A(\underline{\chi})$

	F_1^*	F_3^*	F_5^*	$\cdots$	F_{2d-1}^*
F_0^*	σ_0^1	σ_2^1	0	$\cdots$	0
F_2^*	σ_2^1	σ_2^3	σ_2^5		0
$\cdots$	$\vdots$		$\ddots$	$\vdots$	
F_{2d-2}^*	σ_1^{2d-2}	σ_3^{2d-2}	σ_{2d-2}^5	$\cdots$	0
F_{2d}^*	σ_{2d}^1	σ_2^{2d}	σ_3^{2d}	$\cdots$	σ_{2d-1}^{2d}

$B(\underline{\chi})$

The matrix $A(\chi)$ represents the map from even degrees to odd degrees, while $B(\chi)$ represents the map from odd degrees to even degrees. Alternatively, we may consider the matrix representing the complete map, which is given the following $(2N) \times (2N)$ matrix:

$$C(\chi) = \begin{array}{c|cccccc} & F_0^* & F_1^* & F_2^* & \cdots & F_{2d-1}^* & F_{2d}^* \\ \hline F_0^* & 0 & \sigma_1^0 & 0 & \cdots & 0 & 0 \\ F_1^* & \sigma_1^0 & 0 & \sigma_1^2 & & 0 & 0 \\ F_2^* & 0 & \sigma_2^1 & 0 & & 0 & 0 \\ \cdots & & \vdots & & \ddots & \vdots & \vdots \\ F_{2d-2}^* & 0 & \sigma_{2d-2}^1 & 0 & \cdots & \sigma_{2d-2}^{2d-1} & 0 \\ F_{2d-1}^* & \sigma_{2d-1}^0 & 0 & \sigma_{2d-1}^2 & \cdots & 0 & \sigma_{2d-1}^{2d} \\ F_{2d}^* & 0 & \sigma_{2d}^1 & 0 & \cdots & \sigma_{2d}^{2d-1} & 0 \end{array}$$

Exercise 5.2.8. Show that the matrices $A(\chi)$ and $B(\chi)$ defined in [Construction 5.2.7](#) satisfy

$$A(\chi) \cdot B(\chi) = 0 \quad \text{and} \quad B(\chi) \cdot A(\chi) = 0.$$

Remark 5.2.9. Whenever we are given two $N \times N$ matrices A and B with entries in a ring R such that $AB = 0$ and $BA = 0$, these two matrices determine a $\mathbb{Z}/2$ complex

$$\cdots \xrightarrow{B} R^N \xrightarrow{A} R^N \xrightarrow{B} R^N \xrightarrow{A} \cdots.$$

In particular, by [Exercise 5.2.8](#), the two matrices $A(\chi)$ and $B(\chi)$ in [Construction 5.2.7](#) determine a $\mathbb{Z}/2$ complex over $\mathcal{S} = k[\chi_1, \dots, \chi_n]$.

Theorem 5.2.10. Fix an R -module M and consider the matrices $A(\chi)$, $B(\chi)$, and $C(\chi)$ we obtained in [Construction 5.2.7](#), and set consider the ring of cohomological operators $\mathcal{S} = \mathcal{S}_k = k[\chi_1, \dots, \chi_n]$. Then $V_R(M)$ is the variety defined by the vanishing of the ideal $I_N(C)$ of N -minors of $C(\chi)$.

Equivalently, $V_R(M)$ is the locus where the $\mathbb{Z}/2$ complex determined by $A(\chi)$ and $B(\chi)$ is not exact:

$$V_R(M) = \{a \in \mathbb{A}_k^n \mid \cdots \xrightarrow{B(\chi=a)} \mathcal{S}^N \xrightarrow{A(\chi=a)} \mathcal{S}^N \xrightarrow{B(\chi=a)} \mathcal{S}^N \xrightarrow{A(\chi=a)} \cdots \text{ is not exact}\}.$$

Proof. First, consider the following two matrices $A(\chi)$ and $B(\chi)$ we saw in [Construction 5.2.7](#). Fix $\mathbf{0} \neq a \in \mathbb{A}_k^n$, and $f_a := \widetilde{a}_1 f_1 + \cdots \widetilde{a}_n f_n$, where $\widetilde{a}_i$ is a lift of a_i to Q . By [Exercise 5.1.19](#), the maps

$$\sigma_i = \sum_{|\omega|=i} a_1^{\omega_1} \cdots a_n^{\omega_n} \sigma_\omega$$

form a system of higher homotopies for f_a on F .

Moreover, by [Lemma 5.1.14](#), $\text{pdim}_{Q/(f_a)}(\widehat{M}) < \infty$ if and only if the $\mathbb{Z}/2$ complex

$$\mathcal{P}_a^* := \cdots \longrightarrow \bigoplus_{i \geq 0} F_{2i}^* \xrightarrow{\partial} \bigoplus_{i \geq 0} F_{2i+1}^* \xrightarrow{\partial} \bigoplus_{i \geq 0} F_{2i}^* \longrightarrow \cdots$$

with differential

$$\partial = \sum_{\omega} a_1^{\omega_1} \cdots a_n^{\omega_n} \text{Hom}(\sigma_{\omega}, k)$$

is exact. Thus $a \in V_R(M)$ if and only if $\mathcal{P}_a^*$ is not exact. To simplify the notation, write

$$\mathcal{P}_a^{\text{even}} := \bigoplus_{i \geq 0} F_{2i}^* \quad \text{and} \quad \mathcal{P}_a^{\text{odd}} := \bigoplus_{i \geq 0} F_{2i+1}^*.$$

Note that $\mathcal{P}_a$ is in fact given by

$$\mathcal{P}_a = \cdots \xrightarrow{B(a)} \mathcal{P}_a^{\text{even}} \xrightarrow{A(a)} \mathcal{P}_a^{\text{odd}} \xrightarrow{B(a)} \mathcal{P}_a^{\text{even}} \xrightarrow{A(a)} \cdots$$

where $A(a)$ and $B(a)$ are obtained by setting $\chi_i = a_i$ in $A(\underline{\chi})$ and $B(\underline{\chi})$.

Now let us go back to the matrix C in the statement, and note that after reordering the rows and columns

$$C = \begin{bmatrix} 0 & B \\ A & 0 \end{bmatrix}.$$

Note moreover that

$$C^2 = \begin{bmatrix} BA & 0 \\ 0 & AB \end{bmatrix} = 0,$$

and thus nonexactness of $\mathcal{P}_a$ is equivalent to nonexactness of

$$\cdots \xrightarrow{C(a)} \mathcal{P}_a^{\text{even}} \oplus \mathcal{P}_a^{\text{odd}} \xrightarrow{C(a)} \mathcal{P}_a^{\text{even}} \oplus \mathcal{P}_a^{\text{odd}} \xrightarrow{C(a)} \mathcal{P}_a^{\text{even}} \oplus \mathcal{P}_a^{\text{odd}} \xrightarrow{C(a)} \cdots$$

or equivalently of the differential module determined by $C(a)$.

Any matrix C of size $(2N) \times (2N)$ satisfying $C^2 = 0$ has rank at most N , and C determines an exact complex precisely when it has rank N . Thus

$$a \in V_R(M) \iff \mathcal{P}_a^* \text{ is not exact} \iff \text{rank } C(a) < N.$$

The space of such a is determined by the vanishing of the $N \times N$ minors of C . □

We now take [Theorem 5.2.10](#) as a recipe for computing $V_R(M)$.

Example 5.2.11. Let $Q = k[[x, y]]$, $I = (x^2, xy)$, and $R = Q/I$, and let us find $V_R(R)$. From [Example 5.1.5](#), and [Exercise 5.1.6](#), we take

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q & \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x^2 & xy \end{bmatrix}} & Q \longrightarrow 0 \\ & & \downarrow \cdot x^2 & \swarrow \begin{bmatrix} 0 & -x \end{bmatrix} & \downarrow \cdot x^2 & \swarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \downarrow \cdot x^2 \\ 0 & \longrightarrow & Q & \longrightarrow & Q^2 & \longrightarrow & Q \longrightarrow 0 \end{array} \quad \sigma_{(1,0)}$$

$$\begin{array}{ccccccc} 0 & \longrightarrow & Q & \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} & Q^2 & \xrightarrow{\begin{bmatrix} x^2 & xy \end{bmatrix}} & Q \longrightarrow 0 \\ & & \downarrow \cdot x^2 & \swarrow \begin{bmatrix} x & 0 \end{bmatrix} & \downarrow \cdot x^2 & \swarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix} & \downarrow \cdot x^2 \\ 0 & \longrightarrow & Q & \longrightarrow & Q^2 & \longrightarrow & Q \longrightarrow 0 \end{array} \quad \sigma_{(0,1)}$$

Since our resolution has length 2, there are no other higher homotopies. Let us compute $A(\chi)$ and $B(\chi)$. Write e_1, e_2 for the two basis elements in F_1 with $\partial(e_1) = x^2$ and $\partial(e_2) = xy$, and let v be the basis element for F_2 , so that $\partial(v) = ye_1 - xe_2$. Note that when we tensor down to k , we keep only units. In particular, since we picked F to be minimal, the differential will disappear, and only the units in our homotopies will survive.

Our matrices are

$$\begin{array}{c|cc} & 1 & v \\ \hline e_1 & 0 & 0 \\ e_2 & 0 & 0 \end{array} \quad \begin{array}{c|cc} & e_1 & e_2 \\ \hline 1 & \chi_1 & \chi_2 \\ v & 0 & 0 \end{array}$$

In this case, it is clear that the complex is *never* exact, and thus $V_R(R)$ has full support. Alternatively, one might write the matrix $C(\chi)$ from the statement of [Theorem 5.2.10](#)

$$C(\chi) = \begin{array}{c|cccc} & 1 & e_1 & e_2 & v \\ \hline 1 & 0 & \chi_1 & \chi_2 & 0 \\ e_1 & 0 & 0 & 0 & 0 \\ e_2 & 0 & 0 & 0 & 0 \\ v & 0 & 0 & 0 & 0 \end{array}$$

and note that C has rank 1, so its 2×2 minors vanish. Thus $V_R(M)$ is the variety corresponding to (0) , and M has full support.

Remark 5.2.12. If we transpose the matrices $A(\chi)$, $B(\chi)$, and $C(\chi)$ from [Theorem 5.2.10](#), and follow the same idea as in the proof of [Theorem 5.2.10](#), we instead obtain the $\mathbb{Z}/2$ complex

$$\mathcal{P}_a^* := \cdots \longrightarrow \bigoplus_{i \geq 0} \text{Hom}_Q(F_{2i}, k) \xrightarrow{\partial} \bigoplus_{i \geq 0} \text{Hom}_Q(F_{2i+1}, k) \xrightarrow{\partial} \bigoplus_{i \geq 0} \text{Hom}_Q(F_{2i}, k) \longrightarrow \cdots$$

which computes the stable Ext modules over $Q_a := a_1 f_1 + \cdots + a_n f_n$,

$$\text{Ext}_{Q_a}^{2i}(\widehat{M}, k) \text{ and } \text{Ext}_{Q_a}^{2i+1}(\widehat{M}, k) \text{ for } i \gg 0.$$

Since by [Exercise 0.2.6](#) we can use both Ext and Tor to compute betti numbers, $a \in V_R(M)$ if and only if $\mathcal{P}_a^*$ is not exact. We conclude that [Theorem 5.2.10](#) also holds for the transpose of the matrices A , B , and C .

However, as we will see in [Remark 5.2.13](#), the Ext version of this idea is conceptually more powerful: when R is a complete intersection, both the total Tor and Ext modules $\text{Tor}_*^R(\widehat{M}, k)$ and $\text{Ext}_R^*(\widehat{M}, k)$ can be viewed as graded modules over $\mathcal{S} = k[\chi_1, \dots, \chi_n]$, but while $\text{Tor}_*^R(\widehat{M}, k)$ is only an artinian module, $\text{Ext}_R^*(\widehat{M}, k)$ is noetherian.

Remark 5.2.13. Let k be an algebraically closed field. Let $R = Q/I$ where $(Q, \mathfrak{m}, k)$ is a regular local ring and fix a minimal generating set $\underline{f} = f_1, \dots, f_n$ for I .

Consider the ring of cohomological operators $\overline{\mathcal{S}} = k[\chi_1, \dots, \chi_c]$ for R over k . Let F be any finite free resolution of M over Q and fix a system of higher homotopies $\sigma = \{\sigma_\omega\}$ for $\underline{f}$ on F .

With this data, we construct a $\mathbb{Z}/2$ complex $\mathcal{T}$ as in [Construction 5.2.7](#), determined by the two matrices $A(\chi)$ and $B(\chi)$, given by

$$\mathcal{T}(\chi) = \cdots \longrightarrow \bigoplus_{i \geq 0} \mathcal{S} \otimes_Q \operatorname{Hom}_Q(F_{2i}, k) \longrightarrow \bigoplus_{i \geq 0} \mathcal{S} \otimes_Q \operatorname{Hom}_Q(F_{2i+1}, k) \longrightarrow \cdots.$$

For each $a \in \mathbb{A}_k^n$, consider the $\mathbb{Z}/2$ complex obtained by taking

$$\mathcal{T}_a := \mathcal{T} \otimes_{\mathcal{S}} \mathcal{S}/(\chi_1 - a_1, \dots, \chi_n - a_n).$$

This coincides with the $\mathbb{Z}/2$ complex $\mathcal{P}_a^*$ determined by the matrices $A(\chi = a)$ and $B(\chi = a)$ obtained from $A(\chi)$ and $B(\chi)$ by setting $\chi_i = a_i$, as in the proof of [Construction 5.2.7](#). Thus

$$V_R(M) = \{a \in \mathbb{A}_k^n \mid \mathcal{T} \otimes_{\mathcal{S}} \mathcal{S}/(\chi_1 - a_1, \dots, \chi_n - a_n) \text{ is not exact}\}.$$

Fix $a \in \mathbb{A}_k^n$ and let $\mathfrak{m} = \mathfrak{m}_a = (\chi_1 - a_1, \dots, \chi_n - a_n)$. Using NAK, one can show that

$$\mathcal{T} \otimes_{\mathcal{S}} \mathcal{S}/\mathfrak{m} \cong \mathcal{T}_{\mathfrak{m}} \otimes_{\mathcal{S}_{\mathfrak{m}}} \mathcal{S}_{\mathfrak{m}}/\mathfrak{m}_{\mathfrak{m}}.$$

By [Exercise 5.2.14](#) below,

$$\mathcal{T} \otimes_{\mathcal{S}} \mathcal{S}/\mathfrak{m} \text{ is exact} \iff \mathcal{T}_{\mathfrak{m}} \text{ is exact}.$$

On the other hand, $\mathcal{T}$ computes $\operatorname{Ext}_E(M, k)$ by [Theorem 5.1.36](#), and since localization is exact and thus commutes with taking homology, we conclude that

$$a \notin V_R(M) \iff \mathcal{T} \otimes_{\mathcal{S}} \mathcal{S}/\mathfrak{m} \text{ is exact} \iff \operatorname{Ext}_E(M, k)_{\mathfrak{m}} = 0.$$

Therefore,

$$a \in V_R(M) \iff (\chi_1 - a_1, \dots, \chi_n - a_n) \in \operatorname{Supp}_{\mathcal{S}}(\operatorname{Ext}_E(M, k)).$$

In fact, one most common defines $V_R(M)$ as

$$V_R(M) := \operatorname{Supp}_{\mathcal{S}}(\operatorname{Ext}_E(M, k)).$$

This explains the words *cohomological* and *support* in the name.

Now the key point is that $\operatorname{Ext}_E(M, k)$ is a finitely generated $\mathcal{S}$ -module, which we also saw in [Theorem 5.1.36](#). By Nullstellensatz, we conclude that

$$\sqrt{\operatorname{ann}_{\mathcal{S}} \operatorname{Ext}_E(M, k)} = \bigcap_{a \in V_R(M)} (\chi_1 - a_1, \dots, \chi_n - a_n).$$

Thus the radical ideal in $\mathcal{S}$ defining the variety $V_R(M)$ determines the support of $\operatorname{Ext}_E(M, k)$.

Exercise 5.2.14. Let $(Q, \mathfrak{m})$ be a regular local ring and F be a complex of finitely generated free Q -modules, not necessarily bounded on either side. Show that F is exact if and only if $F \otimes_Q Q/\mathfrak{m}$ is exact.

Remark 5.2.15. It is a well-known fact from dimension theory (see, for example, [Mat89, Theorem 13.4]) that the dimension of a finitely generated graded module is the rate at which the rank of the graded pieces grow. We have thus shown in Remark 5.2.13 that $\dim(V_R(M))$ is the dimension of the graded $\mathcal{S}$ -module $\text{Ext}_E(M, k)$. When R is a complete intersection, note that E is quasiisomorphic to R , so rather than taking $\text{Ext}_E(M, k)$, we may instead take $\text{Ext}_R(M, k)$. Thus when R is a complete intersection (see Avramov–Buchweitz, [AB00b])

$$\text{cx}_R(M) = \dim(V_R(M)).$$

Exercise 5.2.16. Let $\varphi: A \rightarrow B$ be a quasiisomorphism of dg algebras and let M and N be dg B -modules. Show that the induced map

$$\text{Ext}_A(M, N) \rightarrow \text{Ext}_B(M, N)$$

is an isomorphism of dg modules.

Remark 5.2.17. Note that when R is a complete intersection, E is quasiisomorphic to R , and thus by Exercise 5.2.16 we get an isomorphism of dg modules

$$\text{Ext}_R(M, k) \cong \text{Ext}_E(M, k).$$

In summary, we have no need to replace R with E . Thus in this case,

$$V_R(M) = \text{Supp}_{\mathcal{S}}(\text{Ext}_R(M, k)).$$

Lemma 5.2.18. *Our definition of cohomological support varieties is well-defined.*

Proof. We saw in Remark 5.2.4 that the assignment $a \mapsto f_a$ from $\mathbb{A}_n^k$ into I induces a well-defined bijection $a \mapsto [f_a]$ between $\mathbb{A}_n^k$ and $I/\mathfrak{m}I$. It remains to check that if $f - g \in \mathfrak{m}I$, then

$$\text{pdim}_{Q/(f)}(\widehat{M}) = \infty \iff \text{pdim}_{Q/(g)}(\widehat{M}) = \infty.$$

However, since f and g correspond to the same point $a \in \mathbb{A}_n^k$, Remark 5.2.13 gives us that the maximal ideal $\mathfrak{m}_a$ corresponding to a is key to determining whether the point is in $V_R(M)$: indeed,

$$\text{pdim}_{Q/(f)}(\widehat{M}) = \infty \iff \text{Ext}_E(\widehat{M}, k)_{\mathfrak{m}_a} = 0 \iff \text{pdim}_{Q/(g)}(\widehat{M}).$$

□

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Nomenclature

$(-)^{\vee}$	k -dual $\mathrm{Hom}_k(-, k)$
$[x]$	homology class of the cycle x
$\mathrm{ad}(z)$	adjoint map of z , given by $[z, -]$
quadmod	category of quadratic models
$\mathrm{cone}(\varphi)$	the cone of φ
$\mathrm{depth}(M)$	depth of M (wrt the maximal ideal)
$\mathrm{Der}_Q(A, B)$	A -linear derivations $A \rightarrow B$
$\mathrm{Der}_k(A, A)$	k -linear derivations on A
$\mathrm{quad}(A)$	quadratic part of A
$\mathrm{embdim}(R)$	embedding dimension of R , given by $\mu(\mathfrak{m})$
$\mathrm{Kos}(x_1, \dots, x_n)$	the Koszul complex on $x_1, \dots, x_n$
$\mathrm{Lie}(V)$	free Lie algebra on V
$\mathcal{S}$	ring of cohomological operators
$\mu(M)$	minimal number of generators of M , given by $\dim_k(M/\mathfrak{m}M)$
$\mathrm{codepth}(R)$	codepth of R , defined as $\mathrm{embdim}(R) - \mathrm{depth}(R)$
∂_M	differential on M
$\mathrm{pdim}_R(M)$	projective dimension of M over R
$\pi^*(A)$	homotopy Lie algebra of A
$\pi^*(R)$	homotopy Lie algebra of R
$\mathrm{rad}(L)$	radical of the graded Lie algebra L
$\mathrm{Sh}(F, \sigma)$	Shamash construction

$\sim$	We write $f \sim g$ to indicate f and g are homotopic maps
$\simeq$	quasiisomorphism
ΣC	shift of C , with $(\Sigma C)_n = C_{n-1}$ and $\partial^{\Sigma C} = -\partial^C$
$\Sigma^\ell C$	ℓ th shift of C , with $(\Sigma^\ell C)_n = C_{n-\ell}$ and $\partial^{\Sigma^\ell C} = (-1)^\ell \partial^C$
$\text{Syz}_n(M)$	n th syzygy of M
$\underline{x}$	shorthand for $x_1, \dots, x_n$
$V_R(M)$	cohomological support variety of M over R
$\zeta(L)$	center of the graded Lie algebra L
$A[x \mid \partial(x) = z]$	semifree extension of A obtained by adjoining x to kill the cycle z
A^{Lie}	graded Lie algebra on the associated algebra A with bracket the graded commutator
K^R	Koszul complex on a minimal generating set for $\mathfrak{m}_R$
$M^\natural$	underlying graded module
$Q[X]$	semifree extension of Q adjoining divided power variables in even degrees
$Q[X]$	semifree extension of Q , adjoining polynomial variables in even degrees
RLR	regular local ring

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